Khalifa University · Department of Computer Science

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PROJECT TITLE

Intelligent AI-Powered Circular Campus Resource Exchange and Asset Life Cycle Management System

DOCUMENT

Phases 4 & 5 – Design Document (Architecture Design and Detailed Design)

GROUP

Group 6

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Contents

Section	Page	Section	Page
Conventions and section owners	3	8.2 Class diagram	31
1 Context Diagram	4	8.3 Sequence diagrams	33
1.1 External entities and flows	4	8.4 Activity diagrams	71
2 Data-Flow Diagrams	6	8.5 State diagrams	74
2.1 Level 1	6	8.6 Component diagram	78
2.2 Balancing table: context diagram to Level 1	8	8.7 Deployment diagram	80
2.3 Level 2 of process 4.0	8	8.8 AI recommendation workflow	82
2.4 Balancing table: Level 1 process 4.0 to Level 2	10	9 Database Design	84
3 Architecture	10	9.1 Coverage of the brief	88
3.1 Decision record	10	9.2 Data dictionary	89
3.2 Architecture view	11	10 Interface Contracts	90
3.3 How the system calls the OpenAI API	13	10.1 The Suggestion envelope	90
3.4 Prototype 1 scope	13	10.2 Contracts per task	91
4 AI Services	14	10.3 Adapter contracts	92
4.1 Changes from the lab example	15	11 Recommendation Score and Explanation	92
5 Human Approval Points	15	11.1 Factors	92
6 Security	17	11.2 Formula	92
6.1 The nine roles of the brief	17	11.3 Explanation shown with every result	92
6.2 Access matrix (default: deny)	18	12 Sustainability Calculations	93
6.3 Audit log	19	12.1 Worked example	93
6.4 Backup and restore (later increment)	19	13 Model-Monitoring Records	93
6.5 Security boundaries and integration points	19	13.1 Metrics and targets	94
7 Mockups	19	13.2 Review routine	94
8 UML Models	29	AI-Use Statement	94
8.1 Use-case diagram	29	References	94

List of figures

Figures	Content	Page
1 to 3	Data-flow diagrams: context, Level 1, Level 2 of 4.0	5
4	Architecture view	12
5	Human approval gates	16
6 to 13	Mockups of the eight screens	21
14	Use-case diagram	30
15	Class diagram	32
16 to 53	Sequence diagrams, UC-01 to UC-38	34
54, 55	Activity diagrams	72
56 to 58	State diagrams	75
59	Component diagram	79
60	Deployment diagram	81
61	AI recommendation workflow	83
62, 63	Entity-relationship diagram and its table view	85

Conventions and section owners

Sections 1 to 7 cover Phase 4 and Sections 8 to 13 cover Phase 5. UC-xx and P3-FR-xx.n refer to the Phase 3 use cases and functional requirements, P3-NFR and P3-XR to the Phase 3 non-functional and AI requirements, and BR-xx to the Phase 3 business rules. Source files and each member's commits are in the team repository: https://github.com/SAIIV/Asset-Management-Project.

§	Section	Owner
1	Context diagram	Saeed Alkaabi 100063100
2	Data-flow diagrams (Level 1, Level 2 of 4.0, balancing)	Saeed Alkaabi 100063100
3	Architecture: decision record, architecture view, OpenAI API use	Ali Alawadhi 100063186
4	AI services	Humaid Al Ali 100062914
5	Human approval points	Humaid Al Ali 100062914
6	Security: roles, access matrix, audit log, backup, boundaries	Ali Alawadhi 100063186
7	Mockups of the eight screens	Saeed Alkaabi 100063100
8.1, 8.2	Phase 5: use-case and class diagrams	Saeed Alkaabi 100063100
8.3, 8.4	Phase 5: sequence and activity diagrams	Humaid Al Ali 100062914
8.5 to 8.7	Phase 5: state, component and deployment diagrams	Ali Alawadhi 100063186
8.8	Phase 5: AI recommendation workflow	Humaid Al Ali 100062914
9	Phase 5: database design (ERD and data dictionary)	Saeed Alkaabi 100063100
10	Phase 5: interface contracts	Humaid Al Ali 100062914
11	Phase 5: recommendation score and explanation	Ali Alawadhi 100063186
12	Phase 5: sustainability calculations	Saeed Alkaabi 100063100
13	Phase 5: model-monitoring records	Ali Alawadhi 100063186

1 Context Diagram

The context diagram shows the whole Circular Campus system as one process (0) and every person or outside system that sends it data or receives data from it. The eight external roles are the Phase 3 actors. The ninth entity, AI Provider, is the hosted large-language-model service used for image and PDF classification, the resource assistant and drafted summaries. The sentence-embedding model used for text classification and semantic matching runs inside the application (Phase 2 technical feasibility), so it is part of process 0 and is not an external entity. There are no data stores at this level, and no internal steps.

UC-01 Sign in and the account check behind every protected operation (BR-01) are not drawn as separate flows. They apply to every entity equally and would add one identical pair of arrows per role without adding information. The stored accounts are shown as D1 Users on Level 1.

1.1 External entities and flows

External entity	Data into the system	Data out of the system	Phase 3 basis
Requester	resource request, assistant question, receipt confirmation, fault report	ranked matches, assistant answer, request status	UC-08 to UC-12, UC-17, UC-18, UC-20, UC-29, UC-32, UC-34
Asset Custodian	asset details, reservation & release details, lifecycle proposal	suggested classification, action advice, approved request	UC-04 to UC-07, UC-13, UC-15, UC-16, UC-23, UC-31, UC-33, UC-35
Approver	approval decision	item for approval	UC-14, UC-24
Maintenance Officer	inspection & repair results	maintenance task	UC-19, UC-21, UC-22
Finance Officer	benefit inputs	benefit summary	UC-27
Sustainability Officer	report request, emission factors	impact report, activity summary	UC-26, UC-28, UC-38
Administrator	user, reference & AI settings	backup status	UC-02, UC-03, UC-36, UC-37
Auditor	audit query	audit trail	UC-30
AI Provider (LLM API)	suggested labels, draft text	asset content, report facts, user question	P3-FR-31.1, P3-FR-34.1 to 34.3, P3-FR-38.1 to 38.4, BR-16

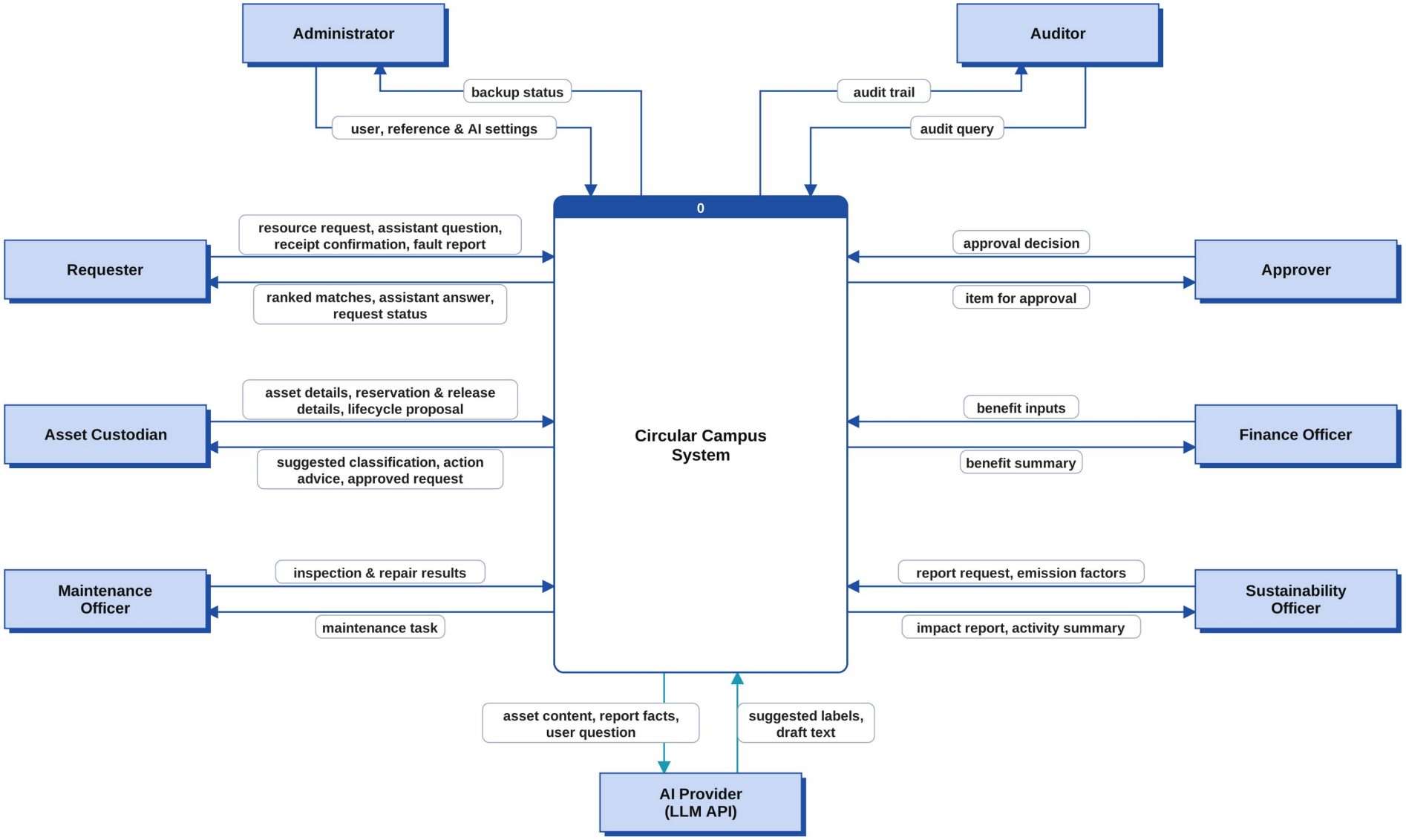


Figure 1. Context diagram (Level 0) of the Circular Campus system.

2 Data-Flow Diagrams

2.1 Level 1

Level 1 splits process 0 into six main processes, one per main job. Each one groups the Phase 3 use cases that change the same records, so each process maps to one module of the architecture in Section 3.

Process	Main job	Use cases	Writes	Reads
1.0 Administer & Audit System	Accounts, reference data, AI settings, backup, audit review	UC-02, UC-03, UC-30, UC-36, UC-37	D1, D6, D7	D7
2.0 Register & Classify Assets	Record an asset, get AI category suggestions, confirm and publish	UC-04 to UC-07, UC-31, UC-35	D2, D7	D6
3.0 Find & Request Assets	Search, semantic matching, assistant answers, submit or revise a request	UC-08 to UC-12, UC-32, UC-34	D1, D3, D7	D2
4.0 Reserve, Approve & Transfer Assets	Reserve units, independent approval, release, receipt and return	UC-13 to UC-18, UC-29	D1, D2, D4, D7	D1, D2, D3
5.0 Maintain & Retire Assets	Faults, inspections, repairs, lifecycle proposals and their authorization	UC-19 to UC-25, UC-33	D1, D2, D5, D7	D2, D6
6.0 Report Impact & Activity	Operational and sustainability reports, benefit inputs, AI activity summaries	UC-26 to UC-28, UC-38	D1, D6, D7	D2, D4, D5, D6

Data stores. A store drawn with an extra vertical line (for example the second D2 Assets and the D7 Audit Logs next to each process) is the same store drawn again to avoid long crossing lines; an external entity with a cut corner is likewise a repeated entity. Every process writes an audit event to D7 because Phase 3 requires a record of the user, action, affected record and time for every important change (BR-14). Notifications are kept in D1 Users (Notification table, Section 9), so every process that sends one (3.0, 4.0, 5.0 and 6.0) writes a notification to D1.

Store	Contents	Written by	Read by
D1 Users	Accounts, roles, departments, notifications	1.0, 3.0, 4.0, 5.0, 6.0	4.0
D2 Assets	One record per physical item: category, condition, lifecycle state, custody, listing flag	2.0, 4.0, 5.0	3.0, 4.0, 5.0, 6.0
D3 Requests	Resource requests and their versions	3.0	4.0
D4 Transfers	Reservations, approvals, release and receipt evidence	4.0	6.0
D5 Maintenance Records	Faults, inspections, repairs, costs, lifecycle decisions	5.0	6.0
D6 Reference Data	Categories, AI thresholds, benefit baselines, emission factors (EPA WARM)	1.0, 6.0	2.0, 5.0, 6.0
D7 Audit Logs	Who did what and when, before and after values, AI model and prompt versions	1.0 to 6.0	1.0

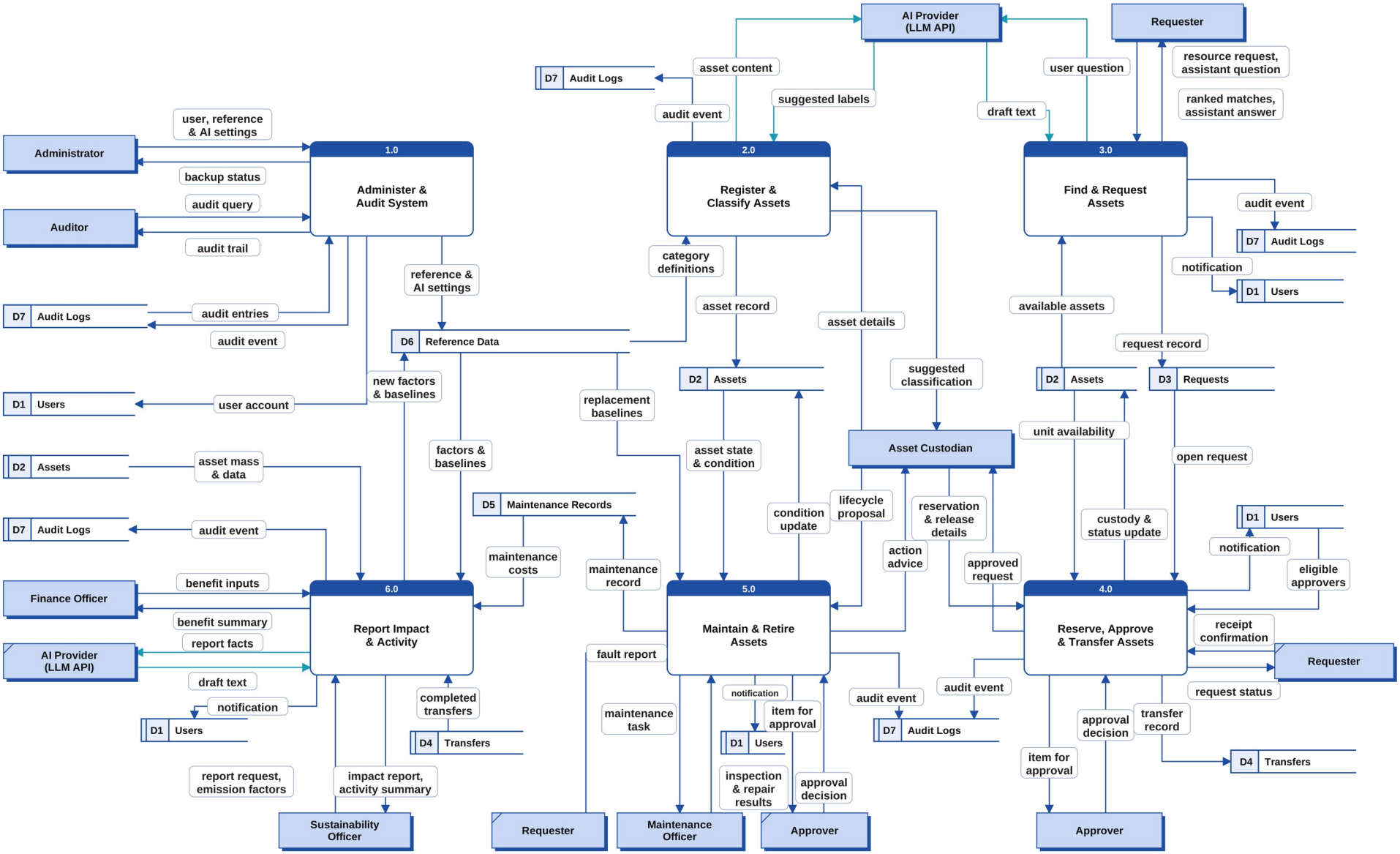


Figure 2. Level 1 data-flow diagram.

2.2 Balancing table: context diagram to Level 1

Every flow on the context diagram appears on Level 1 with the same name, direction and entity. A grouped context flow is split across the Level 1 processes that handle its parts.

#	Context flow	External entity	In/out	Level 1 process	OK
1	resource request, assistant question, receipt confirmation, fault report	Requester	in	3.0 · 4.0 · 5.0 (split)	✔
2	ranked matches, assistant answer, request status	Requester	out	3.0 · 4.0 (split)	✔
3	asset details, reservation & release details, lifecycle proposal	Asset Custodian	in	2.0 · 4.0 · 5.0 (split)	✔
4	suggested classification, action advice, approved request	Asset Custodian	out	2.0 · 5.0 · 4.0 (split)	✔
5	approval decision	Approver	in	4.0 · 5.0 (split)	✔
6	item for approval	Approver	out	4.0 · 5.0 (split)	✔
7	inspection & repair results	Maintenance Officer	in	5.0	✔
8	maintenance task	Maintenance Officer	out	5.0	✔
9	benefit inputs	Finance Officer	in	6.0	✔
10	benefit summary	Finance Officer	out	6.0	✔
11	report request, emission factors	Sustainability Officer	in	6.0	✔
12	impact report, activity summary	Sustainability Officer	out	6.0	✔
13	user, reference & AI settings	Administrator	in	1.0	✔
14	backup status	Administrator	out	1.0	✔
15	audit query	Auditor	in	1.0	✔
16	audit trail	Auditor	out	1.0	✔
17	asset content, report facts, user question	AI Provider	out	2.0 · 6.0 · 3.0 (split)	✔
18	suggested labels, draft text	AI Provider	in	2.0 · 3.0 · 6.0 (split)	✔

18 of 18 context flows match: same name, same direction, same entity. The ins and outs balance: the 9 inbound and 9 outbound context flows are exactly the external inbound and outbound flows of Level 1.

2.3 Level 2 of process 4.0

Process 4.0 Reserve, Approve & Transfer Assets is opened because it is the riskiest process in the system. It is the only place where two users can compete for the same physical unit (BR-05, P3-NFR-P05), where an approval must come from someone other than the requester (BR-02), and where custody and location change (BR-07). A mistake here moves a real item to the wrong department, which is harder to undo than a wrong AI suggestion. It is also on the Prototype 1 demonstration path (UC-13, UC-14, UC-16, UC-17).

Sub-process	What it does	Use cases and rules
4.1 Reserve Eligible Units	Checks each requested unit is still Available and Serviceable and reserves all of them in one step, or none	UC-13, BR-04, BR-05, P3-FR-32.3
4.2 Obtain Independent Approval	Sends the reserved request to an approver who is not the requester and records the decision and reason	UC-14, BR-02, BR-06
4.3 Release Assets	Records the custodian's release of approved units and creates the transfer record	UC-15, UC-16, BR-06
4.4 Confirm Receipt & Custody	Records the requester's receipt and only then changes custody and location	UC-17, UC-18, BR-07, BR-15

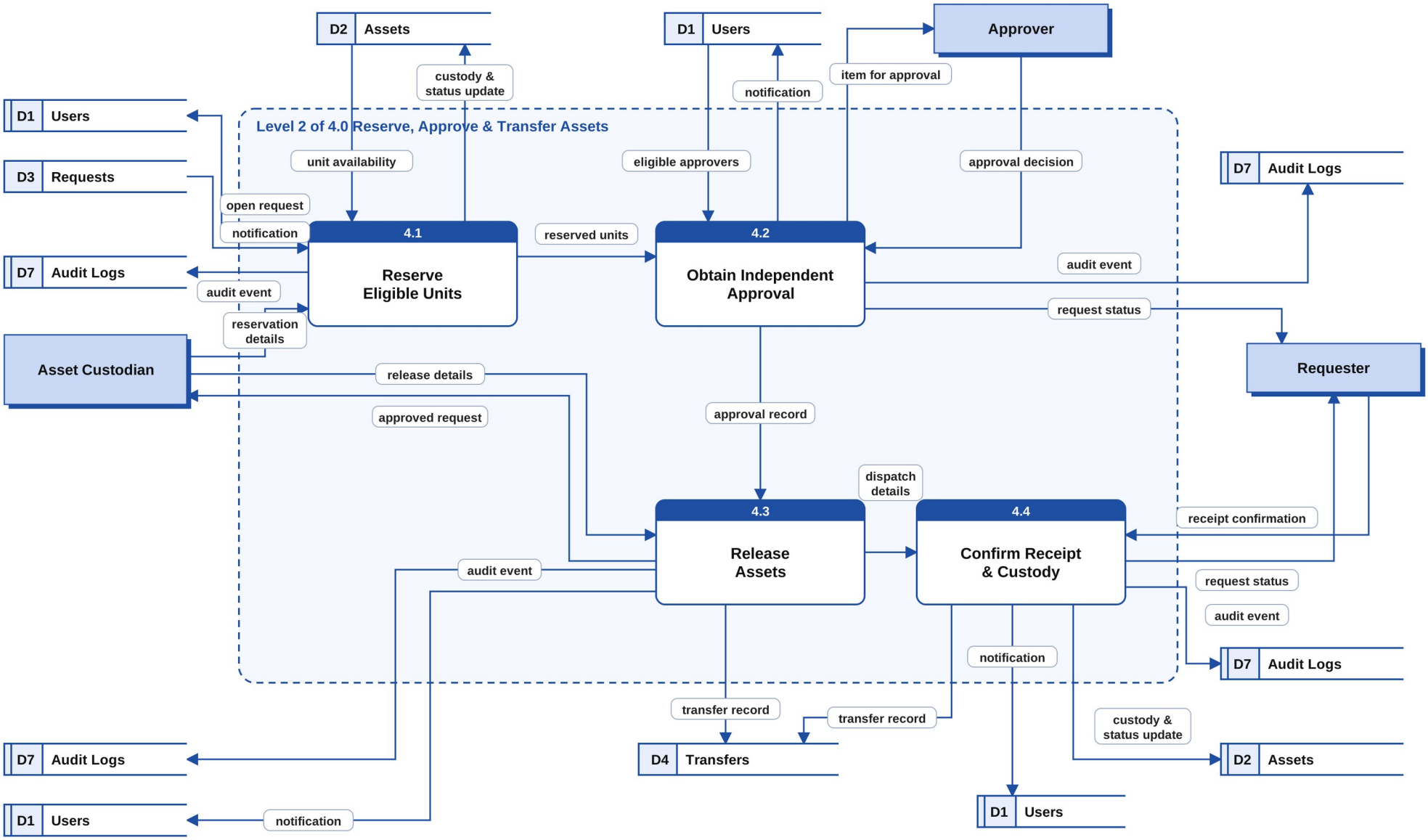


Figure 3. Level 2 data-flow diagram of process 4.0.

2.4 Balancing table: Level 1 process 4.0 to Level 2

#	Level 1 flow of 4.0	Source or destination	In/out	Level 2 process	OK
1	open request	D3 Requests	in	4.1	✔
2	unit availability	D2 Assets	in	4.1	✔
3	eligible approvers	D1 Users	in	4.2	✔
4	reservation & release details	Asset Custodian	in	4.1 · 4.3 (split)	✔
5	approval decision	Approver	in	4.2	✔
6	receipt confirmation	Requester	in	4.4	✔
7	item for approval	Approver	out	4.2	✔
8	approved request	Asset Custodian	out	4.3	✔
9	request status	Requester	out	4.2 · 4.4 (split)	✔
10	custody & status update	D2 Assets	out	4.1 · 4.4 (split)	✔
11	transfer record	D4 Transfers	out	4.3 · 4.4 (split)	✔
12	audit event	D7 Audit Logs	out	4.1 · 4.2 · 4.3 · 4.4 (split)	✔
13	notification	D1 Users	out	4.1 · 4.2 · 4.3 · 4.4 (split)	✔

13 of 13 flows of 4.0 cross the dashed boundary with the same name and direction. The flows reserved units, approval record and dispatch details are internal to 4.0 and do not appear on Level 1.

3 Architecture

Each choice below lists the options we compared, what we chose and why, and what the choice costs us. The Phase 2 report already fixed Python, Django, PostgreSQL and a local sentence-embedding model in one web application, so this section builds on that choice instead of replacing it.

3.1 Decision record

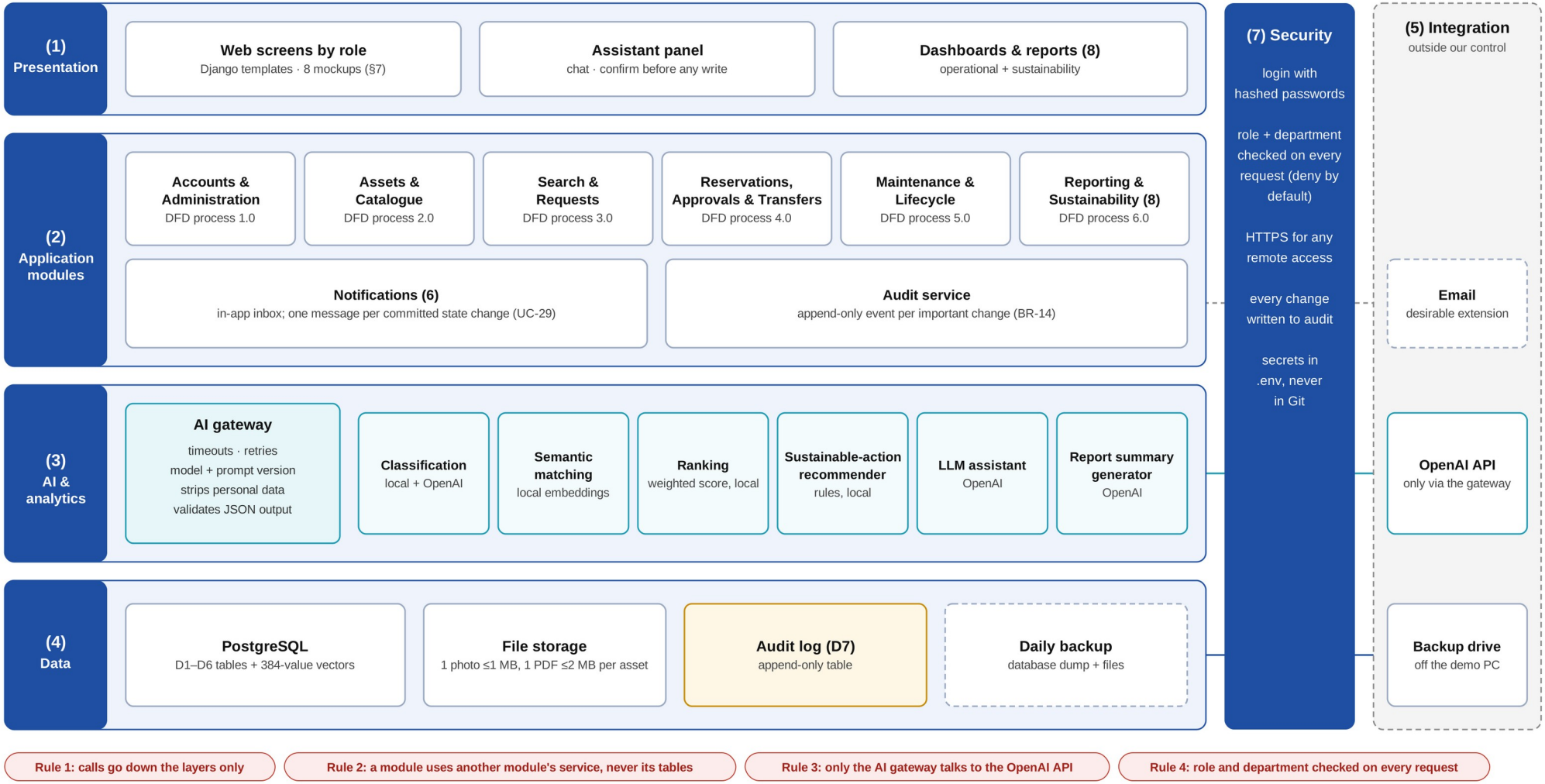
ID	Choice	Options compared	We choose ... because	We give up	Phase 3 links
AD-1	Architecture style	(a) layered monolith; (b) modular monolith with an AI gateway; (c) web app plus a separate AI service; (d) microservices	(b), because four students run one demo PC and Phase 2 chose one Django application. The six business areas match DFD processes 1.0 to 6.0, so each becomes one module with a clear border. The embedding model runs in the same Python process, so a separate AI service (c) adds network calls without a different language or load to justify it.	Deploying or scaling the AI part on its own. Module borders are kept by discipline (Rule 2 in Figure 4), not by the network.	P3-NFR-Q07, P3-NFR-P01, P3-XR-08
AD-2	Text-generation API	(a) OpenAI API; (b) a free open-weight model run locally; (c) no language model, templates only	(a), because one API reads text, photos and PDFs and returns JSON that matches a fixed schema. The demo PC (Saeed's laptop, NVIDIA RTX 3050 Ti with 4 GB) can run only a small local model, which gives weaker classifications and summaries and takes longer to set up. A small OpenAI model fits the AED 150 experimental allowance from Phase 2 [3].	Data leaves the PC, so only synthetic records are sent. We depend on a paid key and the provider being up. If spending is not allowed, the same interface switches to (b), the LocalGenerator, by changing one setting.	BR-16, P3-NFR-SEC06, P3-NFR-SEC08, P3-NFR-Q11
AD-3	Embeddings for classification and matching	(a) local all-MiniLM-L6-v2 (384 values per text); (b) OpenAI embeddings API	(a), because it is free, works offline, keeps asset text on the PC, and is fast enough for the 5-second semantic target.	Some accuracy compared with larger hosted models, and a model warm-up when the server starts.	P3-NFR-P02, P3-XR-02
AD-4	Where vectors are stored	(a) pgvector extension inside PostgreSQL; (b) a separate vector database	(a), because assets and their vectors stay in one database, one transaction and one backup. 500 assets is small.	Specialised vector-database features we do not need at this size.	P3-NFR-Q05, P3-NFR-P02
AD-5	User	(a) server-rendered	(a), because the team uses one	Smoother live updates.	P3-NFR-Q02,

	interface	Django templates; (b) React single-page app with a JSON API	language, forms are validated on the server and keep entered values, and keyboard access is simpler with plain HTML forms.	The assistant panel uses a small amount of JavaScript to fetch answers.	P3-NFR-Q10
AD-6	Running slow AI work	(a) inside the web request; (b) a background worker reading a jobs table in PostgreSQL	(a) for pattern A services with a hard time limit, and (b) for pattern B and C services, because nobody waits for them. Using the database as the queue avoids adding another server.	Instant notification: a background result reaches the inbox on the worker's next cycle (every 30 seconds).	P3-NFR-P03, UC-29

3.2 Architecture view

Figure 4 shows the eight components the project brief names. Every application module is one Level 1 process, so a data flow between two processes in Section 2 is a call between two modules here. Only the AI gateway talks to the OpenAI API.

Module	DFD process	Use cases	Works when the AI is down?
Accounts & Administration	1.0	UC-01 to UC-03, UC-30, UC-36, UC-37	Yes, fully
Assets & Catalogue	2.0	UC-04 to UC-07, UC-31, UC-35	Yes; category chosen by hand
Search & Requests	3.0	UC-08 to UC-12, UC-32, UC-34	Yes; rule search instead of ranking; assistant off
Reservations, Approvals & Transfers	4.0	UC-13 to UC-18	Yes, fully (no AI in this module)
Maintenance & Lifecycle	5.0	UC-19 to UC-25, UC-33	Yes; options assessed by hand
Reporting & Sustainability	6.0	UC-26 to UC-28, UC-38	Yes; numbers and charts without a drafted summary
Notifications	all	UC-29	Yes, fully



(1) to (8) are the eight components the project brief names: presentation, application, AI and analytics, data, integration, notification, security, reporting.
Each application module matches one Level 1 process, so module borders and data-flow borders are the same.

Figure 4. Architecture view: layers, modules, AI services and integration points.

3.3 How the system calls the OpenAI API

Every module asks the AI gateway for a result; no module calls OpenAI directly. One call goes through these steps:

1. **Request.** The module calls the gateway with a task name (for example classify_asset) and only the records the user is allowed to see that the task needs.
2. **Prompt.** The gateway fills a versioned prompt template kept in GitHub (for example classify v1.2). Asset text and documents go in a separate block marked as data, never as instructions.
3. **Clean.** Names, e-mail addresses and staff numbers are removed. A record not marked synthetic is not sent.
4. **Send.** One HTTPS request through the official openai Python library asks for JSON in a fixed schema. The time limit comes from Section 4. A network error gets one retry; a timeout gets none.
5. **Check.** The reply is checked against the schema and the business rules: a category must exist in D6, and every number in a summary must equal the database result. A reply that fails is treated as unsure and never saved as fact.
6. **Record.** An AI record goes to D7: task, input record versions, model name, prompt version, output, score and time taken. The person's later decision is added to it.
7. **Return.** The module shows the result as a suggestion. Nothing is written to D2 to D5 until a person confirms it (Section 5).

Setting	Value
Interface	TextGenerator.generate(task, facts, schema, time_limit). Two versions: OpenAIGenerator and LocalGenerator. One setting chooses which one runs.
Key	Stored in the server's .env file, which is not in GitHub (P3-NFR-SEC02)
Cost limit	Monthly spending limit set in the OpenAI account so the total stays within the AED 150 allowance
Switch off	The Administrator can turn each AI feature off in AI settings (UC-36). The conventional screens keep working.
Model name	Kept in D6 AI settings and written with every AI record, so results from different models are not mixed (P3-NFR-Q11)

3.4 Prototype 1 scope

Prototype 1 implements the core features and AI features below, as set in our Phase 6 plan. It has four roles: Requester, Custodian, Approver and Administrator. Until the Sustainability Officer role is added, the Administrator views the sustainability dashboard and edits the report summary before it is published.

ID	Prototype 1 feature	What it covers
C1	Login and roles	Requester, Custodian, Approver and Administrator; every view checks the access matrix (6.2)
C2	Departments and assets	Register an asset with category, condition, department and status
C3	Publish and search	Publish surplus items; filter by keyword, category, condition and department
C4	Request and track	Submit a request and follow its status
C5	Approve and transfer	Independent approval, release, receipt; custody changes only on receipt
C6	States and history	Only the moves of the asset state diagram (Figure 56); who did what and when
C7	Sustainability dashboard	Items reused, cost avoided, waste diverted and kg CO2e, each with its formula (Section 12)
C8	Notifications	In-app messages when a request is made, approved, rejected or received
C9	Basic report	Table of requests and transfers with CSV export
AI #1	Classification and tagging	Section 4, service 1

AI #2	Matching and ranking	Section 4, services 2 and 3
AI #5	Generated report summary	Section 4, service 6

Everything else in this document is fully designed but planned for a later increment, and is labelled "(later increment)" where it appears:

Later increment	Designed in
Resource assistant (UC-34)	Sections 4, 7 (Mockup 8), 8.3, 10.2
Sustainable-action recommender (UC-33)	Sections 4, 8.3, 10.2
Lifecycle actions: propose, authorize, retire (UC-23 to UC-25)	Sections 5, 8.3, 8.5
Inspection and maintenance workflow (UC-19 to UC-22)	Sections 7 (Mockup 5), 8.3
Finance Officer, Maintenance Officer, Sustainability Officer and Auditor roles (UC-27, UC-30)	Sections 6.1, 6.2, 8.3
Scheduled backups and the restore test (UC-37)	Sections 6.4, 8.3, 8.7
E-mail notifications	Section 6.5

4 AI Services

The system has six AI services, not one AI box. Patterns: A = ask and wait, B = compute ahead, C = queue and notify. Similarity and ranking scores are not probabilities, so each "unsure" rule gives the score where low confidence starts. Thresholds are kept in D6 AI settings and only the Administrator can change them; each change gets a new version.

Service	Runs where	Pattern and why	Time limit	If down	If unsure	Phase 3
1 Classification (category, tags, material)	Text: local embeddings. Photo or PDF: OpenAI through the gateway	A. The custodian is waiting at the form, but can pick the category without it.	5 s for text; 20 s with a photo or PDF	"Suggestions unavailable"; fields stay empty and the custodian picks from the list. Entered data is kept.	Top similarity below 0.60, or less than 0.10 ahead of the next category: "Low confidence, please choose" and nothing is preselected.	P3-XR-01, UC-31, UC-35
2 Semantic matching	Local embeddings stored in PostgreSQL	B. Each asset vector is computed once when it is saved and read by every search.	Vector within 60 s of saving; search within 5 s	Keyword and rule search, labelled conventional.	Similarity below 0.40: the asset is listed after the rule matches as a weak match.	P3-NFR-P02, UC-32
3 Ranking with reasons	Local weighted score in the application	A. The requester is waiting for the list.	5 s	Eligible units in Asset ID order with no scores, labelled "not ranked".	Score below 70 of 100: "Weak match" and no top pick. A factor missing for any candidate is left out for all and the weights are rescaled.	P3-XR-02, P3-XR-03, UC-32
4 Sustainable-action recommender (later increment)	Local rules on condition, repair cost, replacement baseline and safety	C. It runs after an inspection is recorded; nobody waits, and a person reviews it later.	2 min; retried on the next worker cycle	"No advice available. Assess the options manually."	Mass or cost missing: "Medium confidence" or "Low confidence" with the missing inputs named. Never applied by itself.	P3-XR-04, UC-33, BR-10
5 Resource assistant (LLM) (later increment)	OpenAI through the gateway	A. The user is waiting in the chat.	15 s; "still working" shown after 5 s	"Assistant unavailable. Use Search or New request." Other screens work.	No supporting record: it says it does not know. No policy record: it says the policy is unavailable. Any write opens the normal form for confirmation.	P3-XR-05, UC-34
6 Report summary generator (LLM)	OpenAI through the gateway	C. The draft is written in the background after a report runs; the officer is notified.	30 s per draft; 2 retries	Numbers and charts still come from database queries; "Draft summary unavailable".	Any number in the draft that differs from the query result blocks publishing. Missing data is listed.	P3-XR-06, UC-38

4.1 Changes from the lab example

Service	Lab example	Ours	Why
Classification	5 s	5 s for text, 20 s with a photo or PDF	Reading a photo or PDF needs the hosted model, which is slower; still under the 30 s limit of P3-NFR-P03
Semantic matching	search < 100 ms	search within 5 s	P3-NFR-P02 sets 5 s for 95 of 100 semantic queries
Ranking	no score above 0.70	score below 70 of 100	Phase 3 defines the score as 100 × the weighted sum
Assistant and reports	one row, A·C, 8 s	two rows: A 15 s and C 30 s	They use different patterns, so each gets its own row; 30 s is the P3-NFR-P03 limit

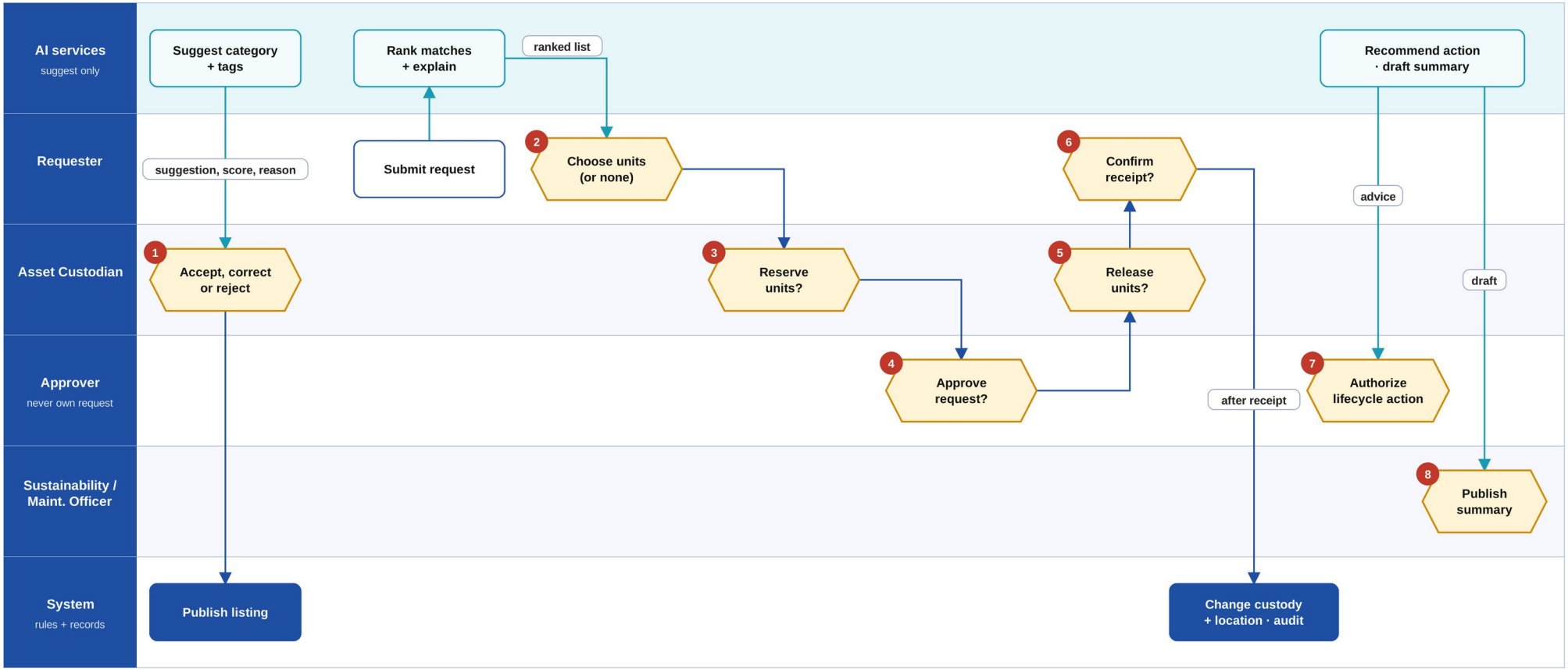
In every case the conventional screen works without the AI, and the fallback is labelled conventional rather than counted as a working AI feature (P3-XR-08, P3-NFR-Q06).

5 Human Approval Points

Final approval is never given to the AI. Every AI output is advice, and a person decides at each of the eight gates below (BR-10, BR-11, P3-NFR-S03). A person is placed wherever a wrong answer is costly or hard to undo: physical items moving between departments, items leaving the university, and claims published about the university. When the person and the AI disagree, the person's decision stands and the reason is recorded.

Gate	Who decides	What they see	AI's part	Why a person	Log entry	Phase 3
1 Accept, correct or reject the classification	Asset Custodian	Suggested category and tags, similarity, reason, which input it came from	Suggests	A wrong category hides the asset from the people who need it	Decision, correction, reason, model and prompt version	UC-35, BR-11, P3-XR-07
2 Choose units	Requester	Ranked list, score, factor bars, warnings	Ranks	Only the requester knows the real need	Chosen units, their rank, ranking version	UC-32, P3-XR-03
3 Reserve units	Asset Custodian	Request and eligibility checks for each unit	None	Reserving blocks other departments	Units reserved, request version	UC-13, BR-05
4 Approve request	Approver who is not the requester	Request, reserved units, checks, what the AI contributed	None (shows the AI score that was used)	University property moves between departments	Decision, reason, AI score seen	UC-14, BR-02, BR-06
5 Release units	Asset Custodian	Approved request and current checks	None	Physical hand-over	Release time and units	UC-15, UC-16, BR-06
6 Confirm receipt	Requester	Dispatch details and condition at release	None	Custody changes only after receipt	Receipt or dispute, condition	UC-17, BR-07, P3-NFR-S04
7 Authorize lifecycle action	Approver who did not propose it	Proposal, inspection evidence, AI advice, missing inputs	Recommends	Donation, recycling and disposal cannot be undone	Decision, reason, advice version	UC-24, BR-10, P3-XR-04
8 Publish summary	Sustainability Officer	Draft, number check, data cut-off date	Drafts	Published claims speak for the university	Published or discarded text, edits	UC-38, P3-XR-06

The resource assistant follows the same rule: it can fill in a request, but the request is only submitted from the normal form after the user confirms it (P3-XR-05).



Gate = a person decides. Each gate records who decided, when, what they saw (including the AI score and model version) and the reason for any override.
Check: no path from the AI lane reaches the System lane without passing a gate. Gate 7 uses an Approver who did not propose the action (BR-02, BR-10).

Figure 5. Human approval gates. No path from the AI lane reaches the System lane without passing a gate.

6 Security

6.1 The nine roles of the brief

The brief names nine roles. Phase 3 represents them with eight actors, so some brief roles share an actor's permissions.

Brief role	Phase 3 actor	Note
Department representative	Requester	Phase 3: department representatives use the Requester role
Requester	Requester	
Asset custodian	Custodian	
Administrator (department)	Approver	Heads of department and department administrators approve requests and lifecycle actions
Procurement officer	Finance Officer	Phase 3: procurement is represented by validating purchase alternatives; no purchasing is built
Finance officer	Finance Officer	
Maintenance staff	Maintenance Officer	
Sustainability officer	Sustainability Officer	
System administrator	Administrator	Accounts, reference data, AI settings and backup; no business approval (BR-02)
(also) Auditor	Auditor	Phase 3 role that reads the audit history (UC-30)

6.2 Access matrix (default: deny)

Action	Dept. rep	Requester	Custodian	Dept. admin (Approver)	Procure-ment	Finance	Mainte-nance	Sustain-ability	System admin	Auditor	AI services
Register or edit an asset	—	—	own dept.	—	—	—	—	—	—	—	suggest only
Accept, correct or reject an AI category	—	—	own dept.	—	—	—	—	—	—	—	never
Search listed assets	✔	✔	✔	✔	✔	✔	✔	✔	—	—	rank only
Submit or revise a request, choose a match	own dept.	own dept.	—	—	—	—	—	—	—	—	rank, draft only
Reserve units	—	—	own assets	—	—	—	—	—	—	—	never
Approve a request	—	—	—	assigned, not own	—	—	—	—	—	—	never
Release units	—	—	own assets	—	—	—	—	—	—	—	never
Confirm receipt	own request	own request	—	—	—	—	—	—	—	—	never
Report a fault	✔	✔	✔	—	—	—	✔	—	—	—	—
Record inspection and repair	—	—	—	—	—	—	assigned	—	—	—	never
Propose a lifecycle action	—	—	own assets	—	—	—	—	—	—	—	recommend only
Authorize donation, recycling, disposal	—	—	—	not own proposal	—	—	—	—	—	—	never
Validate avoided-purchase baselines	—	—	—	—	✔	✔	—	—	—	—	—
Run operational reports	—	—	own dept.	own dept.	✔	✔	—	✔	—	✔	—
Publish a sustainability summary	—	—	—	—	—	—	—	✔	—	—	draft only
Manage users, roles, AI settings	—	—	—	—	—	—	—	—	✔	—	never
Run or restore a backup	—	—	—	—	—	—	—	—	✔	—	—
Read the audit log	—	—	—	—	—	—	—	—	read only	✔	—
Edit or delete an audit entry	—	—	—	—	—	—	—	—	—	—	—

"own dept." means records of the user's department only. Role, department and record relationship are checked on the server for every request, including a changed record ID in the address bar (BR-01, P3-NFR-SEC01). A user may hold more than one role but can never approve their own request or proposal (BR-02).

6.3 Audit log

The audit log (D7) is append-only. In Prototype 1 the application enforces this: it has no operation that updates or deletes an audit row. The database-level permission (the application's database account may insert and read audit rows but not update or delete them) is applied in a later increment. A correction adds a new event linked to the original (BR-14, P3-NFR-SEC07).

Field	Example
Event ID and time	E-55812 · 15 Oct 2026 10:42:07 (UAE time)
Who	User ID and the role they acted in
What	Action and record: approve · request R-2041 version 1
Before and after	status Pending → Approved
Reason	Required for any reject or AI override
AI fields (when AI was used)	Task, model name, prompt or ranking version, score, input record versions, the person's decision
Linked event	The original event a correction refers to
Never stored	Passwords, API keys, full prompts with personal data

6.4 Backup and restore (later increment)

Item	Plan
What	PostgreSQL dump (pg_dump) and the uploaded-files folder. Secrets are not in the backup; the .env file is kept separately by the system administrator.
How often	Daily at 18:00 during development, run automatically by a scheduled task
Where	An encrypted external drive kept off the demo PC; the last 7 daily and 4 weekly copies are kept
Restore test	Restore into a separate test database and folder (UC-37). Compare record counts per table, check every foreign key, and compare SHA-256 checksums of the files (P3-NFR-Q05).
Result shown to	System administrator, as the backup status flow in Figure 2

6.5 Security boundaries and integration points

Boundary	What crosses it	Protection
Browser to server	Pages, forms, uploads	HTTPS for any remote access; local demo on loopback only (P3-NFR-SEC04); session cookie and CSRF token; logout ends the session (P3-NFR-SEC03)
Server to OpenAI API	Minimal synthetic facts; suggestions back	Only through the AI gateway; key in .env; personal data removed; planned test: 5 prompt-injection tests on the Prototype 1 AI features that use the API (photo and PDF classification, report summary) (P3-NFR-SEC06, P3-NFR-SEC08)
Application to database	All records	Local connection; least-privilege database account; passwords hashed (P3-NFR-SEC02)
Uploads	One photo ≤ 1 MB, one PDF ≤ 2 MB per asset	File type checked from content; served only after a permission check (P3-NFR-SEC05)
Server to backup drive (later increment)	Database dump and files	Encrypted drive; restore tested
E-mail (desirable)	Notifications	Not in Prototype 1; the in-app inbox is used (UC-29)

7 Mockups

The eight screens the brief lists. Every screen shows the signed-in role, real fields with fictional sample data, the AI result with its confidence and reason, the person's decision, and a dashed box showing the AI-down state. Numbered red markers (1) to (5) point to: the AI result, its score, the reason, the person's decision, and the version recorded in the audit log. All data is fictional (P3-NFR-S05).

Screen	Role	AI result, confidence and reason	Person's decision	If the AI is down	Phase 3
Asset registration	Custodian	Suggested category and tags, similarity 0.82, reason (title, description, spec sheet)	Accept, Correct or Reject	Fields stay empty; category chosen from the list	UC-04, UC-31, UC-35
Request submission	Requester	3 eligible matches found before submitting, with scores	Submit request; choose candidates	Rule-based results by date added	UC-10, UC-32
Recommendation review	Requester	Score 84/100, five factor bars, plain-words reason, weak-match warning below 70	Choose these units or Not suitable (with reason)	Eligible list in Asset ID order, not ranked	UC-32, UC-35
Approval workflow	Approver	What the AI contributed: rank and reason, marked advice only	Approve, Reject (reason) or ask a question	Only the AI panel changes; approval never depends on AI	UC-14, BR-02, BR-06
Maintenance tracking (later increment)	Maintenance Officer	Repair, donate or recycle with bars, reason, medium confidence because mass is unknown	Record repair and return to service	"No advice available"; form works as normal	UC-19, UC-21, UC-22, UC-33
Sustainability dashboard	Sustainability Officer	Draft summary; 4 of 4 numbers match the report query	Publish, Edit or Discard	Numbers and charts still shown from the database	UC-28, UC-38
Reports	Custodian	Activity summary marked AI draft; every number matches the table	Run and export; approve the draft	Table, totals and exports work; summary box says unavailable	UC-26, UC-38
AI assistant (later increment)	Requester	Answer with source listings; "Unsure" when no record supports it	Open in request form, Change or Cancel	"Assistant unavailable. Use Search or New request."	UC-34, UC-10

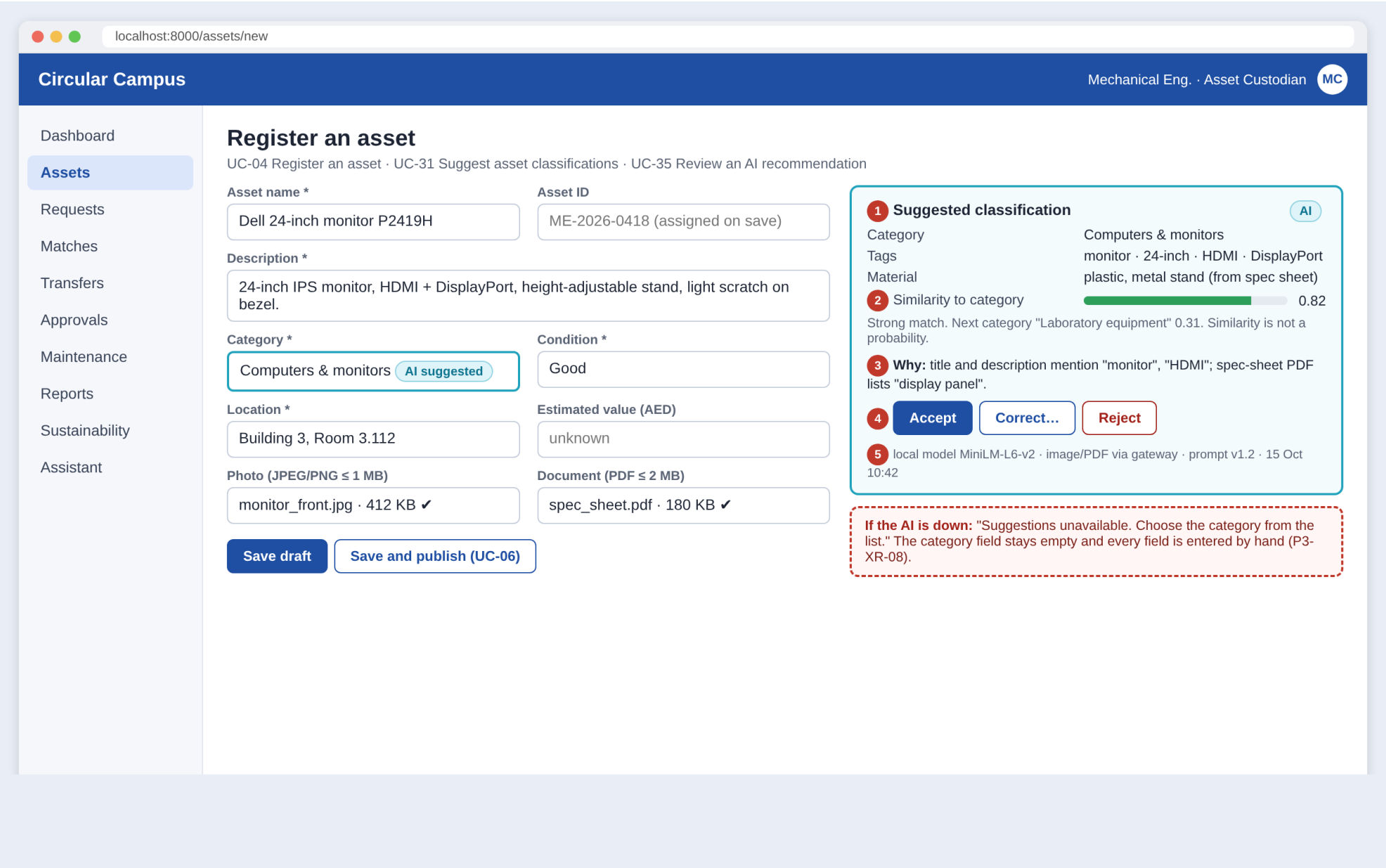


Figure 6. Mockup 1: asset registration (Custodian).

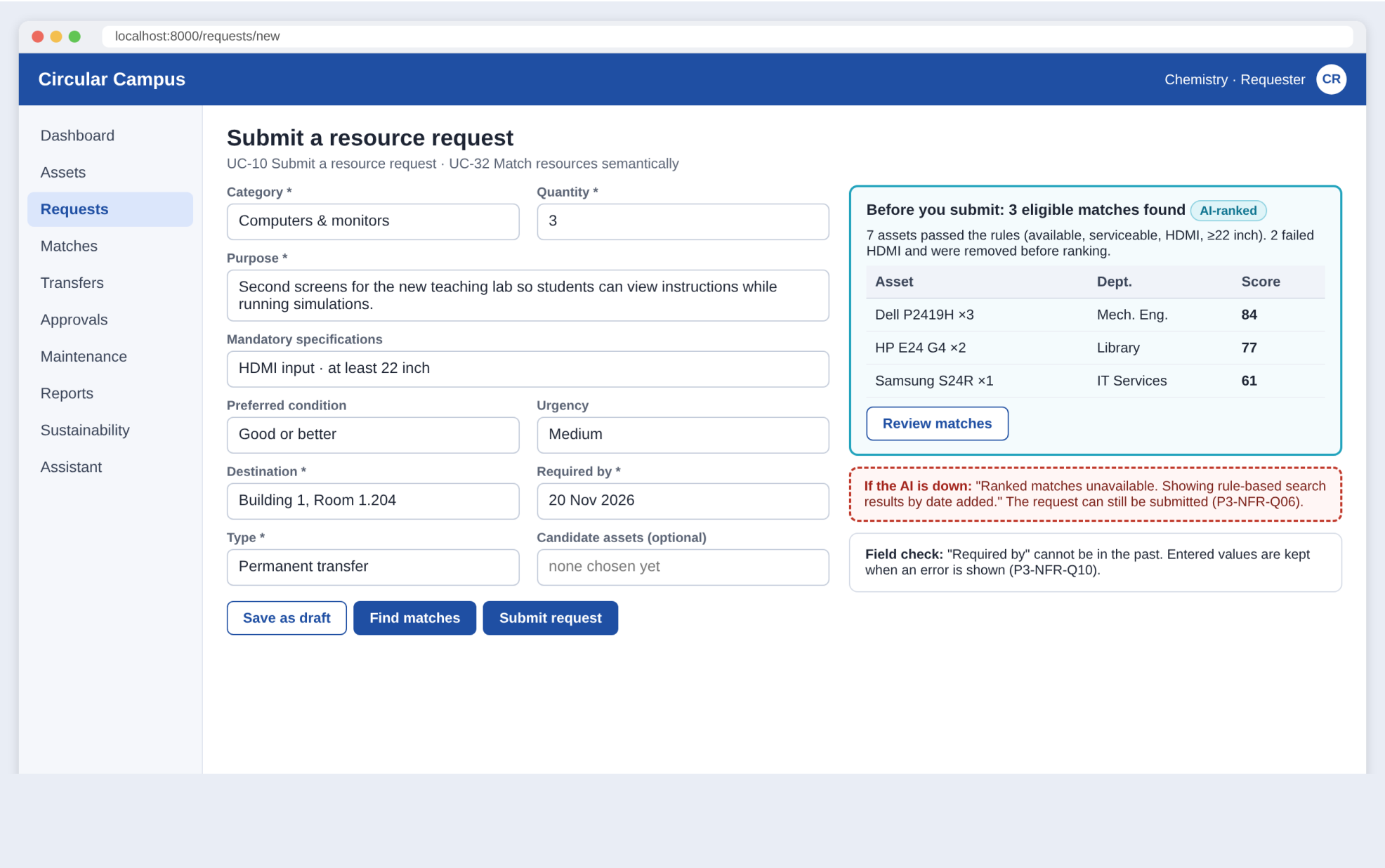


Figure 7. Mockup 2: request submission (Requester).

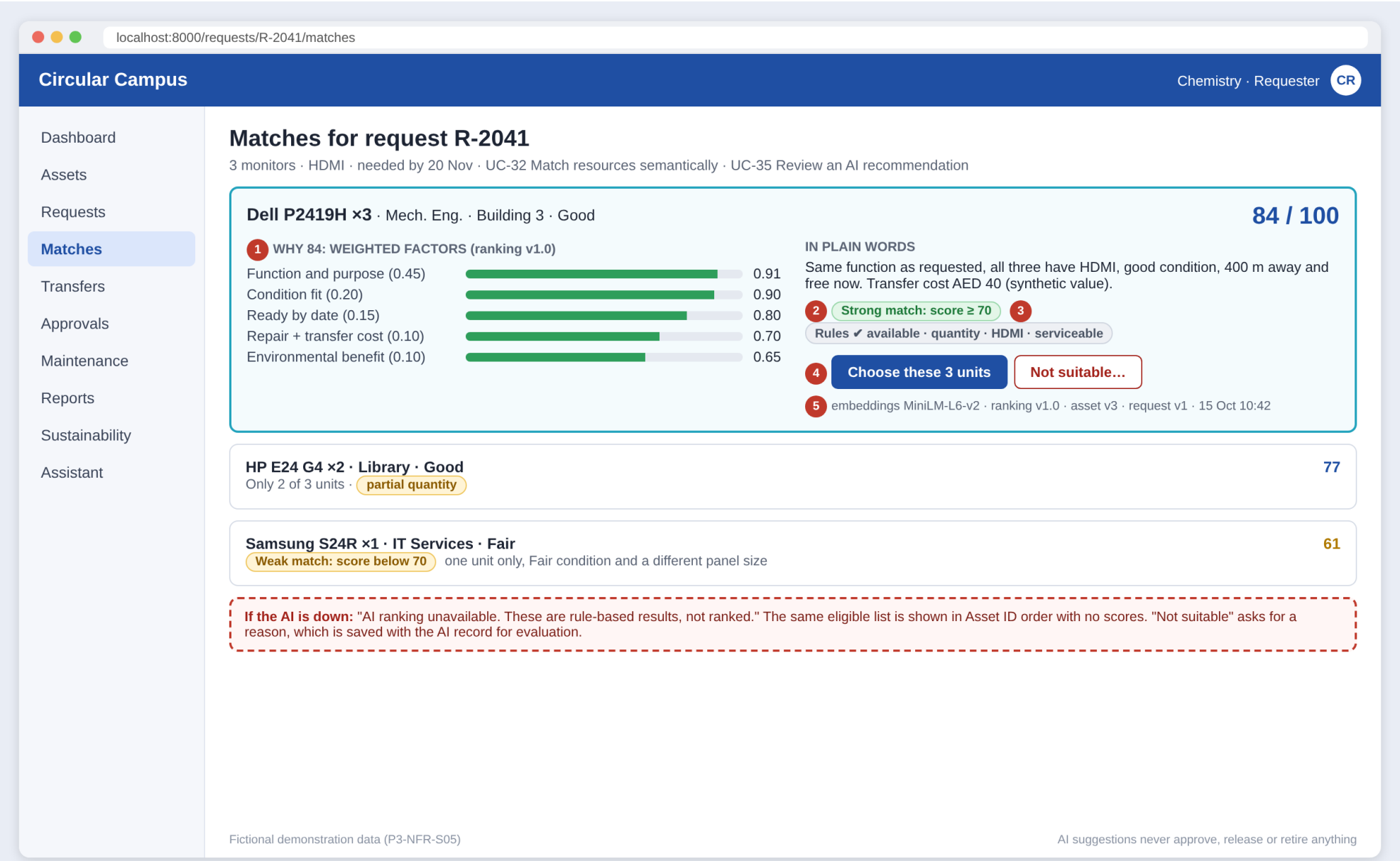


Figure 8. Mockup 3: recommendation review (Requester).

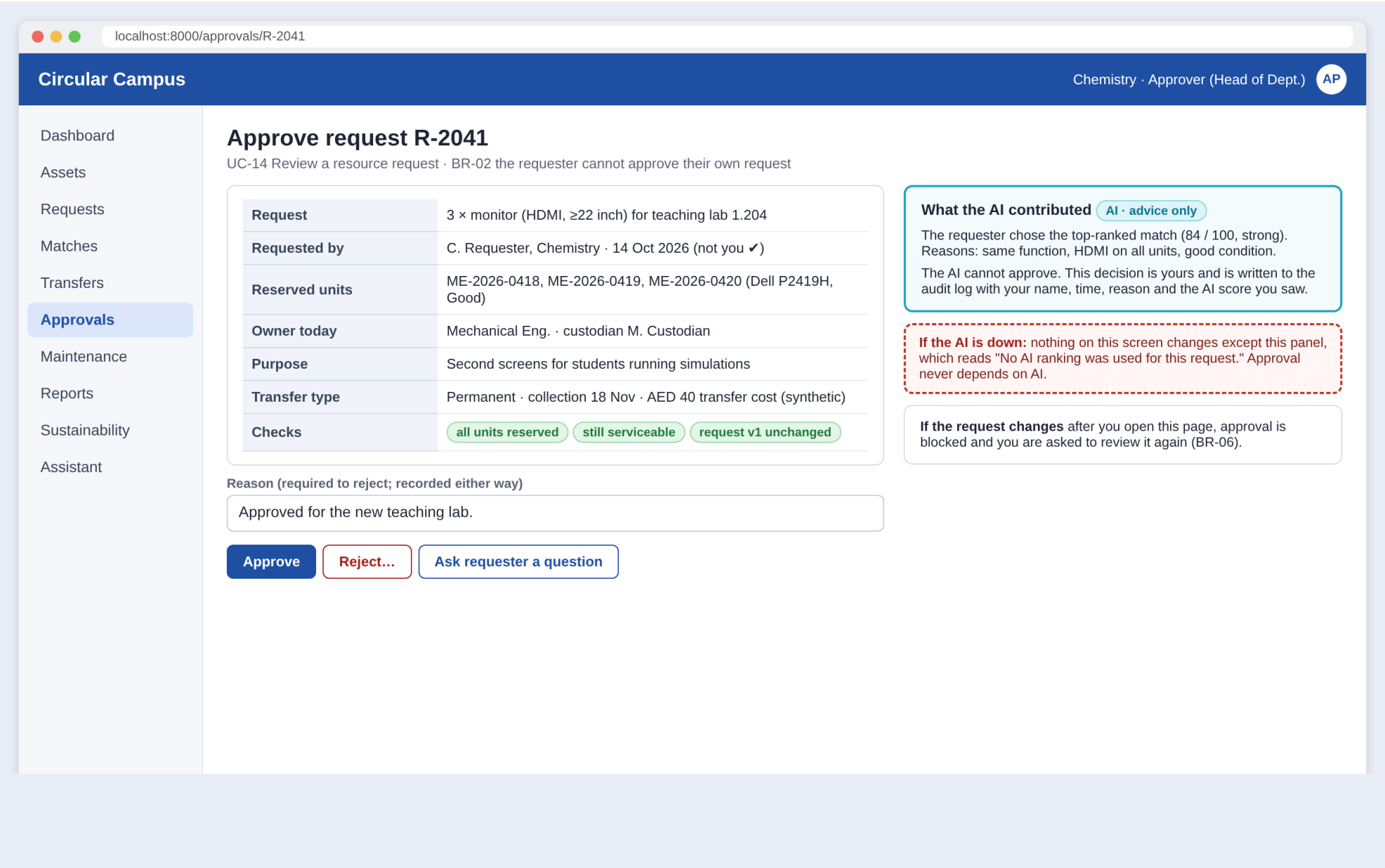


Figure 9. Mockup 4: approval workflow (Approver).

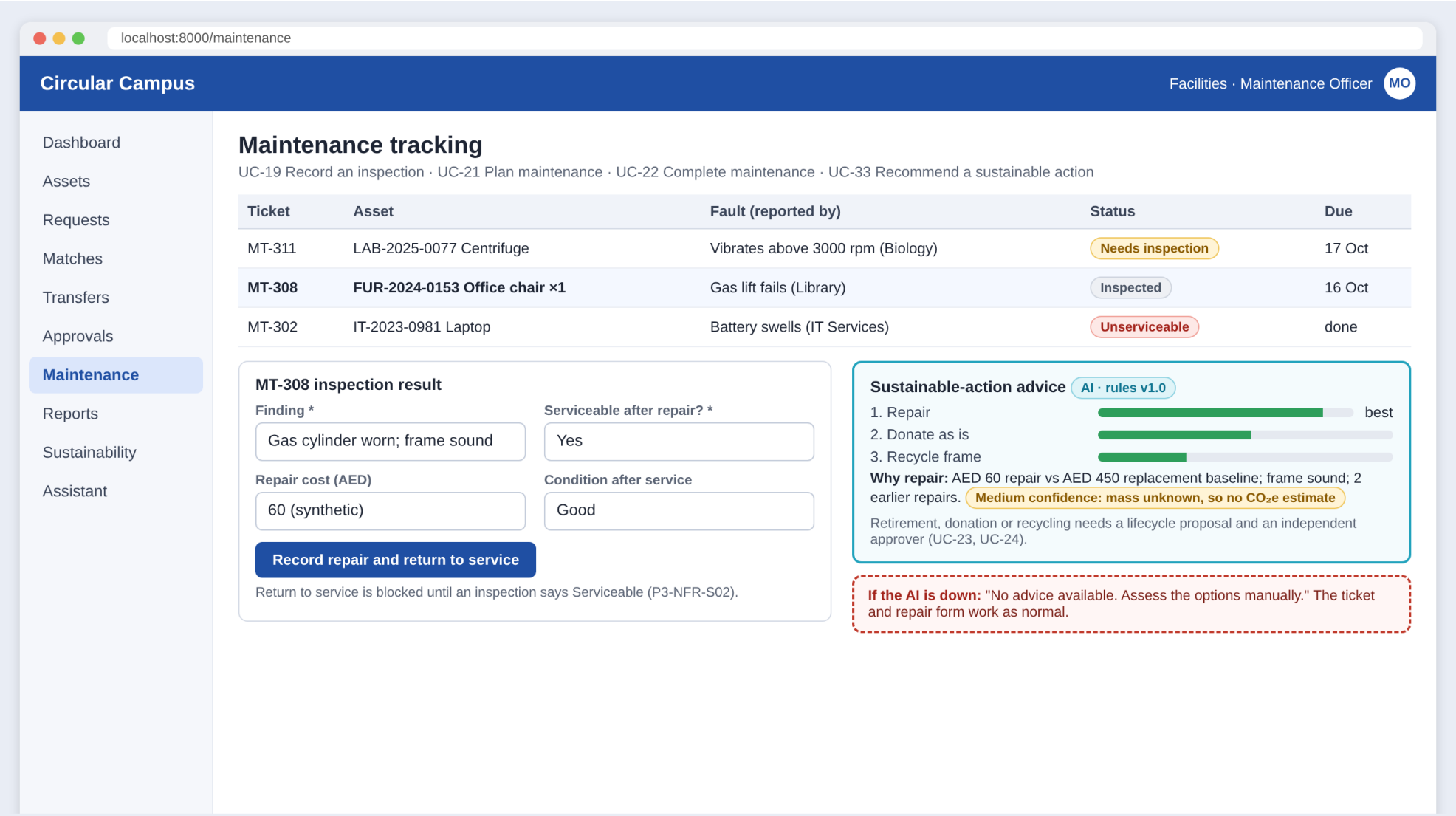


Figure 10. Mockup 5: maintenance tracking (later increment) (Maintenance Officer).

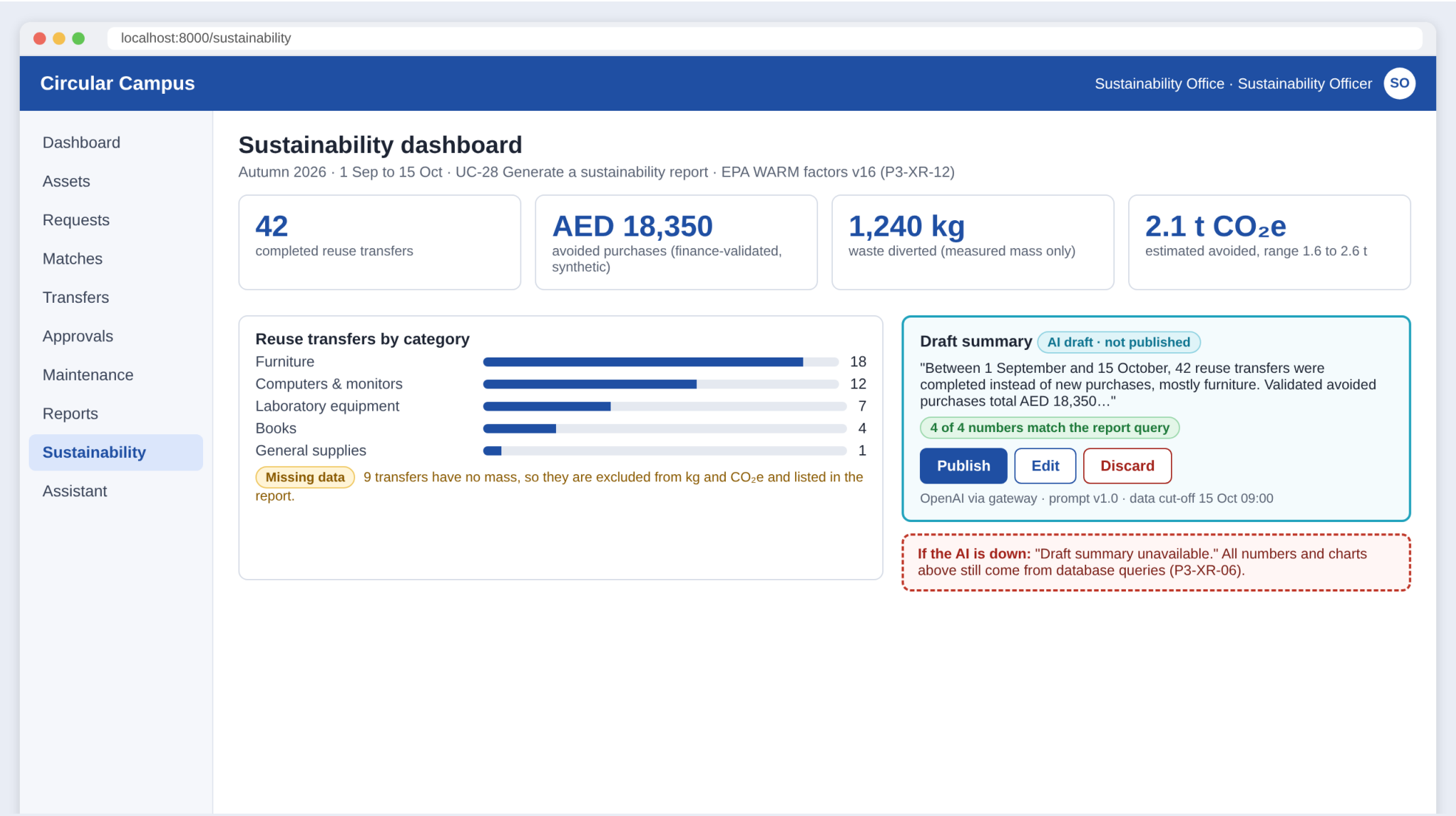


Figure 11. Mockup 6: sustainability dashboard (Sustainability Officer).

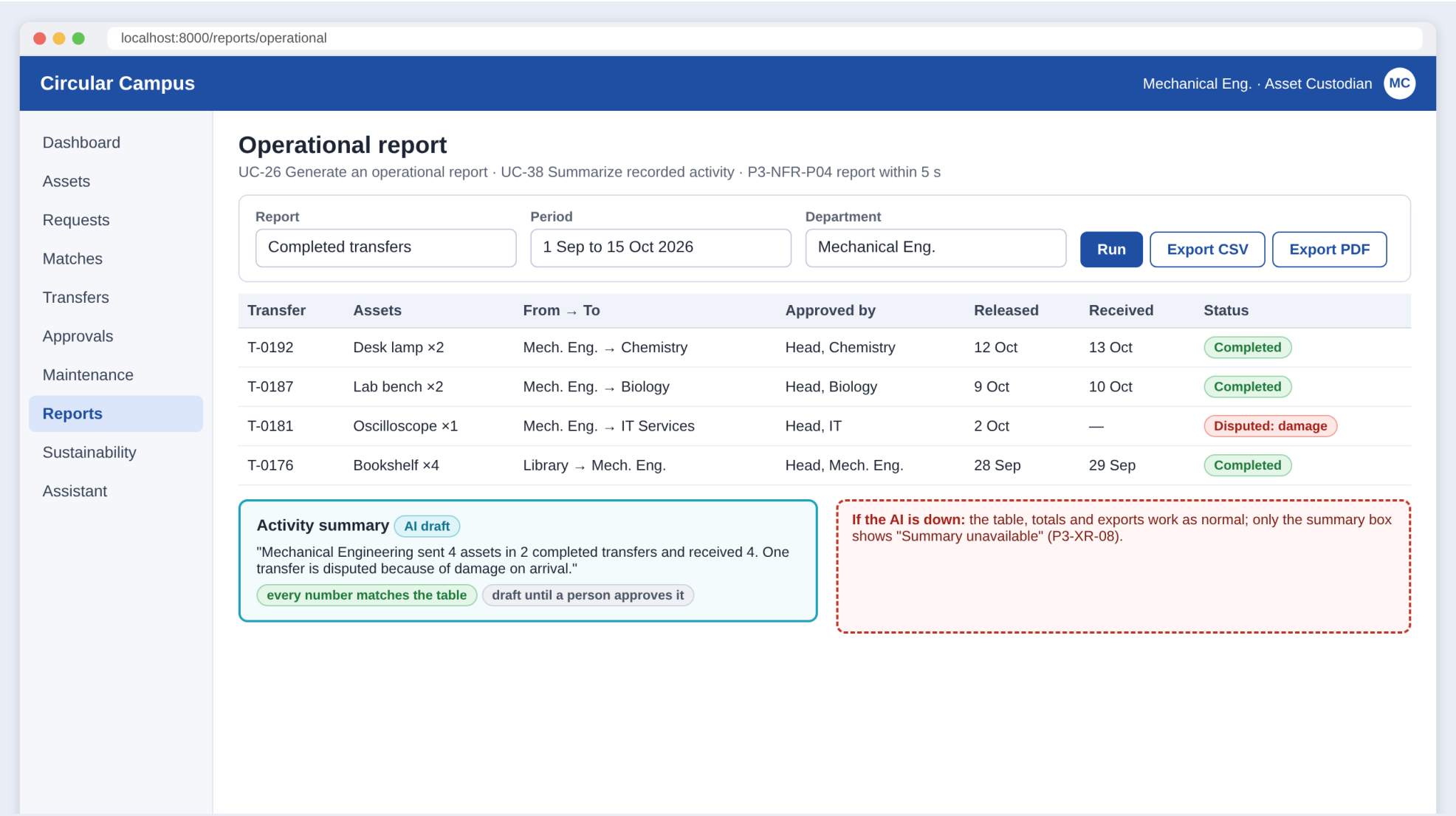


Figure 12. Mockup 7: reports (Custodian).

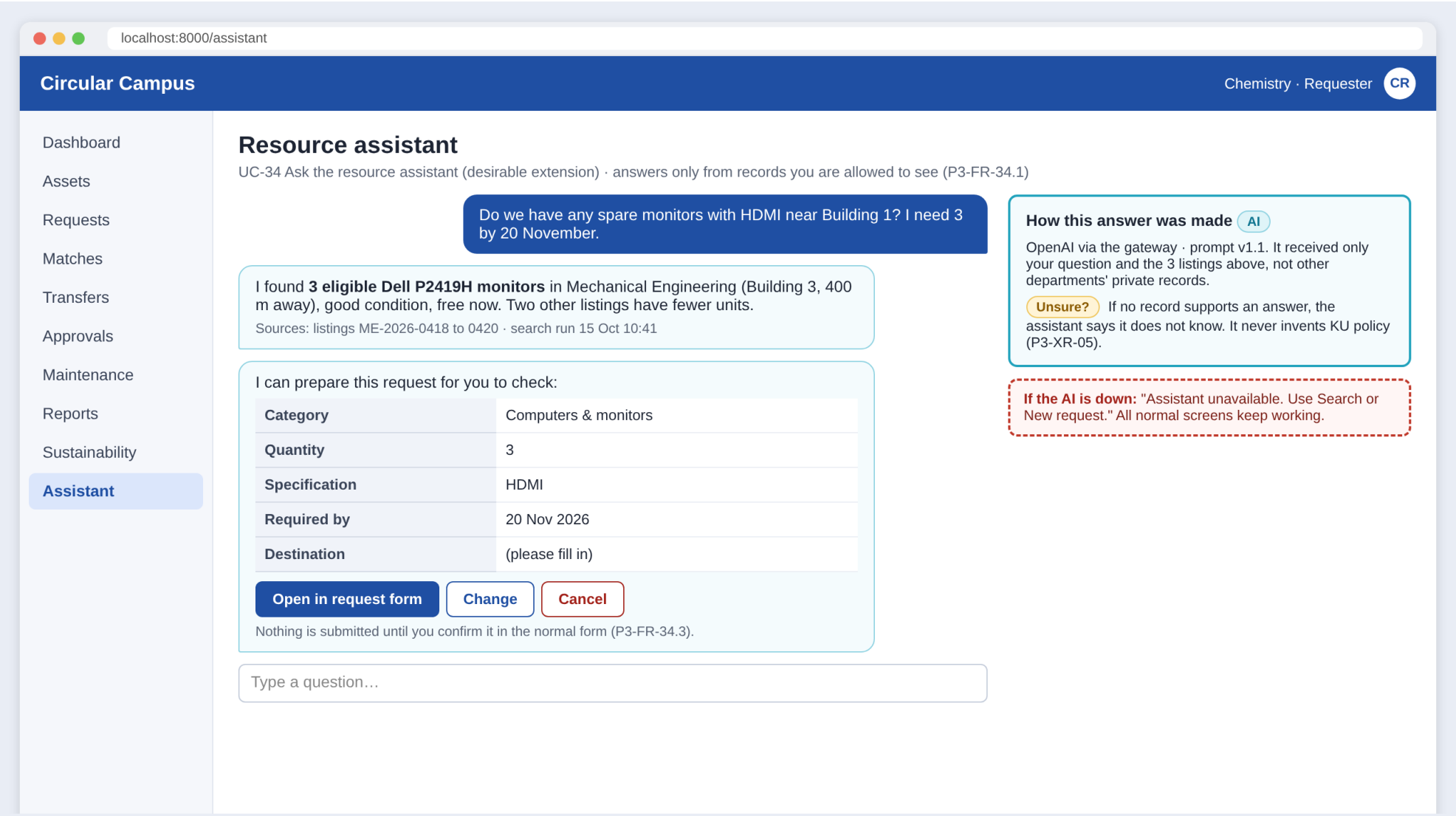


Figure 13. Mockup 8: ai assistant (later increment) (Requester).

8 UML Models

Phase 5 turns the architecture of Sections 1 to 7 into detailed models. Figures 14 to 63 continue the numbering of Sections 1 to 7.

Model	Figure	Section	What it answers
Use-case diagram	14	8.1	Who uses the system for which of the 38 goals
Class diagram	15	8.2	Which classes the code will have and how the AI services plug in
Sequence diagrams	16 to 53	8.3	For each use case, which part calls which, in what order, and what happens otherwise
Activity diagrams	54, 55	8.4	Who does each step of the two main business flows
State diagrams	56 to 58	8.5	Which status values a record can have and what moves it
Component diagram	59	8.6	Which modules exist and which interfaces they use
Deployment diagram	60	8.7	Which software runs on which machine
AI recommendation workflow	61	8.8	How a ranked, explained recommendation is produced and monitored
Entity-relationship diagram	62, 63	9	Which records are stored and how they relate

8.1 Use-case diagram

Figure 14 is the Phase 3 use-case diagram, kept unchanged so that every Phase 5 model uses the same numbers. The table shows which Phase 5 models realize each group.

Use cases	Group	Main actors	Realized in
UC-01 to UC-03, UC-30, UC-36, UC-37	Accounts, reference data, AI settings; audit review and backup (later increment)	Administrator, Auditor	Figures 16 to 18, 45, 51 and 52; class User, Role; ER UserRole, AuditLog, ModelVersion
UC-04 to UC-07, UC-31, UC-35	Register and publish assets; AI classification	Custodian	Figures 19 to 22, 46 and 50; Figure 56; class Asset; ER Asset, its subtype tables and Attachment
UC-08 to UC-12, UC-32, UC-34	Search, request, AI matching; assistant (later increment)	Requester	Figures 23 to 27, 47 and 49; Figures 54, 57(a) and 61
UC-13 to UC-18	Reserve, approve, release, receive, return	Custodian, Approver, Requester	Figures 28 to 33; Figures 55, 57(b) and 58
UC-19 to UC-25, UC-33	Inspection, maintenance, lifecycle actions (later increment)	Maintenance Officer, Custodian, Approver	Figures 34 to 40 and 48; Figure 58(c); ER Inspection, MaintenanceRecord, LifecycleAction
UC-26 to UC-29, UC-38	Reports, notifications, summaries; benefit inputs (later increment)	Sustainability Officer, Finance Officer	Figures 41 to 44 and 53; Section 12; ER SustainabilityImpact, Notification

Circular Campus overall use case diagram

All 38 goals within one system boundary. Solid lines show initiating actor associations; dashed arrows show include and extend relationships.

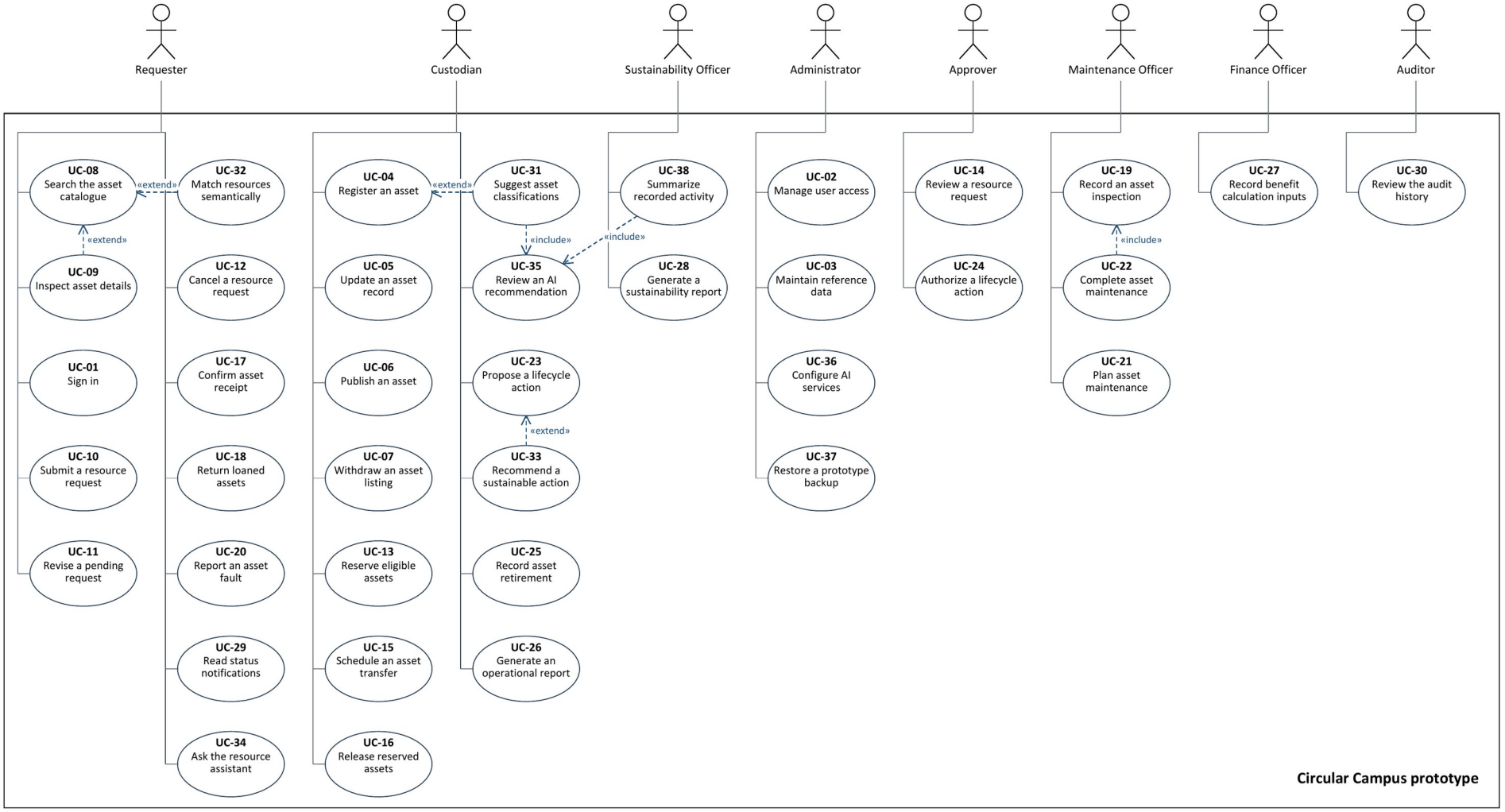


Figure 14. Use-case diagram: eight actors and UC-01 to UC-38 (from Phase 3, unchanged).

8.2 Class diagram

Figure 15 is the class diagram of the whole system: the business classes in the upper part and the AI gateway, AI services and model adapters in the lower part. Class names are the same as the entity names in Section 9.

Design choice	Why
Asset is a superclass of ITAsset, LabEquipment and FurnitureAsset	Each kind has its own fields (for example a calibration date for lab equipment) but every rule about custody and status is written once
ResourceRequest is composed of RequestSpecification	A specification has no meaning without its request and is deleted with it
AssetCollection groups assets	A collection (for example the set a department offers for an event) can be deleted without deleting its assets; it is stored with the linking table AssetCollectionItem (Section 9)
ResourceRequest knows its requester User	A request can always find who asked for it; a user does not need to hold its requests
MatchRecommendation links one request to one asset	It resolves the many-to-many relation and stores the score, the reasons and the model version; it does not authorize a transfer, only an Approval does
Condition, AssetStatus and RequestStatus are enumerations	A field can only hold one of the listed values, the same values used in Sections 8.5 and 9
AIService is an interface with six classes that realize it	The gateway treats every AI task the same way: one call, one Suggestion, one AIRecord
TextGenerator has two versions: OpenAIGenerator and LocalGenerator	Decision AD-2: a single setting moves the project off the paid API if paid use is not allowed
Suggestion has no save method	The AI cannot write business records; only a service method called after a person confirms can (Section 5)

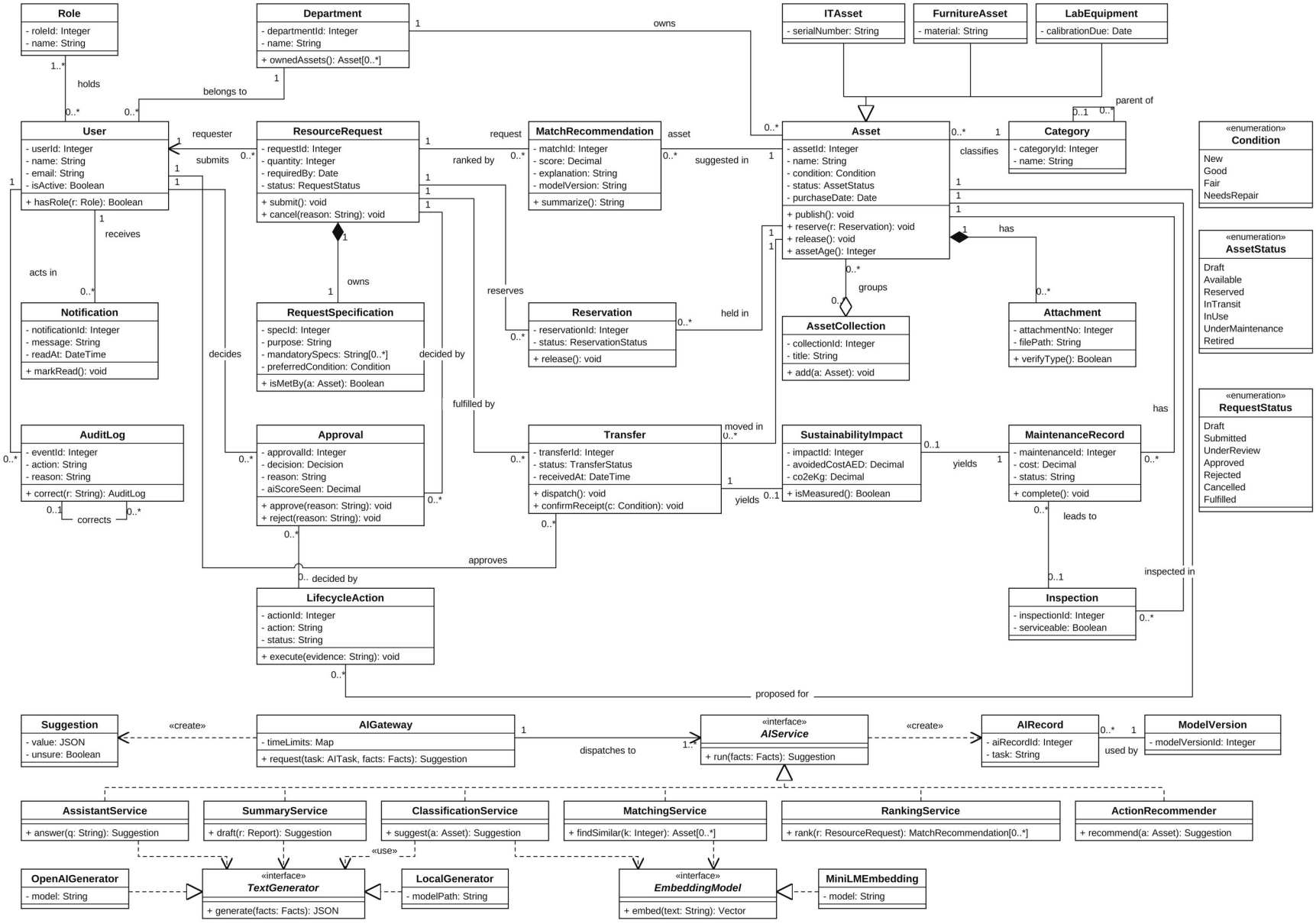


Figure 15. Class diagram of the whole system.

8.3 Sequence diagrams

There is one sequence diagram for each use case, UC-01 to UC-38 (Figures 16 to 53). Each diagram shows the main flow of the Phase 3 use case and, in its Alternative frame, the main exception. The participants are the parts of Figure 59: the Web UI, the application module that owns the use case, the AI Gateway where AI is used, Notifications and the Database. A notification is sent without waiting for an answer.

Figure	Use case	Primary actor	Alternative shown
16	UC-01 Sign in	Requester	Password wrong or account disabled
17	UC-02 Manage user access	Administrator	Role, department or admin right not valid
18	UC-03 Maintain reference data	Administrator	Blank field or broken link
19	UC-04 Register an asset	Custodian	Mandatory field missing or negative value
20	UC-05 Update an asset record	Custodian	Record changed or allocated meanwhile
21	UC-06 Publish an asset	Custodian	Asset not eligible
22	UC-07 Withdraw an asset listing	Custodian	Active reservation or transfer
23	UC-08 Search the asset catalogue	Requester	Invalid filter
24	UC-09 Inspect asset details	Requester	Record outside the user's scope
25	UC-10 Submit a resource request	Requester	Invalid field or quantity
26	UC-11 Revise a pending request	Requester	Request reserved or changed meanwhile
27	UC-12 Cancel a resource request	Requester	Units already dispatched
28	UC-13 Reserve eligible assets	Custodian	A unit is no longer eligible: nothing reserved
29	UC-14 Review a resource request	Approver	Approver is the requester
30	UC-15 Schedule an asset transfer	Custodian	Date passed or reservation incomplete
31	UC-16 Release reserved assets	Custodian	Approval no longer valid
32	UC-17 Confirm asset receipt	Requester	Missing or damaged units: transfer disputed
33	UC-18 Return loaned assets	Requester	Loan closed or another transfer exists
34	UC-19 Record an asset inspection (later increment)	Maintenance Officer	Evidence incomplete
35	UC-20 Report an asset fault (later increment)	Requester	No description
36	UC-21 Plan asset maintenance (later increment)	Maintenance Officer	Invalid plan
37	UC-22 Complete asset maintenance (later increment)	Maintenance Officer	No passing inspection or negative cost
38	UC-23 Propose a lifecycle action (later increment)	Custodian	No rationale or negative input
39	UC-24 Authorize a lifecycle action (later increment)	Approver	Approver proposed the action
40	UC-25 Record asset retirement (later increment)	Custodian	Evidence missing or asset still moving
41	UC-26 Generate an operational report	Custodian	Query cannot complete
42	UC-27 Record benefit calculation inputs (later increment)	Finance Officer	Event already claimed
43	UC-28 Generate a sustainability report	Sustainability Officer	Factor without source, date or units
44	UC-29 Read status notifications	Requester	No access to the linked record
45	UC-30 Review the audit history (later increment)	Auditor	Record outside the auditor's scope
46	UC-31 Suggest asset classifications	Custodian	AI fails or answers late: category list
47	UC-32 Match resources semantically	Requester	AI late: units in Asset ID order
48	UC-33 Recommend a sustainable action (later increment)	Custodian	Not enough evidence
49	UC-34 Ask the resource assistant (later increment)	Requester	A claim has no source
50	UC-35 Review an AI recommendation	Custodian	Record changed or no permission
51	UC-36 Configure AI services	Administrator	Test response invalid
52	UC-37 Restore a prototype backup (later increment)	Administrator	Restored data inconsistent
53	UC-38 Summarize recorded activity	Sustainability Officer	A number does not match the report

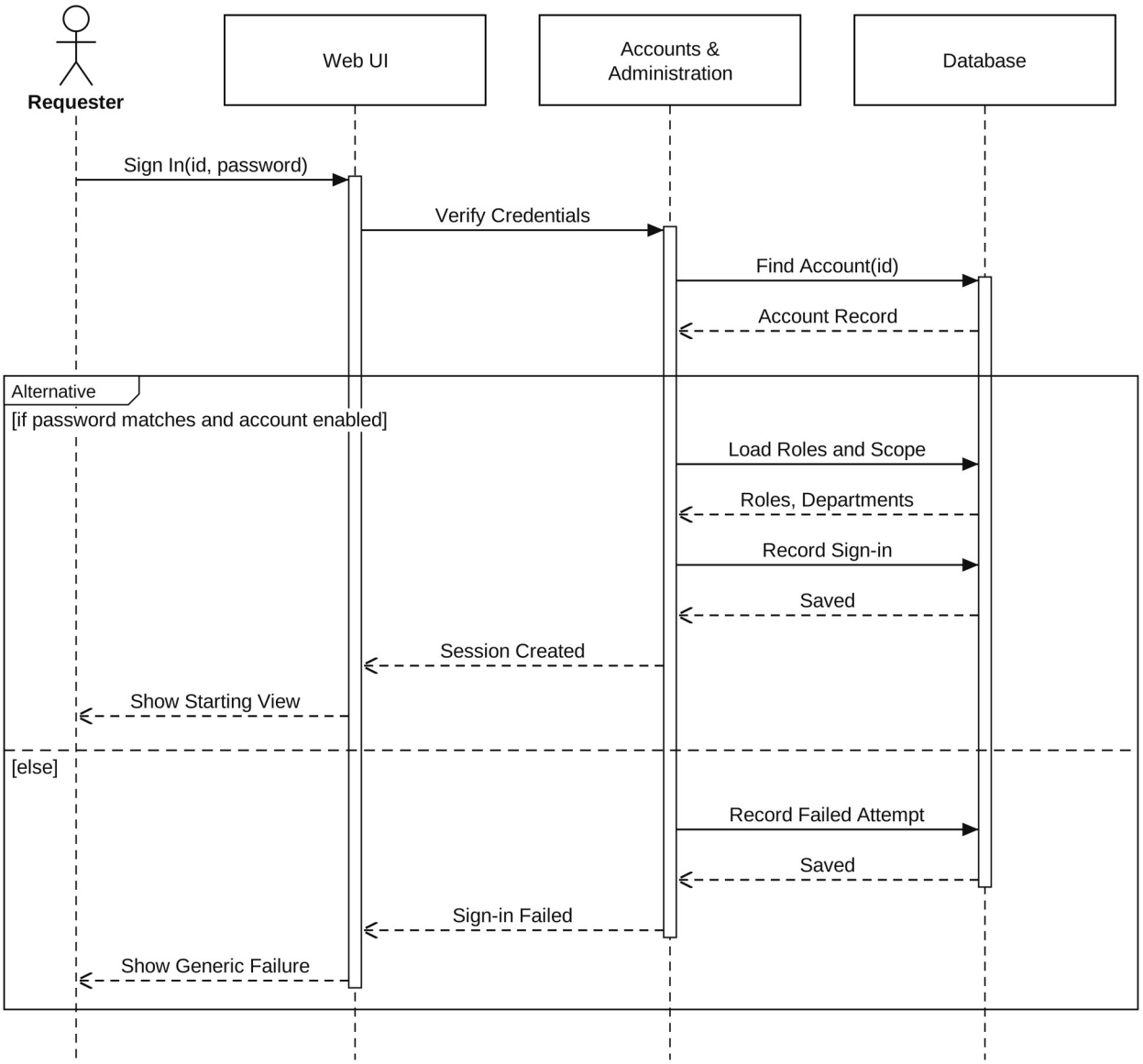


Figure 16. Sequence diagram: UC-01 Sign in.

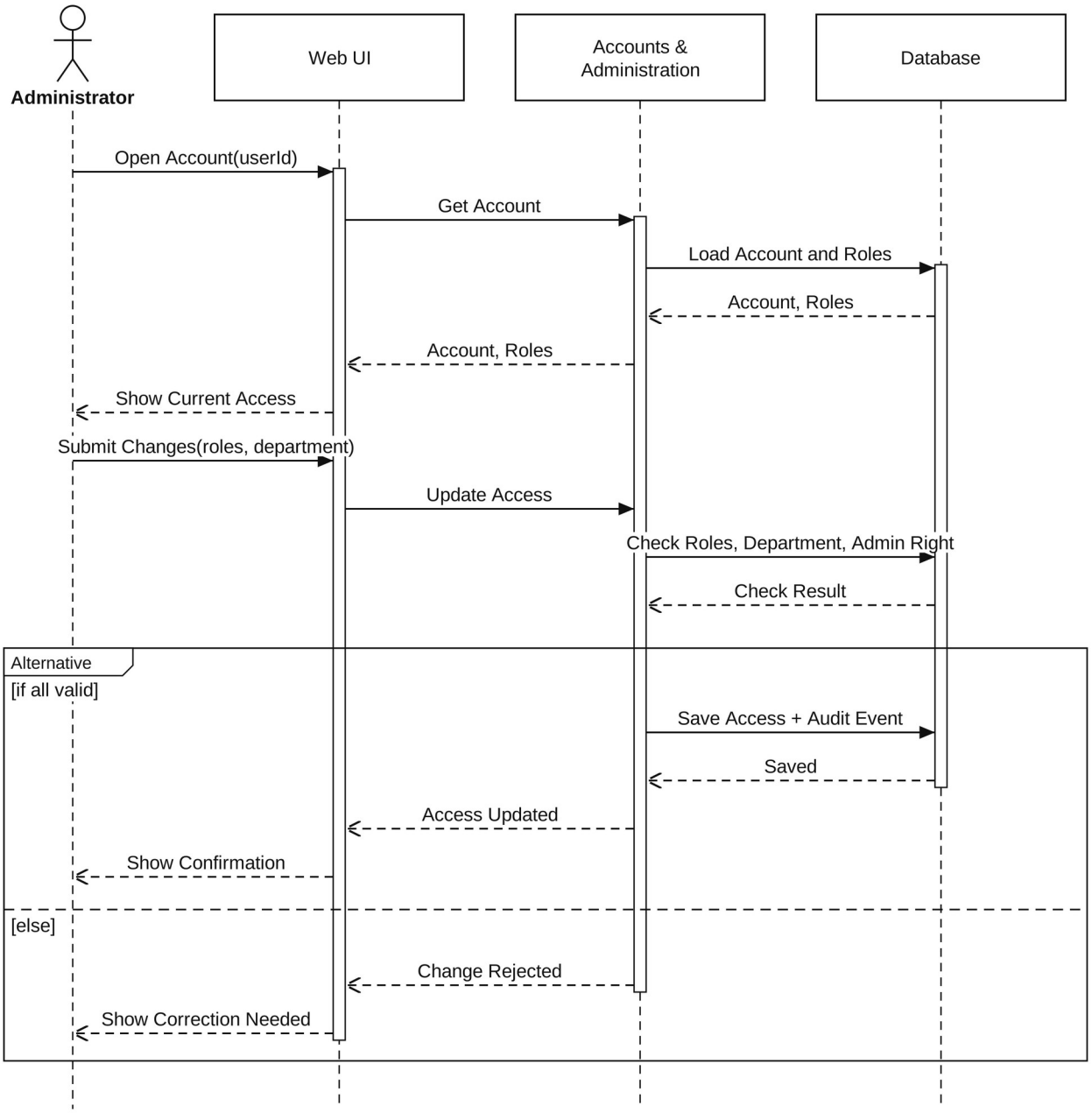


Figure 17. Sequence diagram: UC-02 Manage user access.

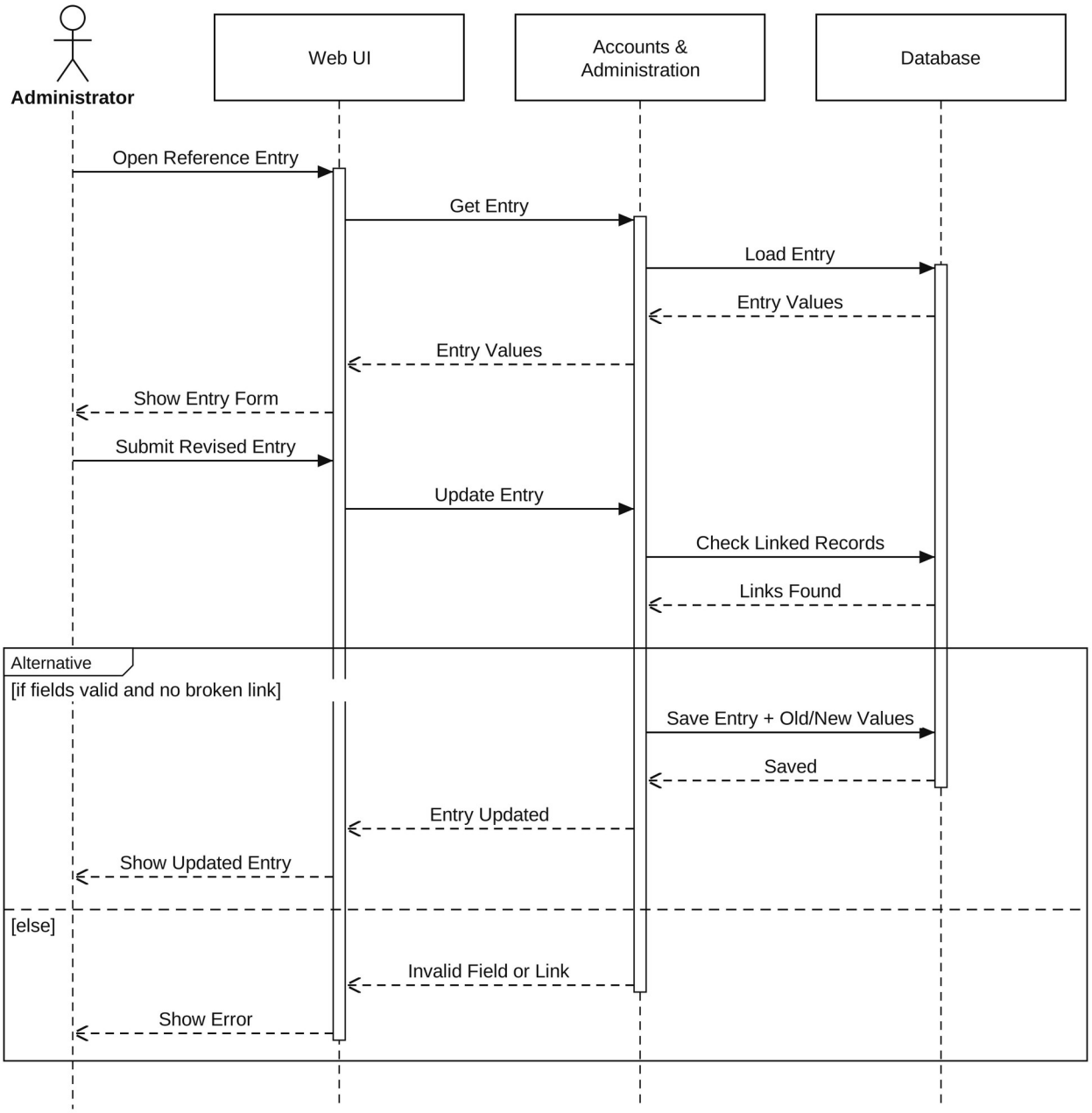


Figure 18. Sequence diagram: UC-03 Maintain reference data.

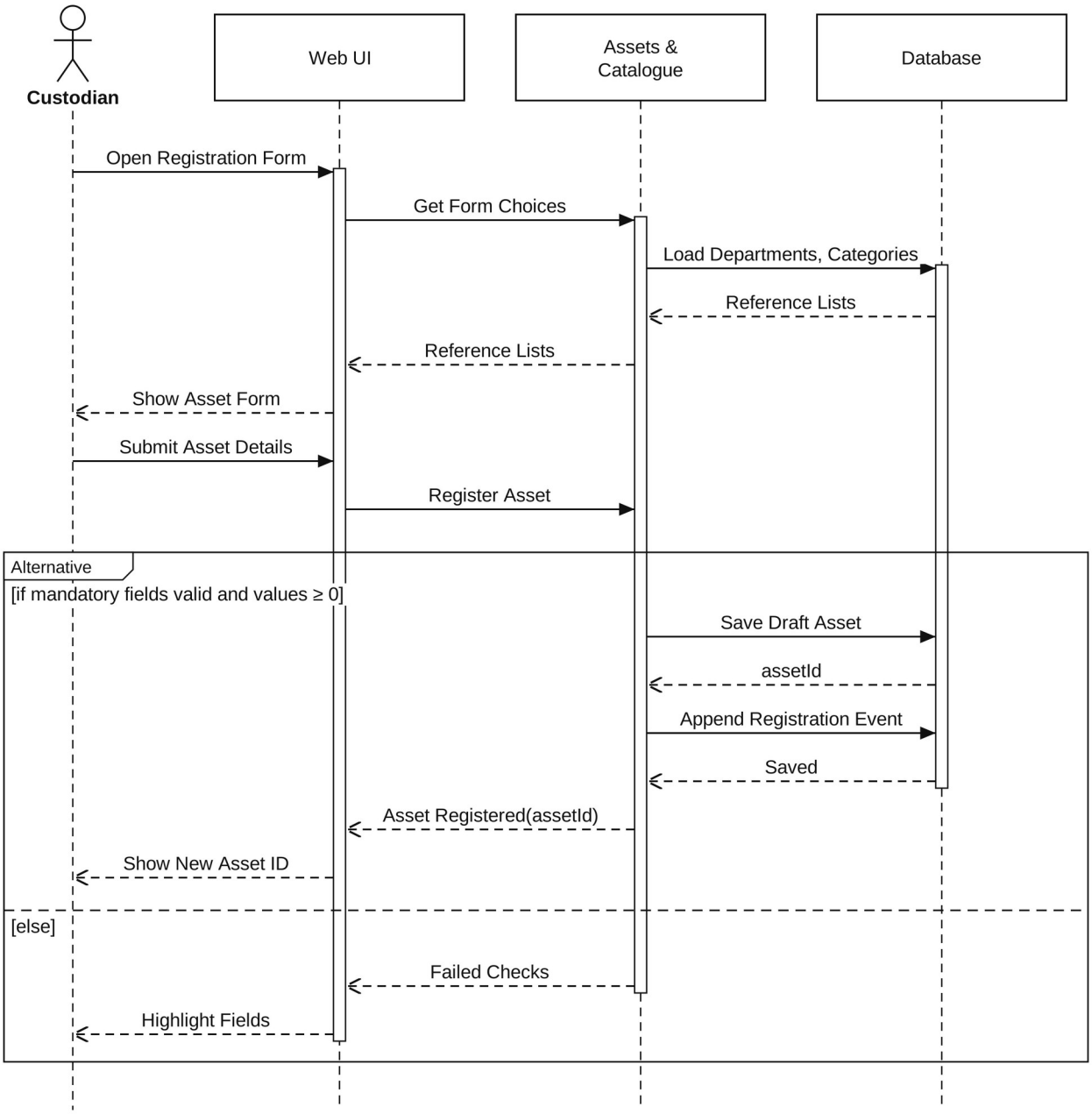


Figure 19. Sequence diagram: UC-04 Register an asset.

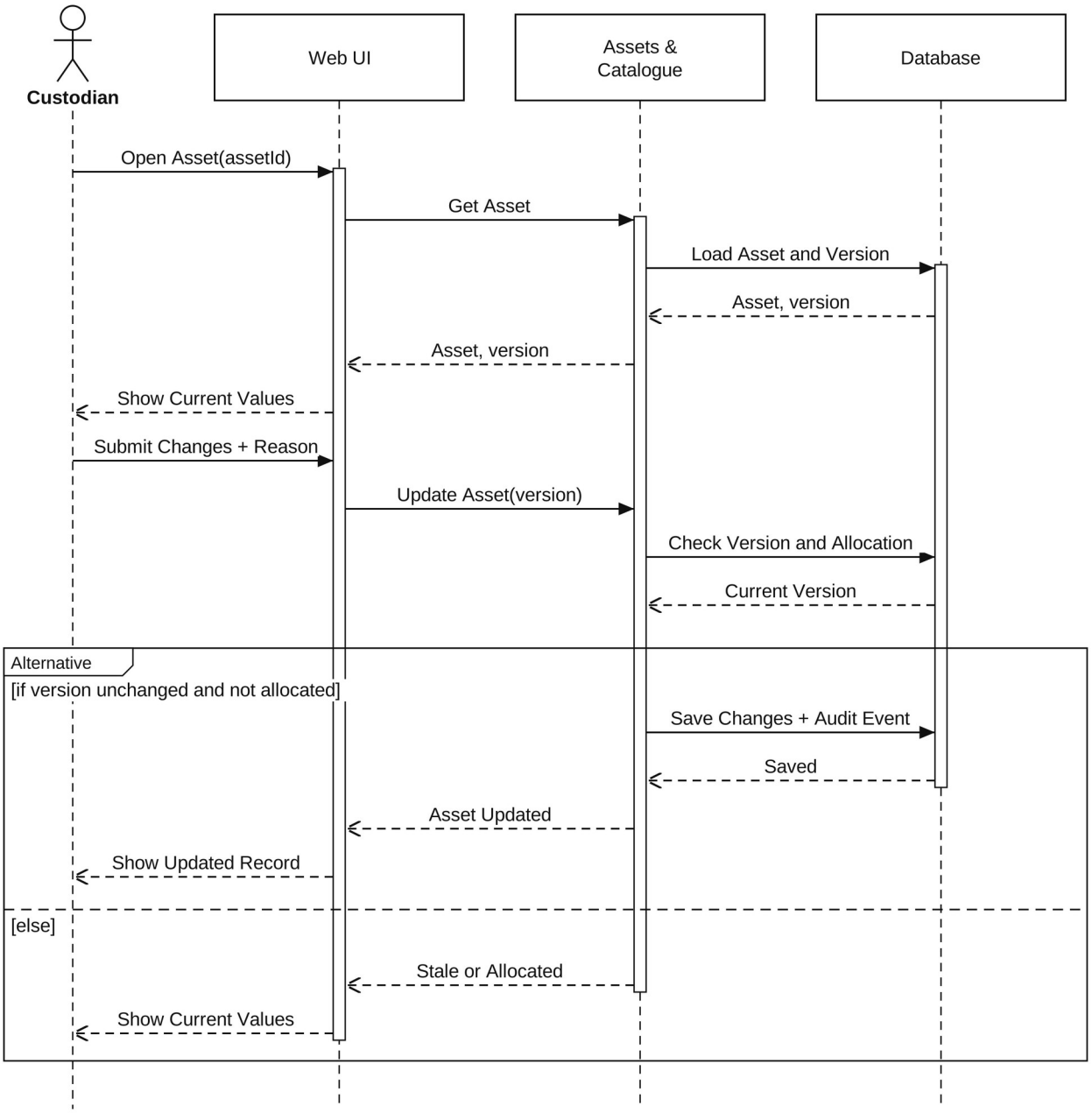


Figure 20. Sequence diagram: UC-05 Update an asset record.

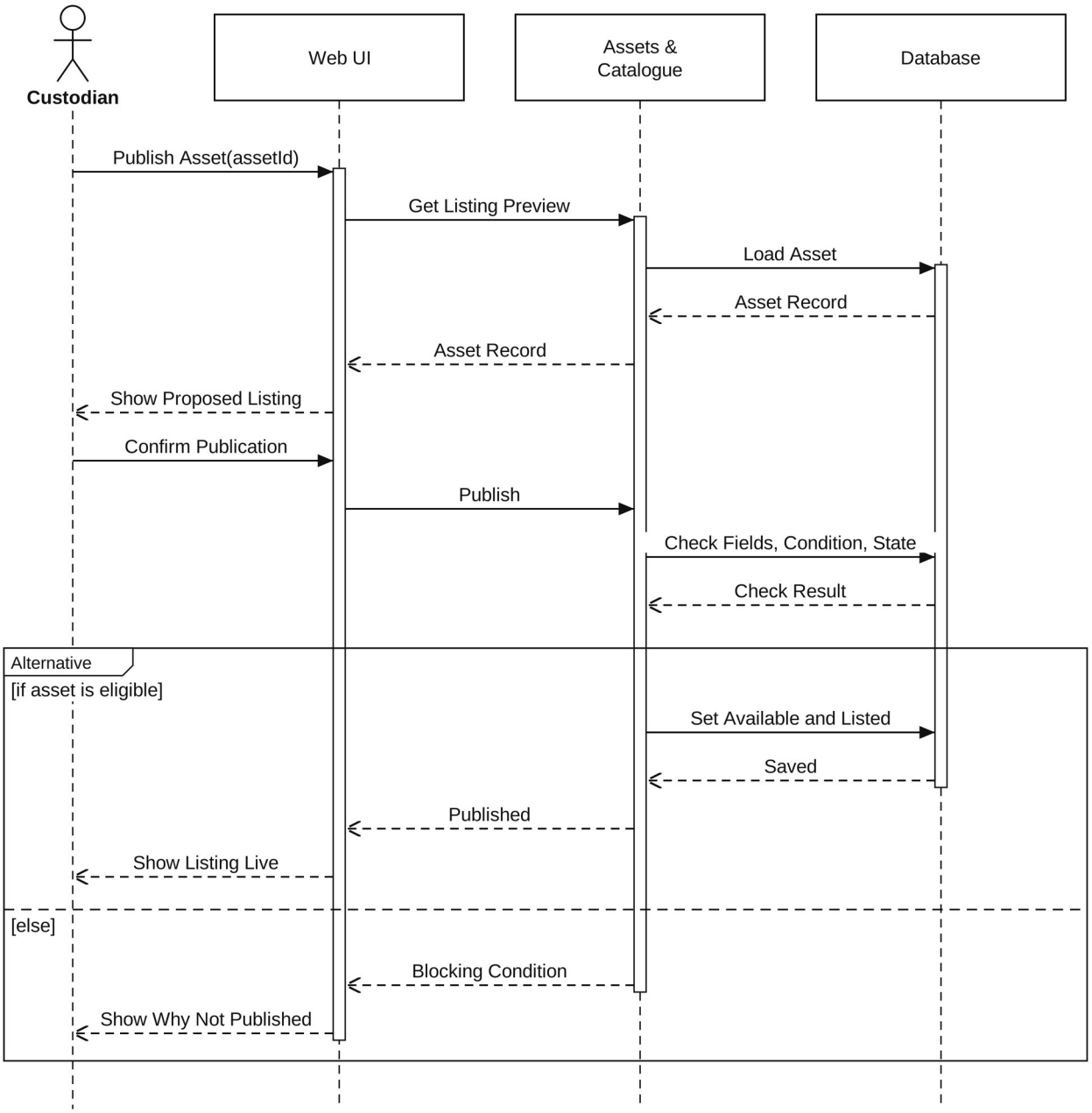


Figure 21. Sequence diagram: UC-06 Publish an asset.

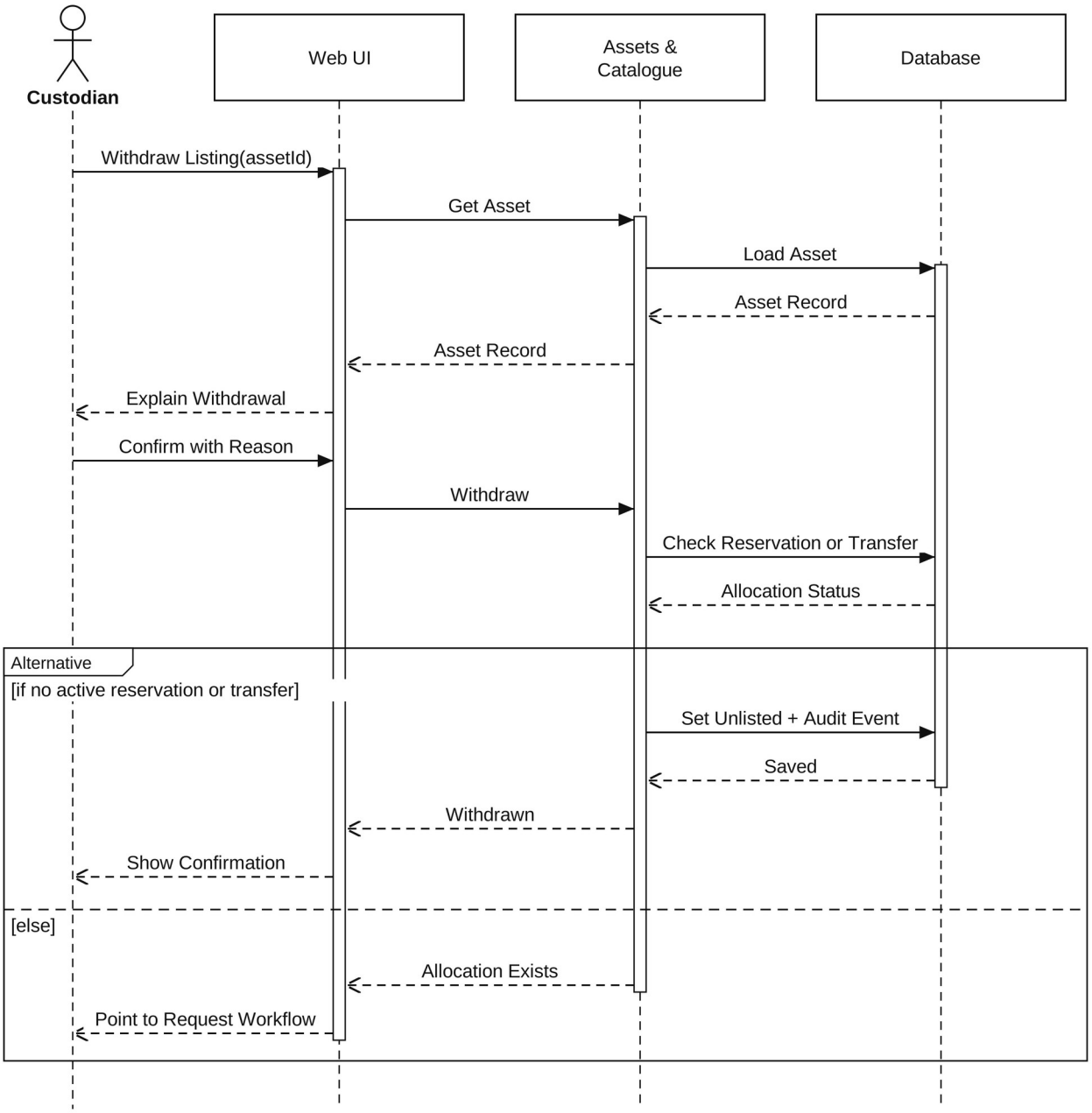


Figure 22. Sequence diagram: UC-07 Withdraw an asset listing.

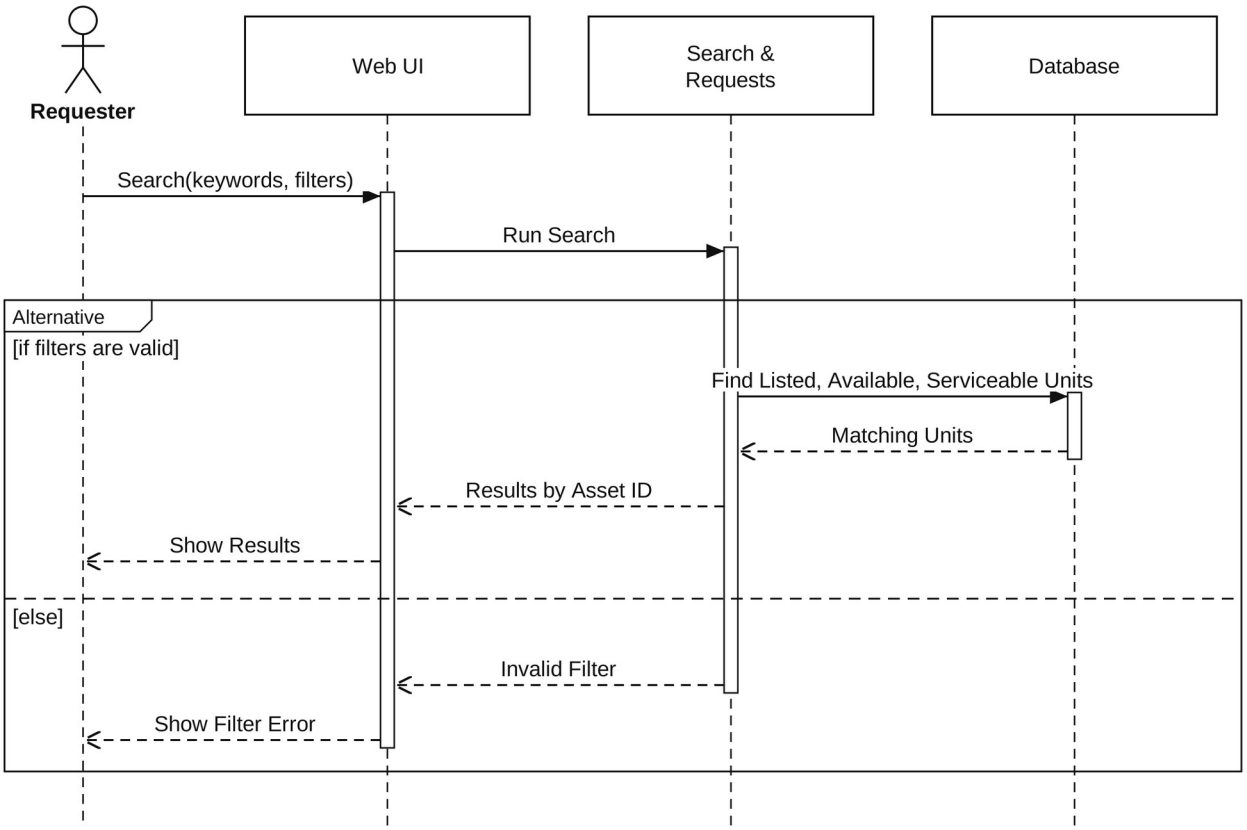


Figure 23. Sequence diagram: UC-08 Search the asset catalogue.

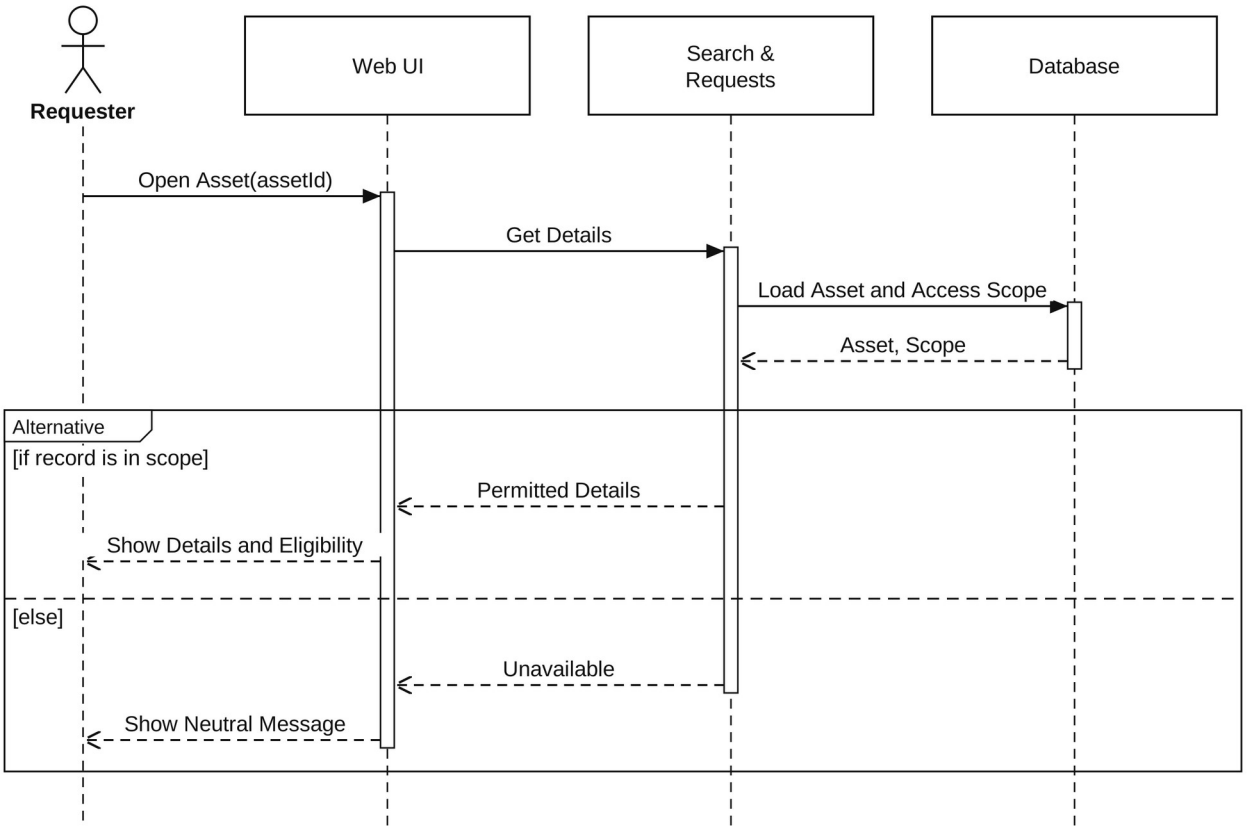


Figure 24. Sequence diagram: UC-09 Inspect asset details.

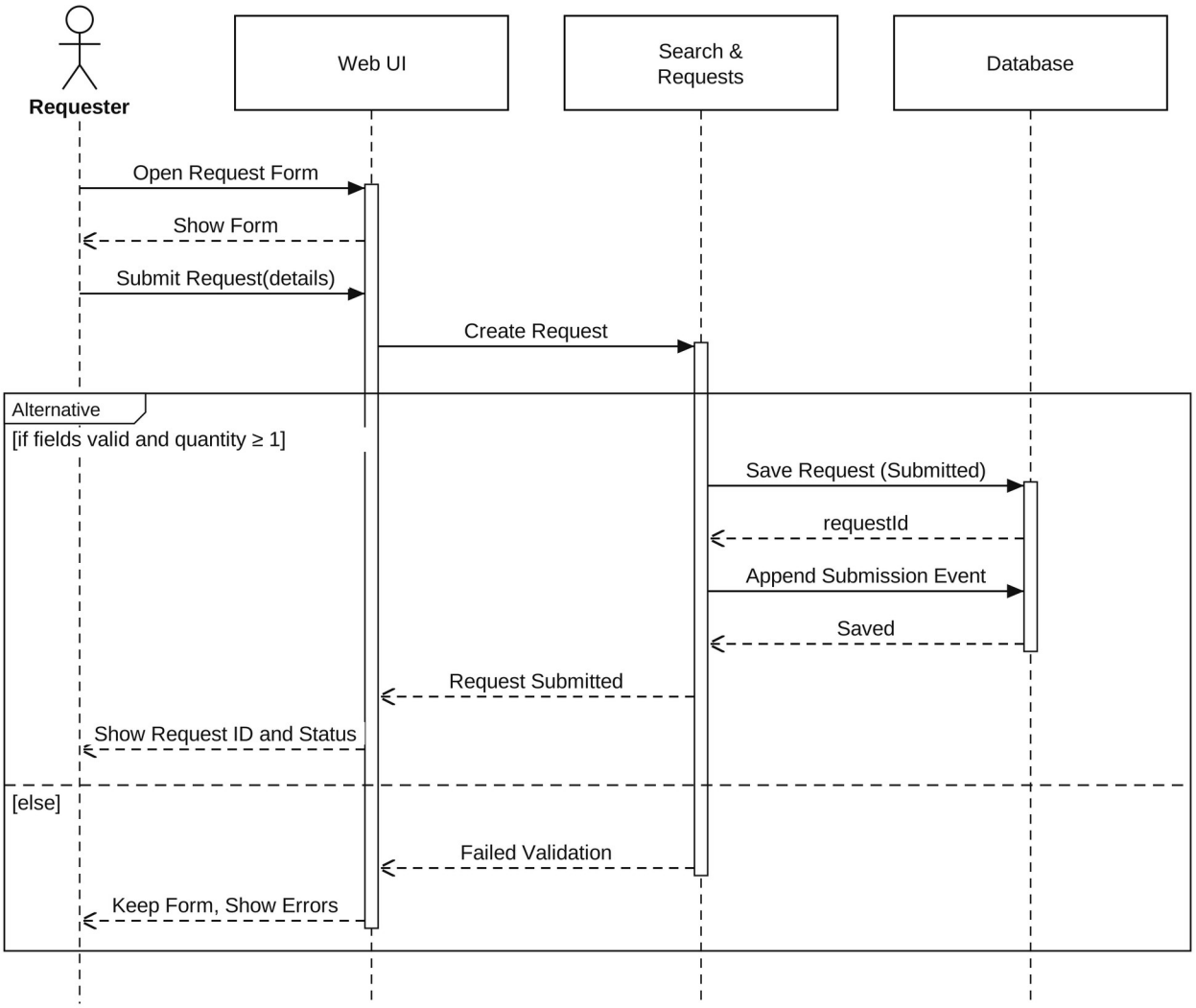


Figure 25. Sequence diagram: UC-10 Submit a resource request.

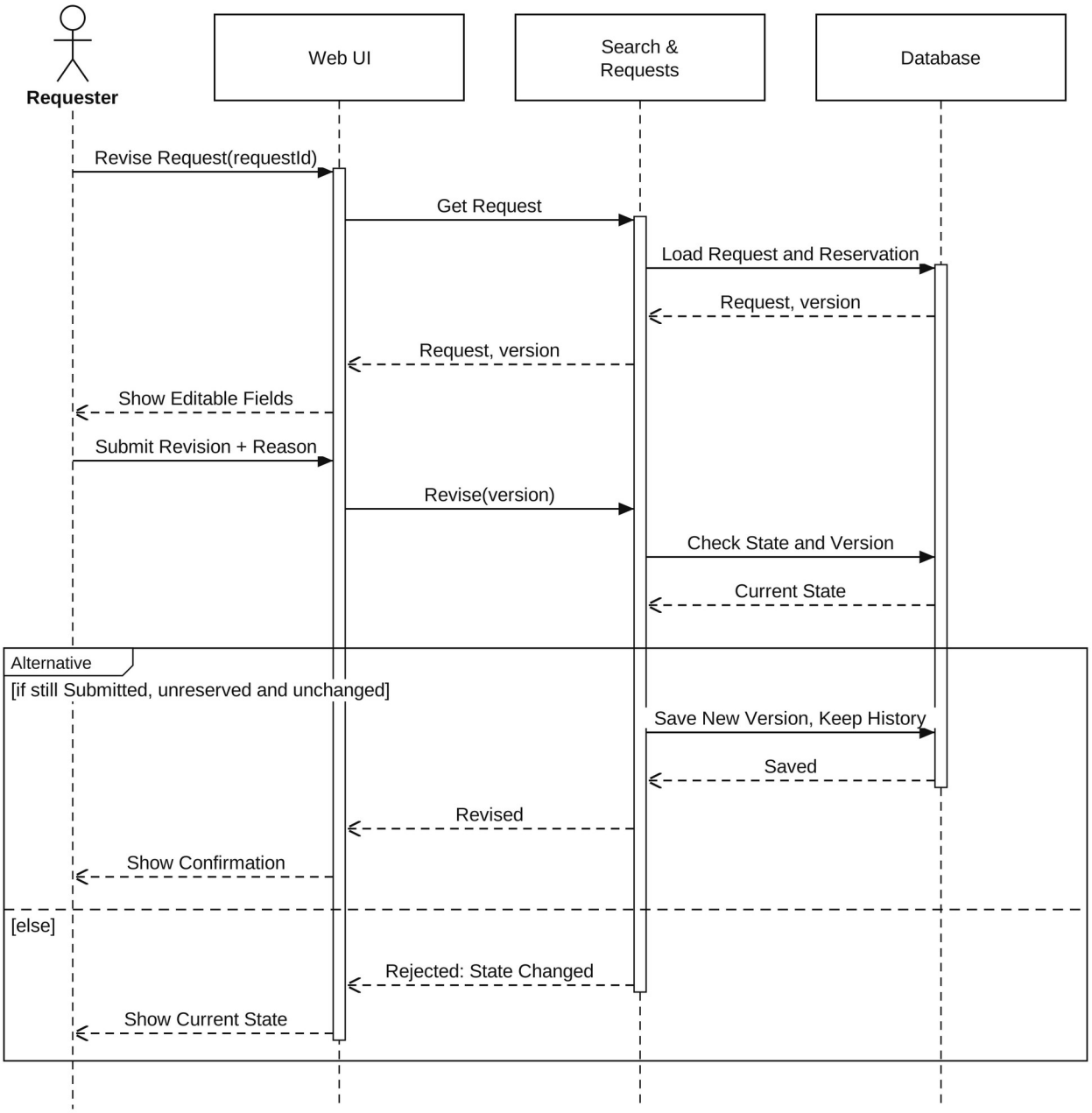


Figure 26. Sequence diagram: UC-11 Revise a pending request.

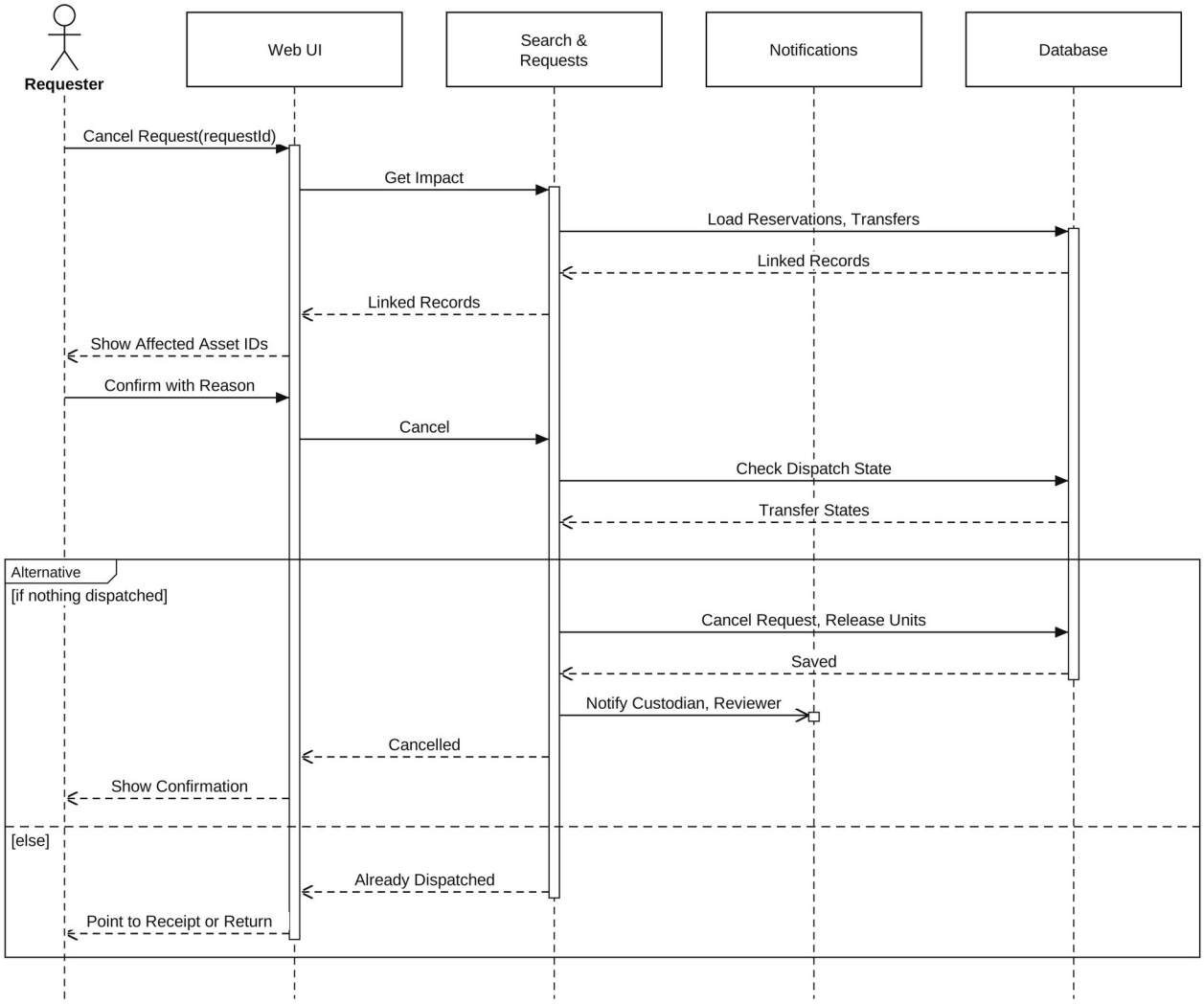


Figure 27. Sequence diagram: UC-12 Cancel a resource request.

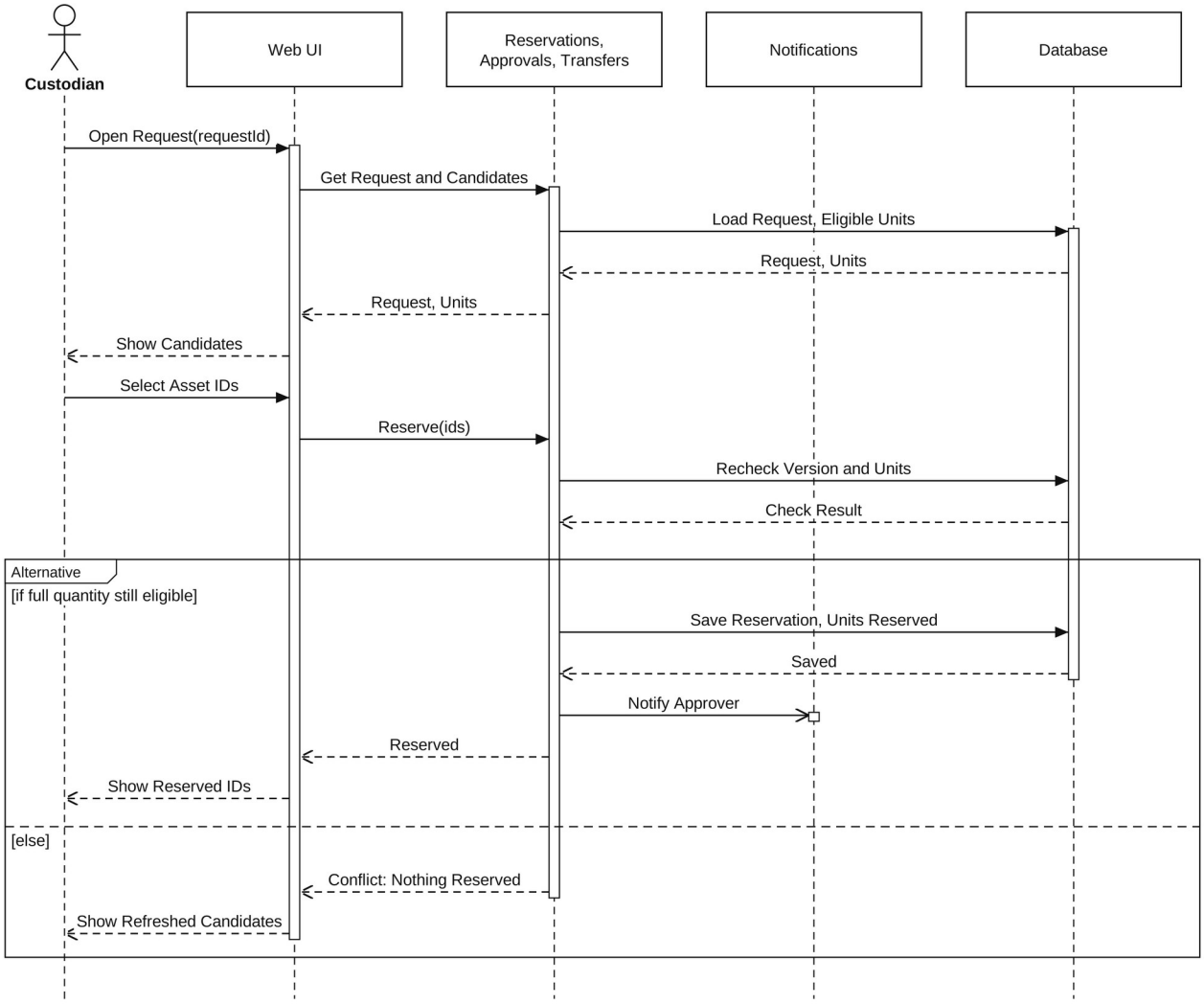


Figure 28. Sequence diagram: UC-13 Reserve eligible assets.

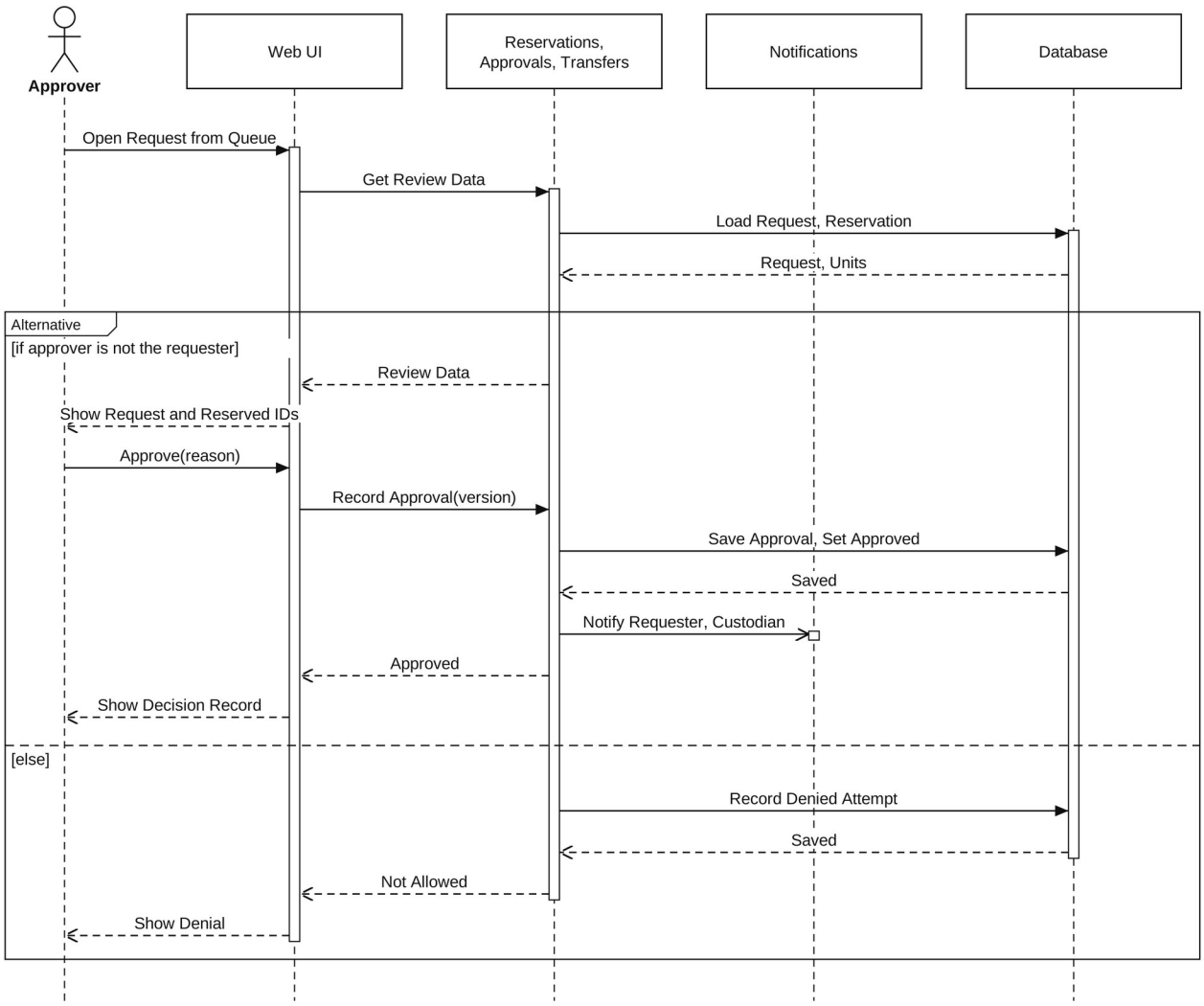


Figure 29. Sequence diagram: UC-14 Review a resource request.

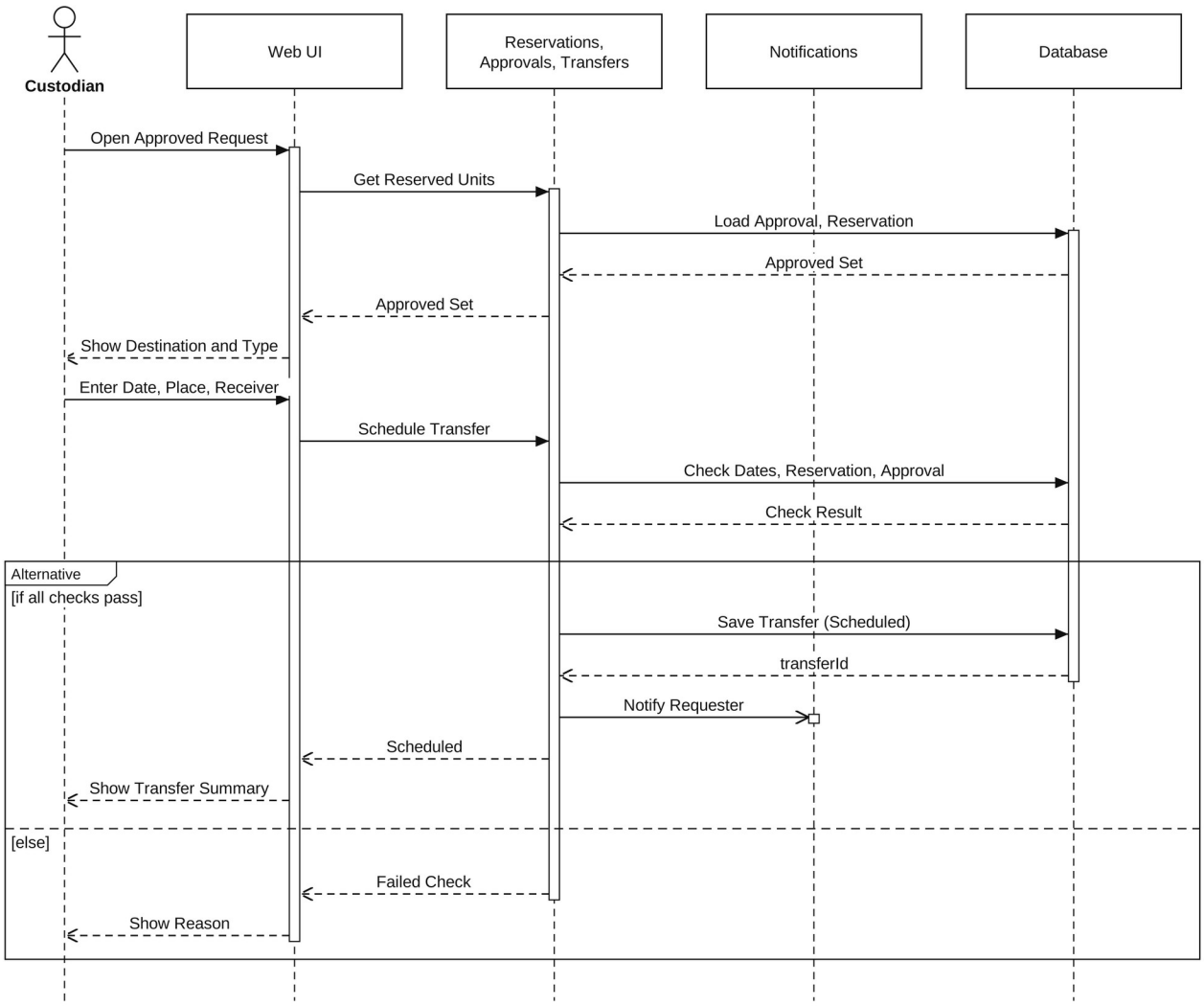


Figure 30. Sequence diagram: UC-15 Schedule an asset transfer.

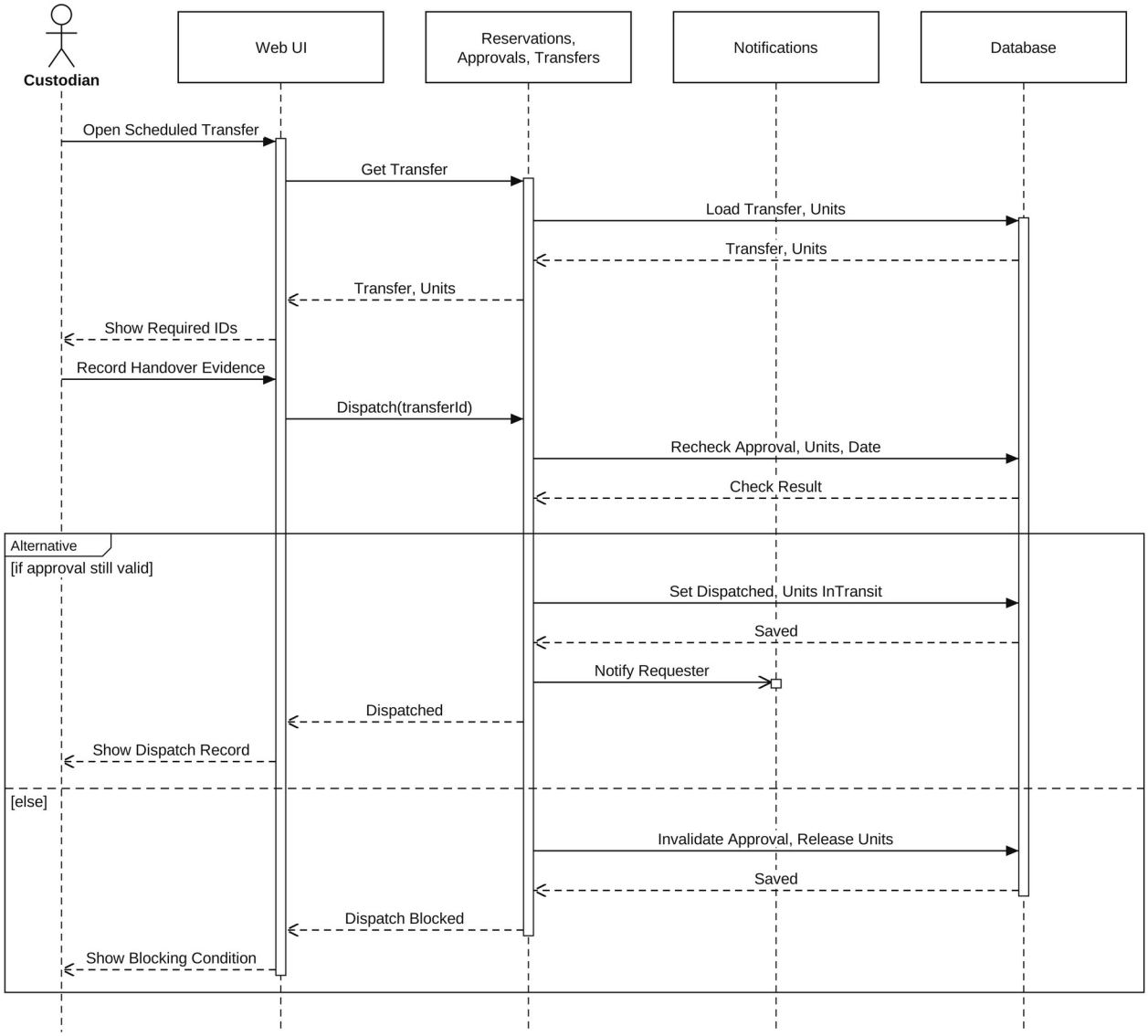


Figure 31. Sequence diagram: UC-16 Release reserved assets.

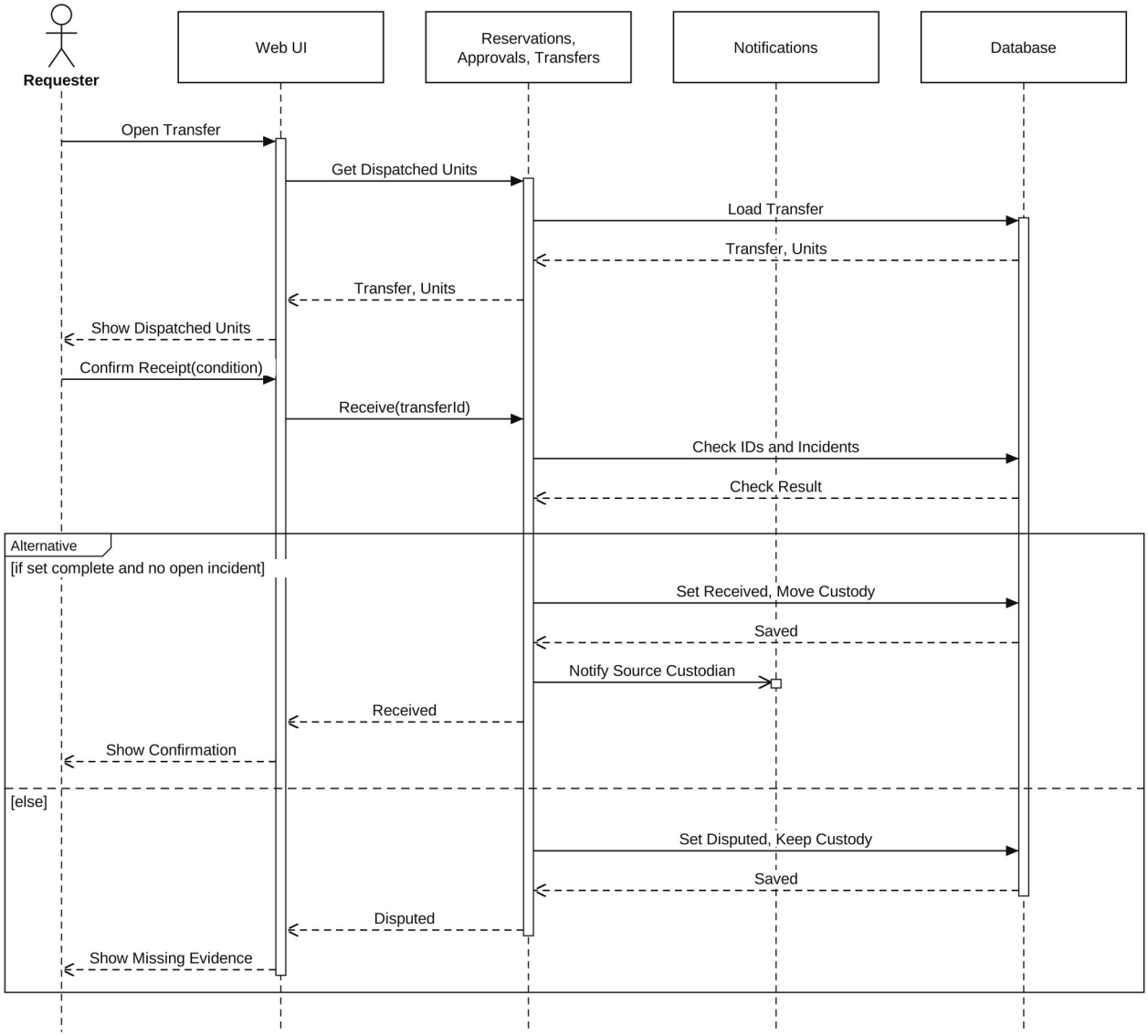


Figure 32. Sequence diagram: UC-17 Confirm asset receipt.

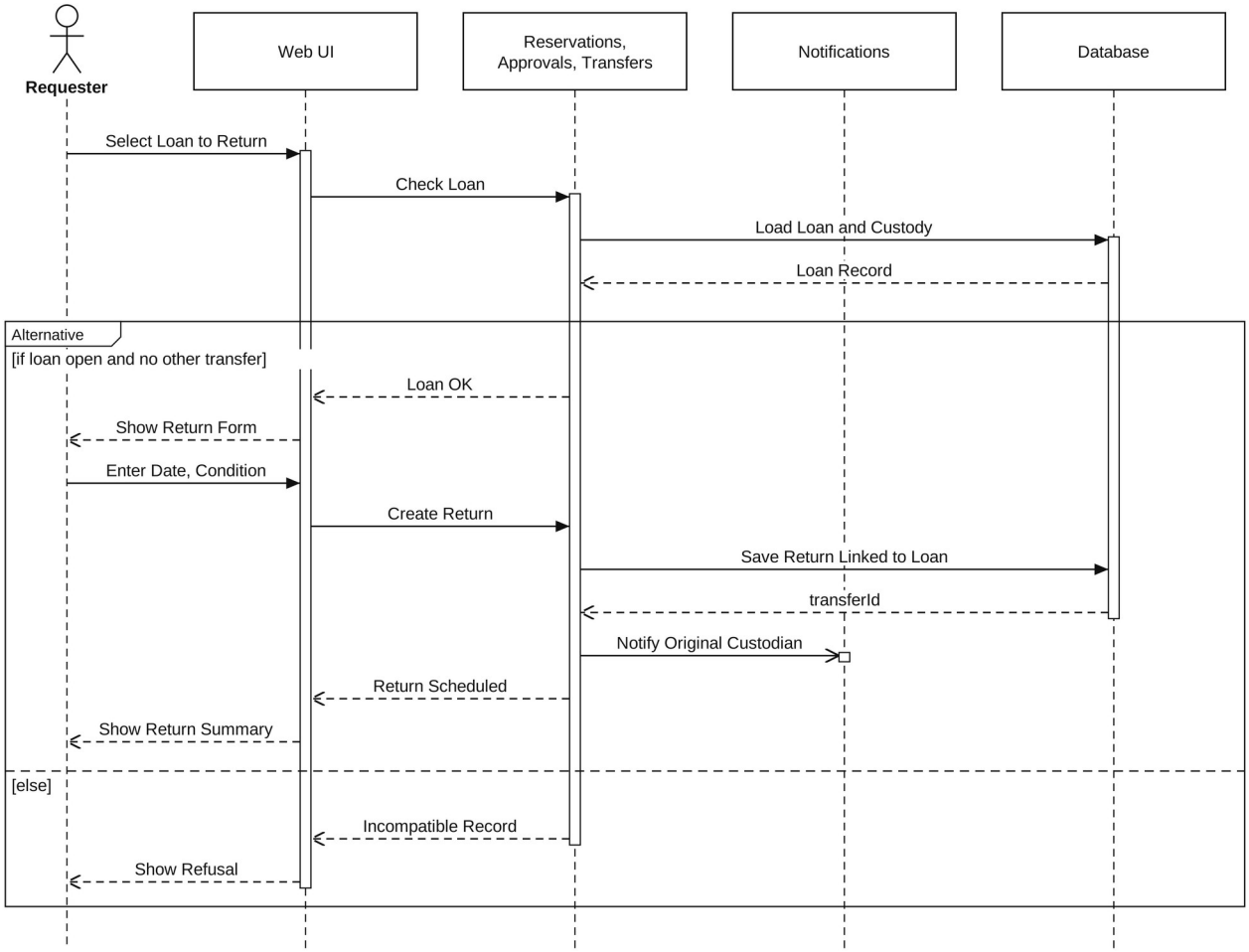


Figure 33. Sequence diagram: UC-18 Return loaned assets.

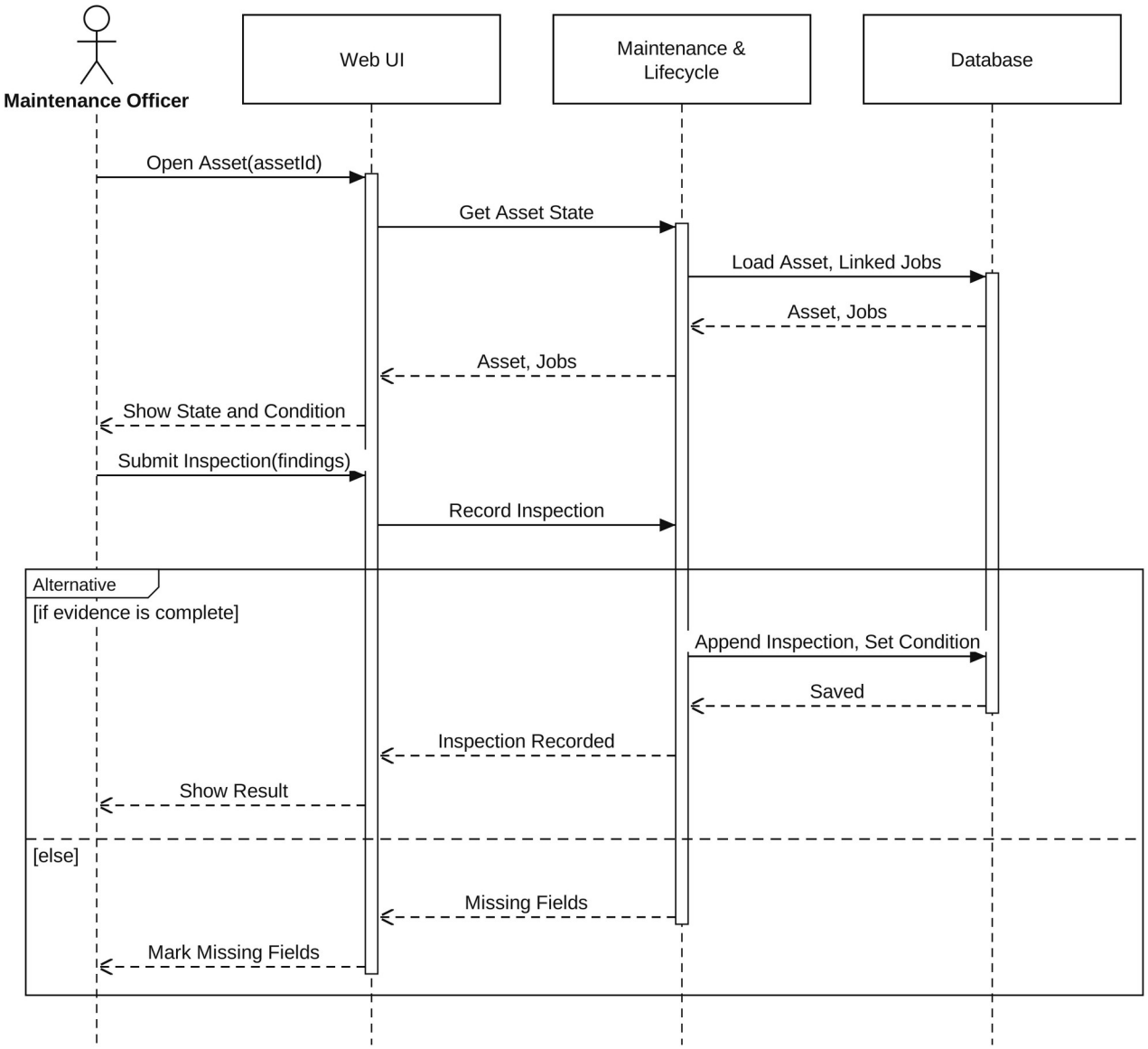


Figure 34. Sequence diagram: UC-19 Record an asset inspection (later increment).

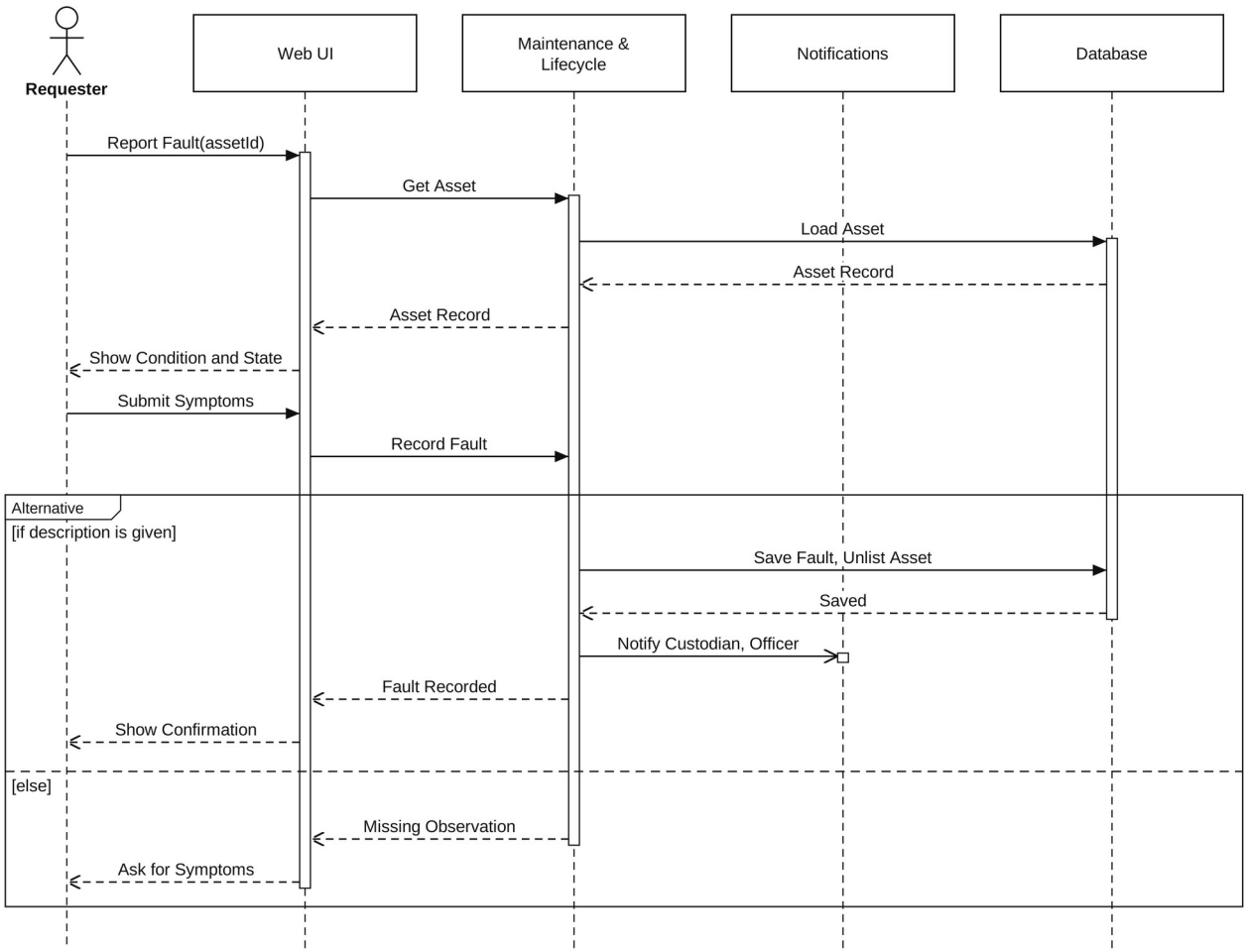


Figure 35. Sequence diagram: UC-20 Report an asset fault (later increment).

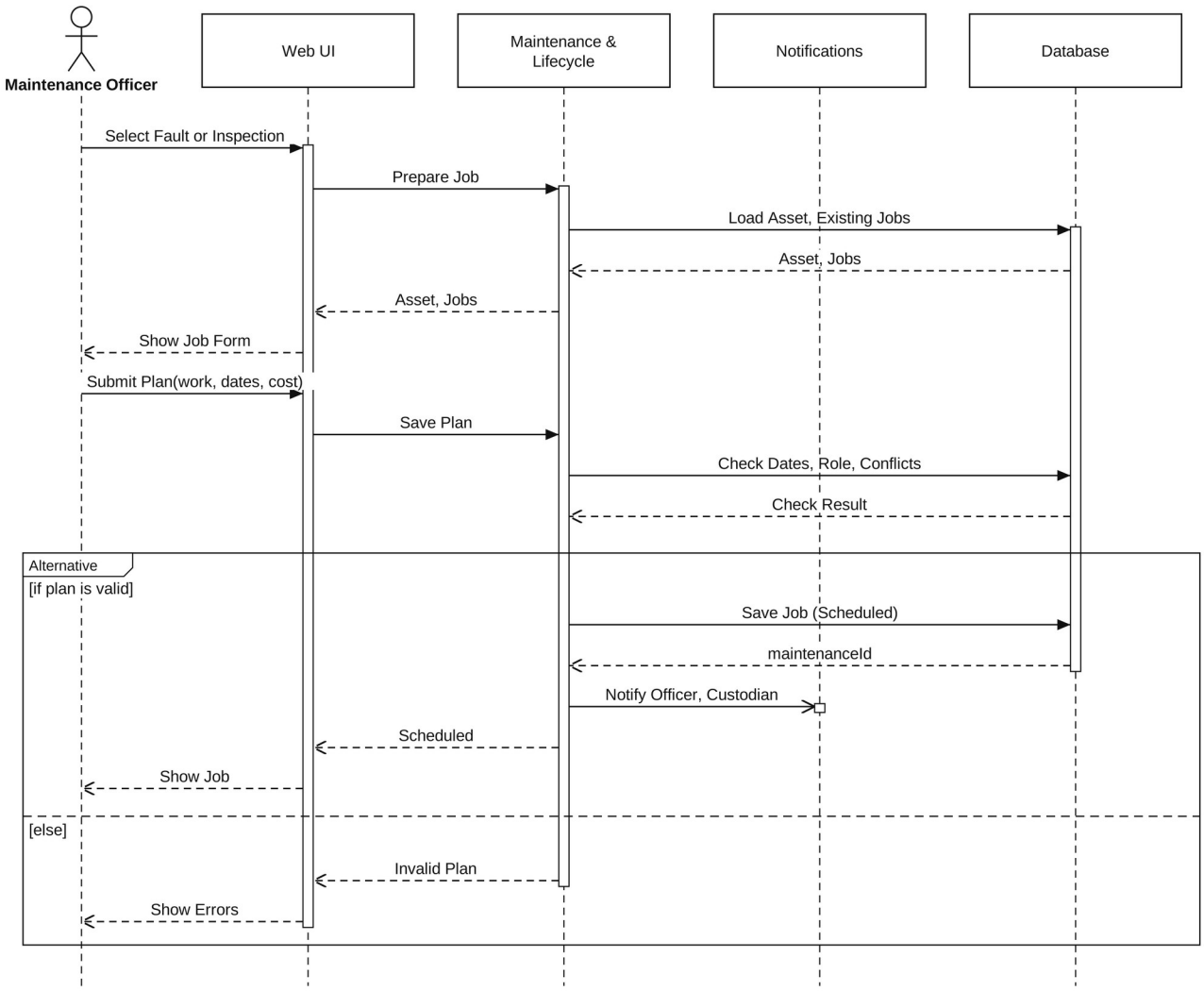


Figure 36. Sequence diagram: UC-21 Plan asset maintenance (later increment).

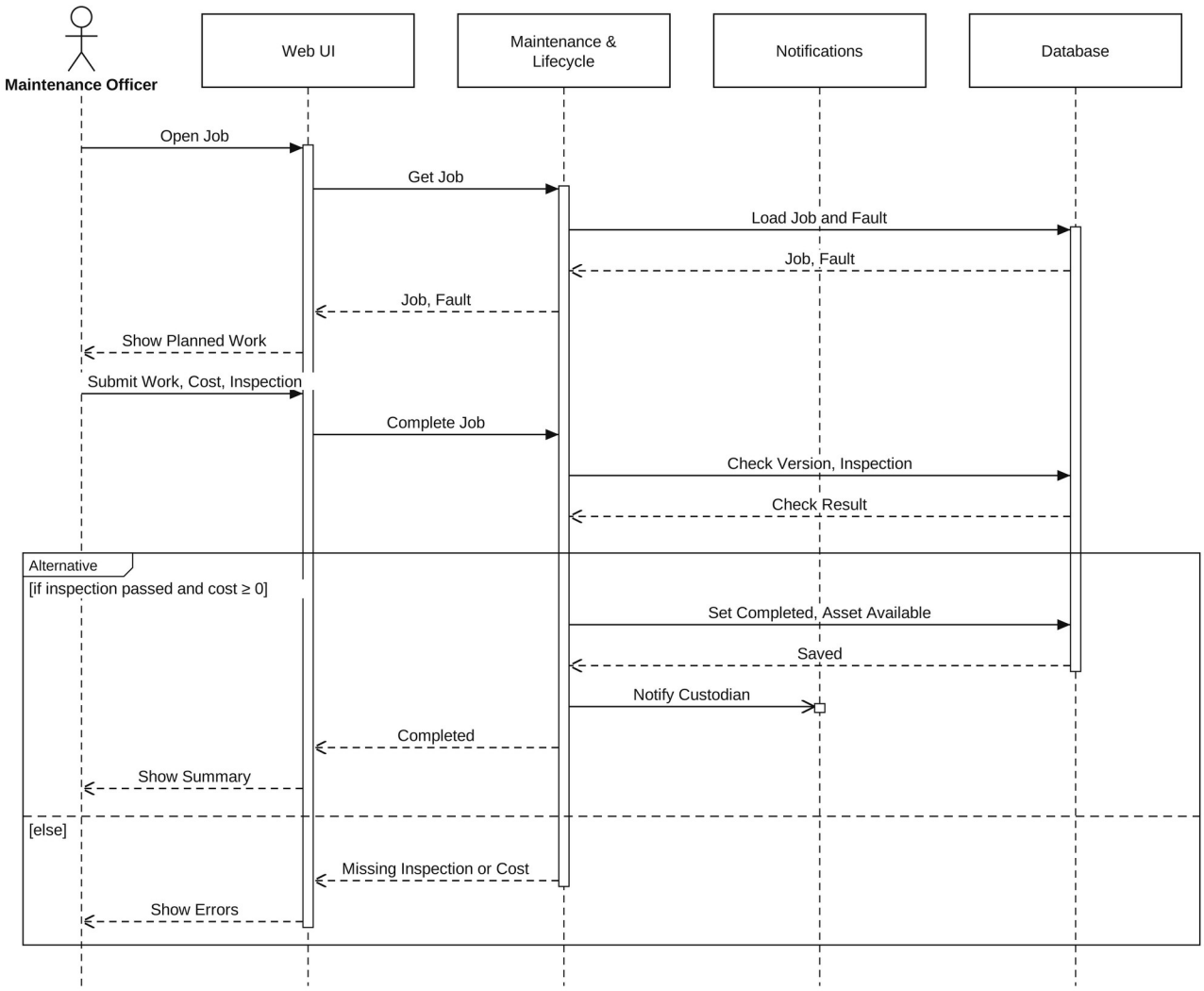


Figure 37. Sequence diagram: UC-22 Complete asset maintenance (later increment).

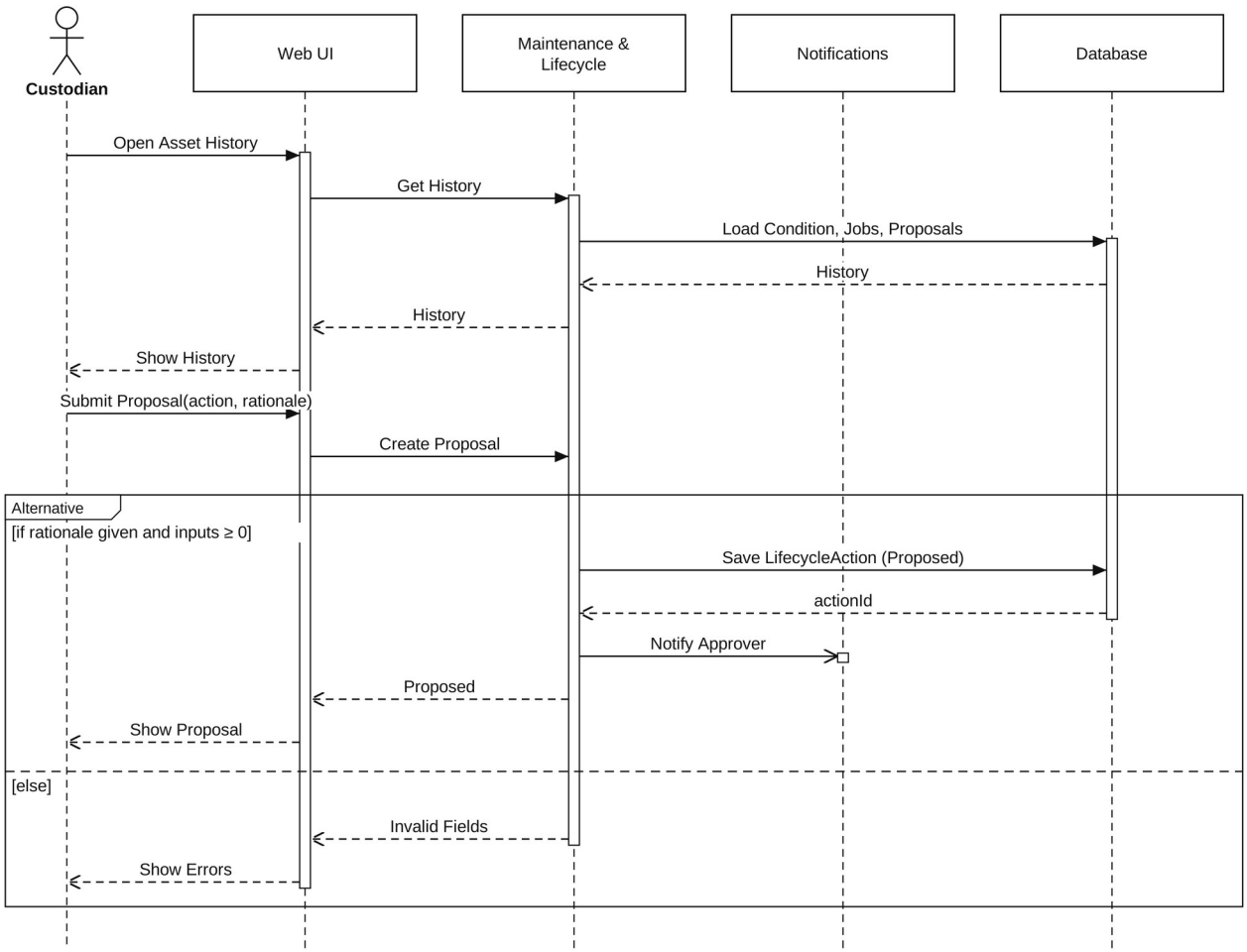


Figure 38. Sequence diagram: UC-23 Propose a lifecycle action (later increment).

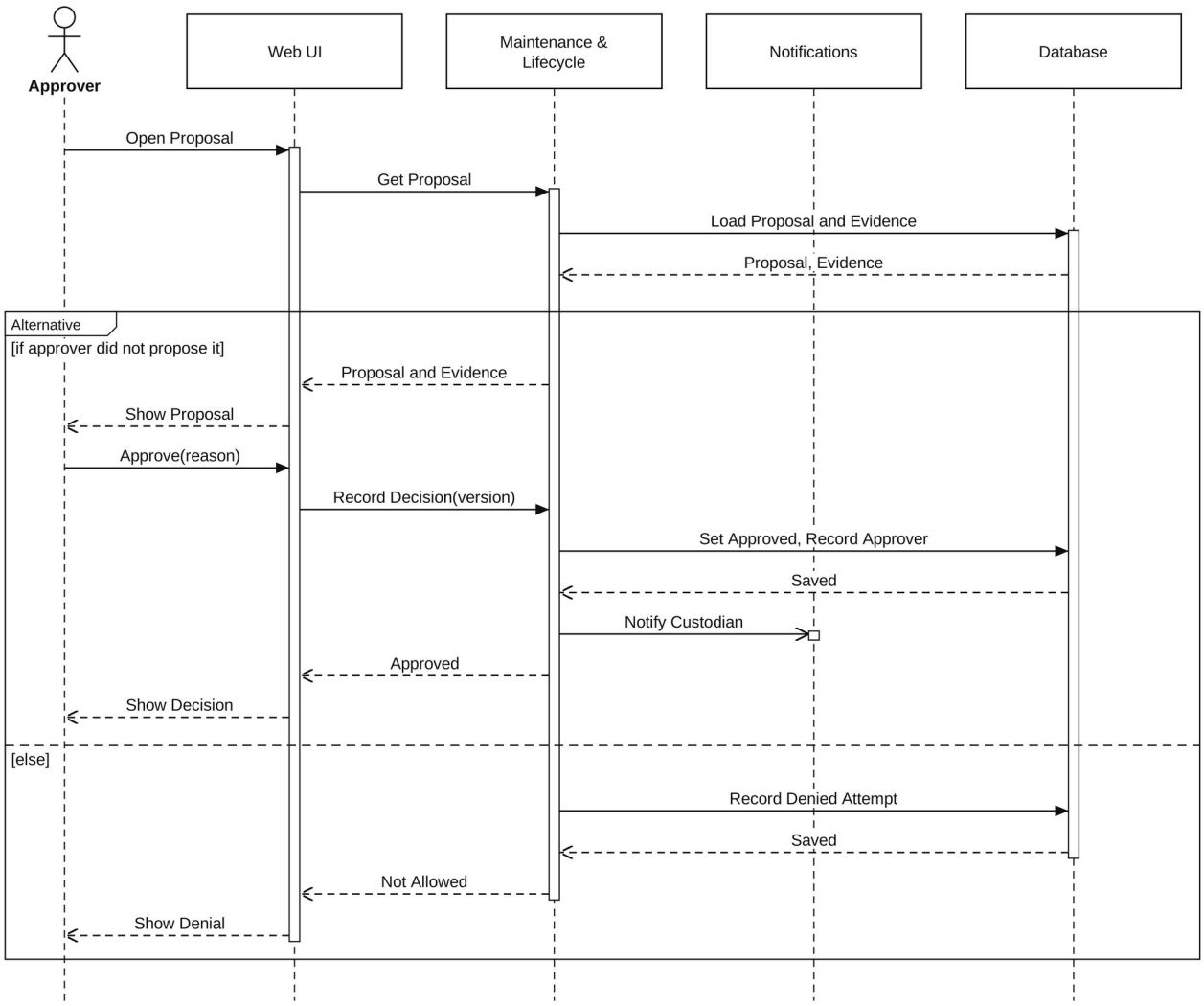


Figure 39. Sequence diagram: UC-24 Authorize a lifecycle action (later increment).

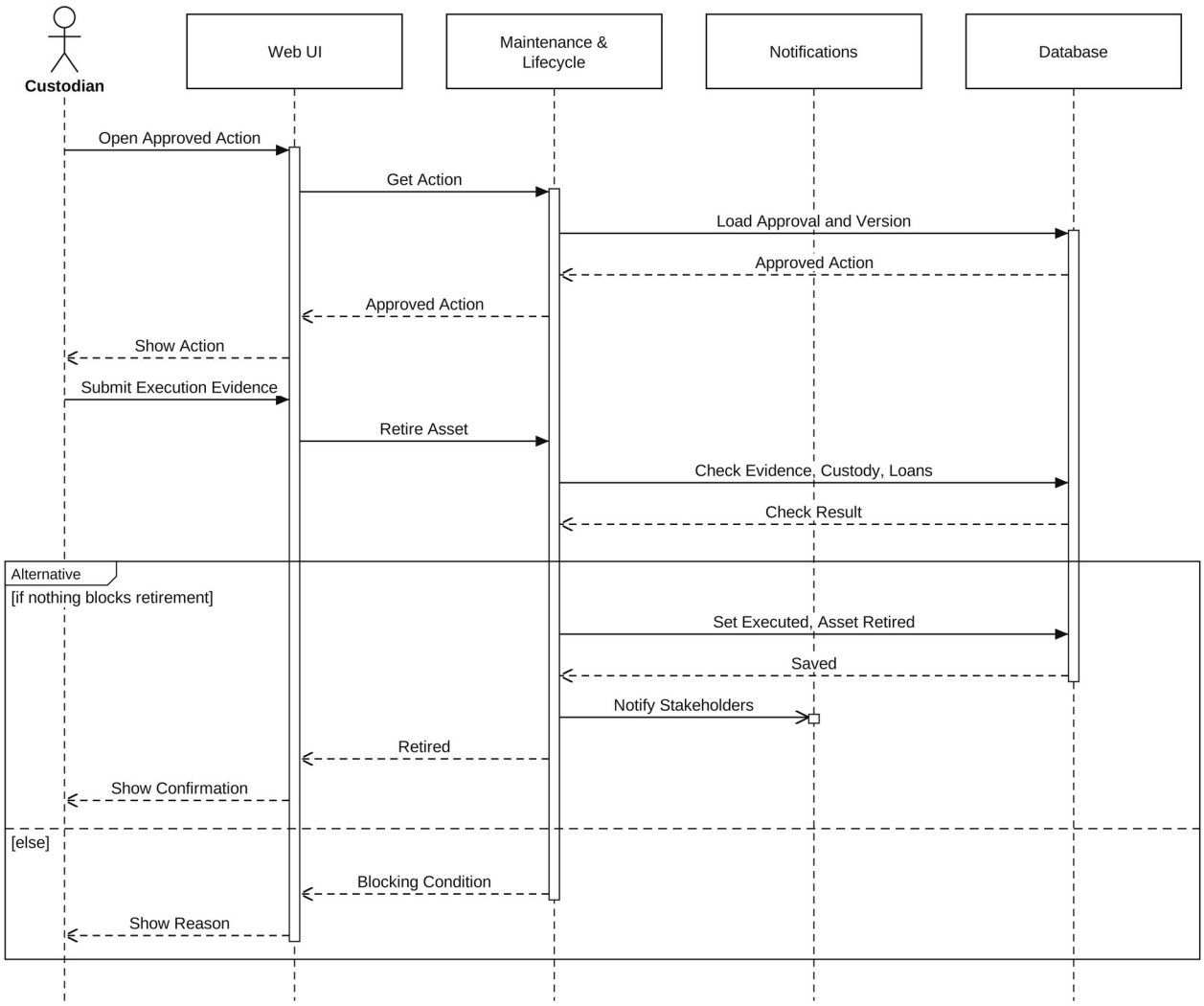


Figure 40. Sequence diagram: UC-25 Record asset retirement (later increment).

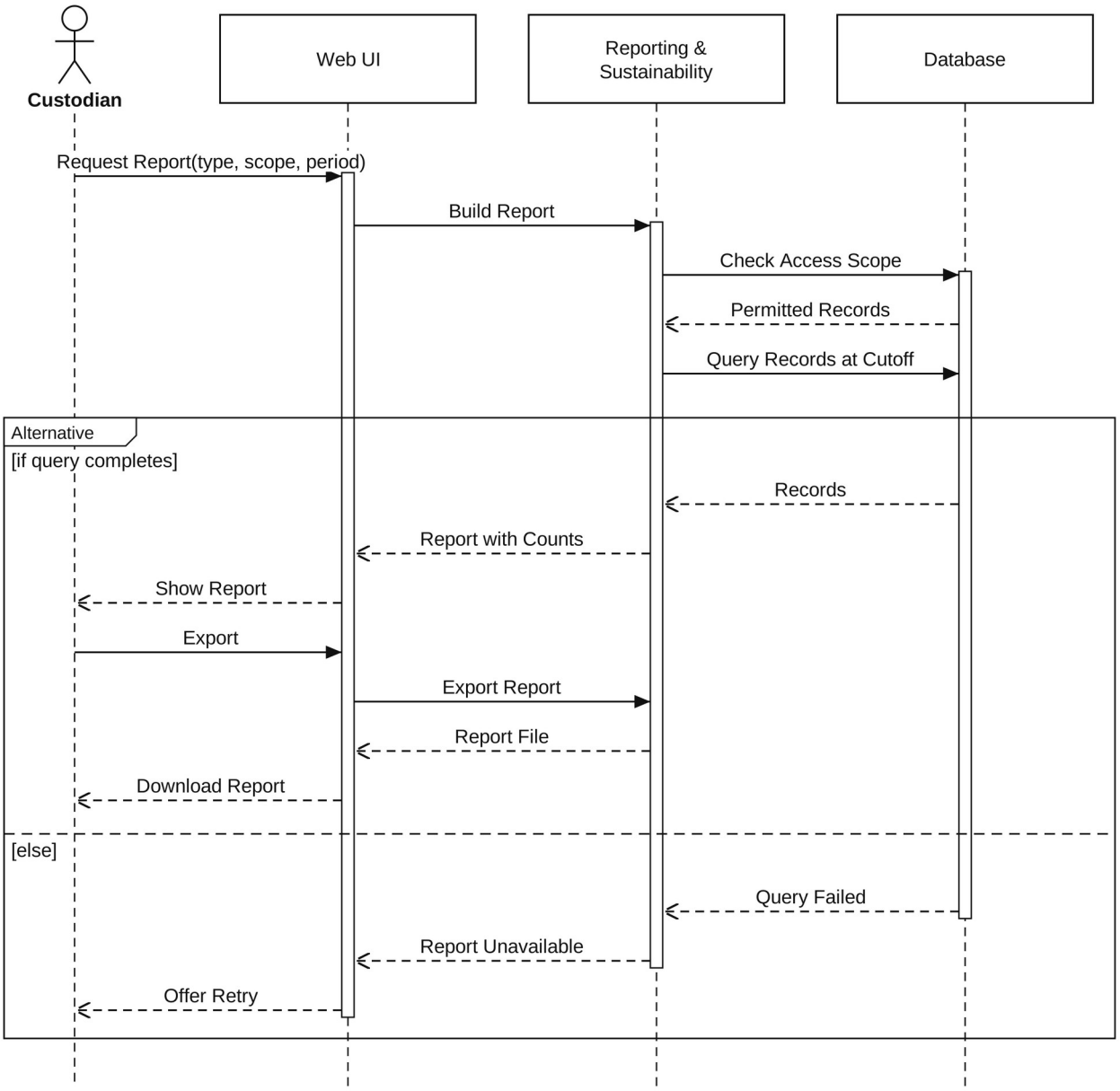


Figure 41. Sequence diagram: UC-26 Generate an operational report.

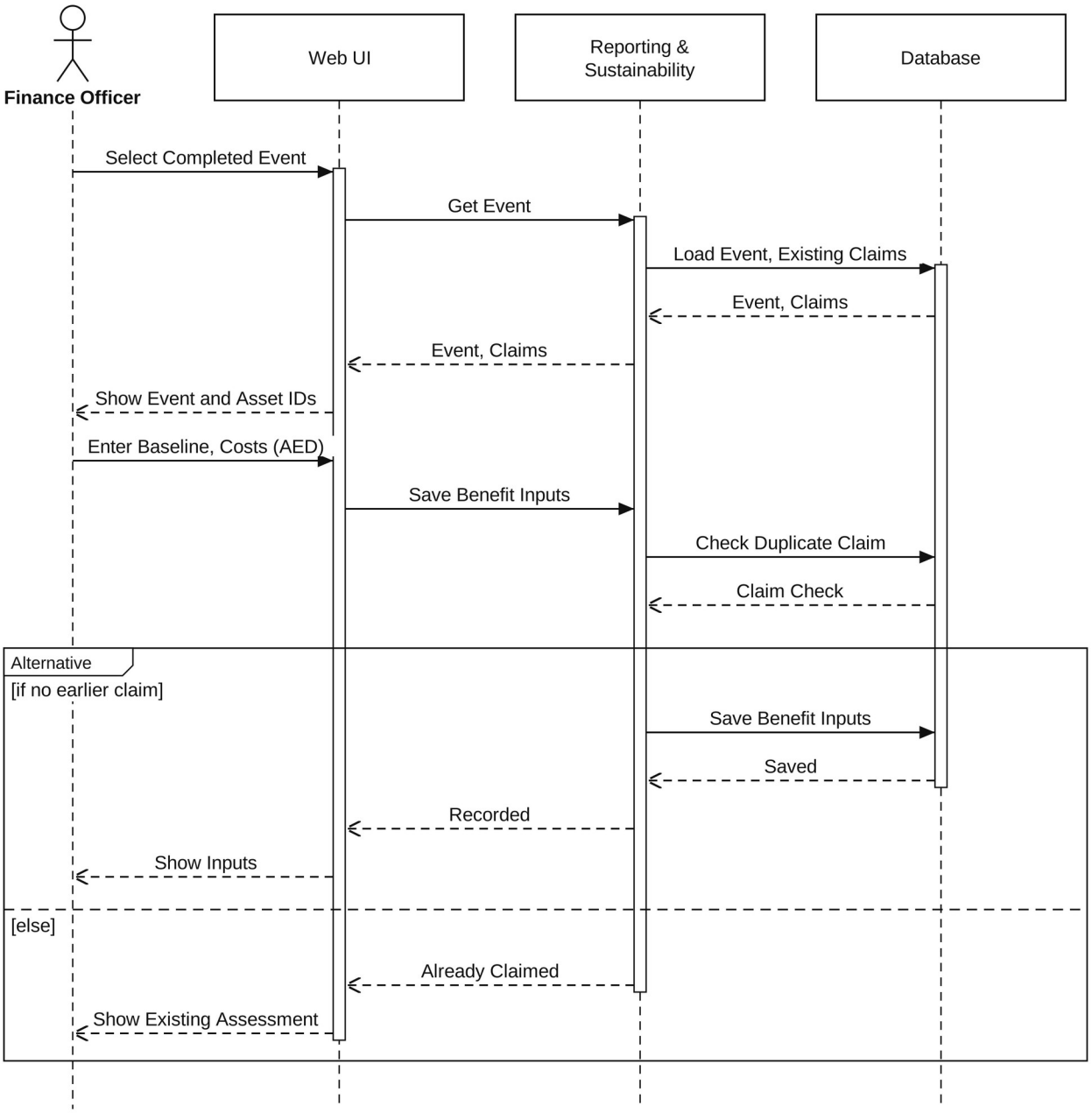


Figure 42. Sequence diagram: UC-27 Record benefit calculation inputs (later increment).

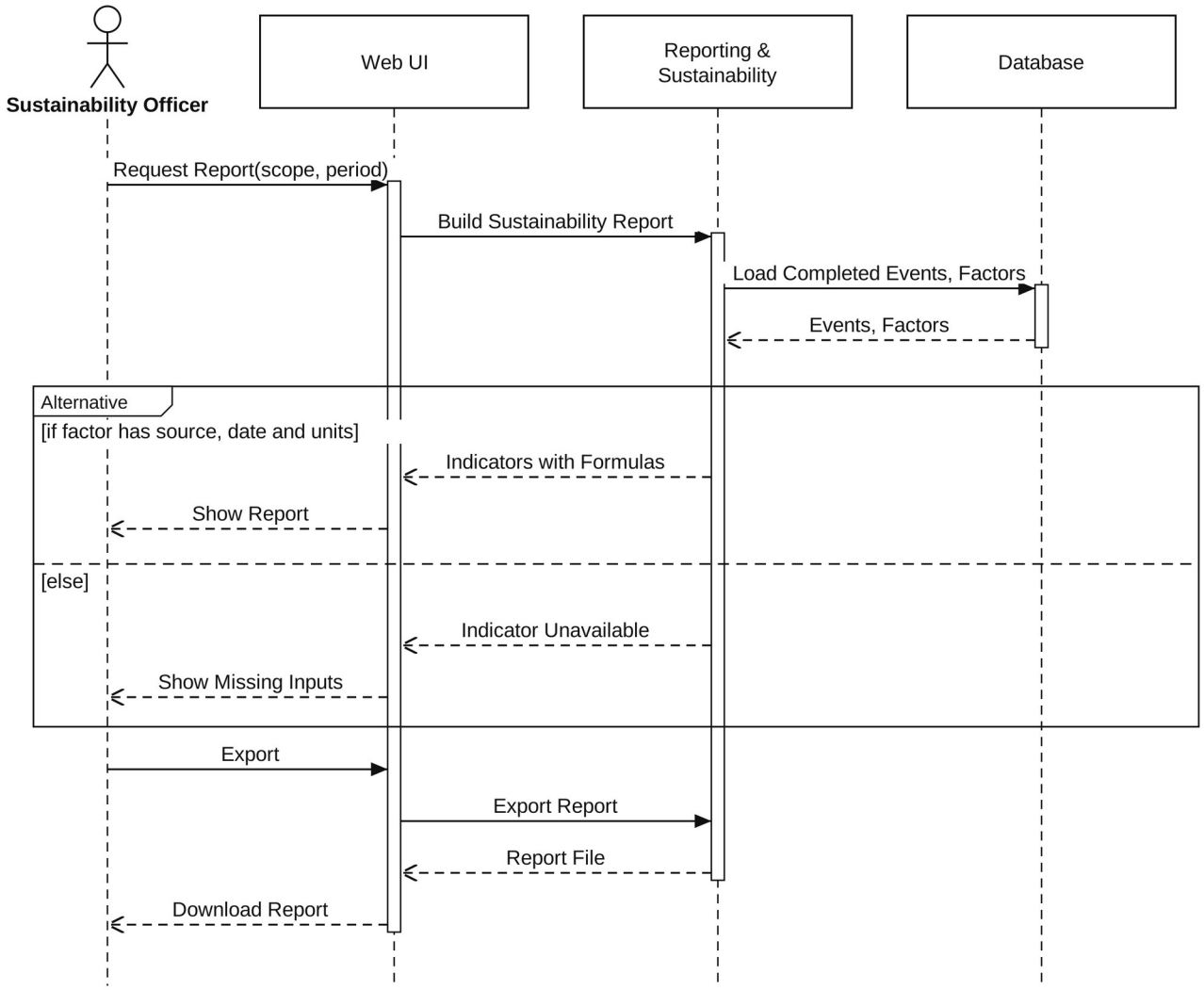


Figure 43. Sequence diagram: UC-28 Generate a sustainability report.

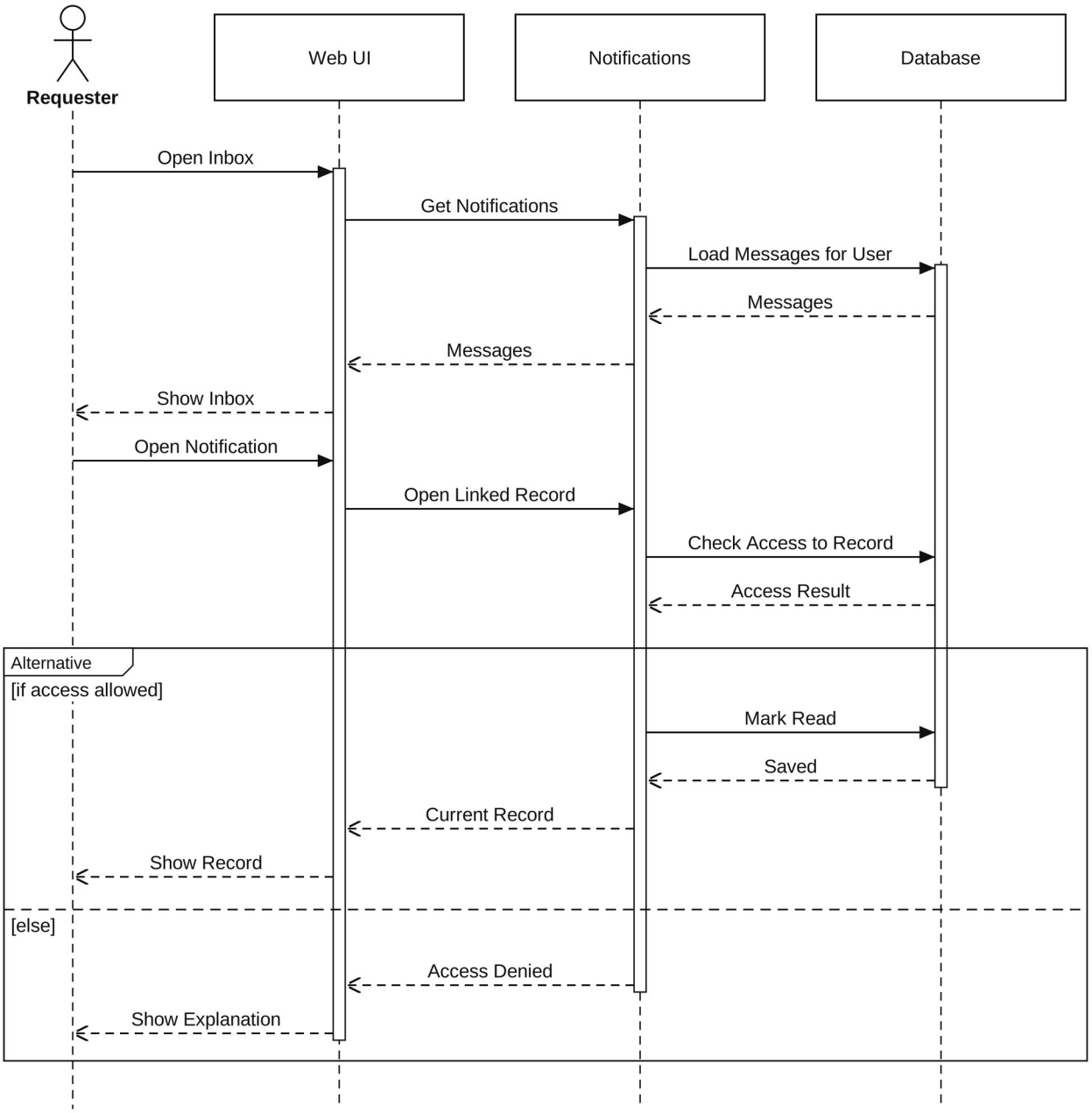


Figure 44. Sequence diagram: UC-29 Read status notifications.

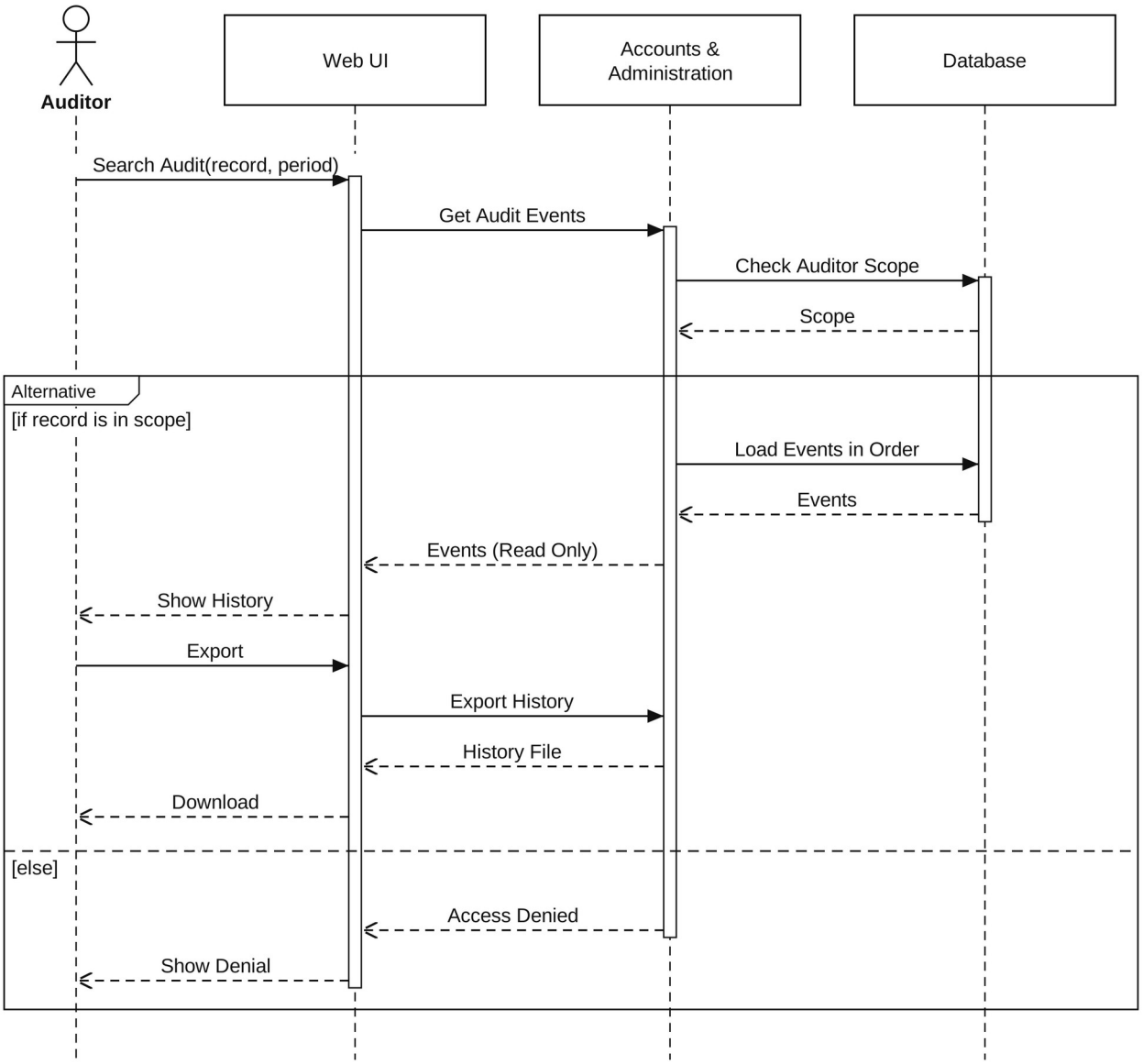


Figure 45. Sequence diagram: UC-30 Review the audit history (later increment).

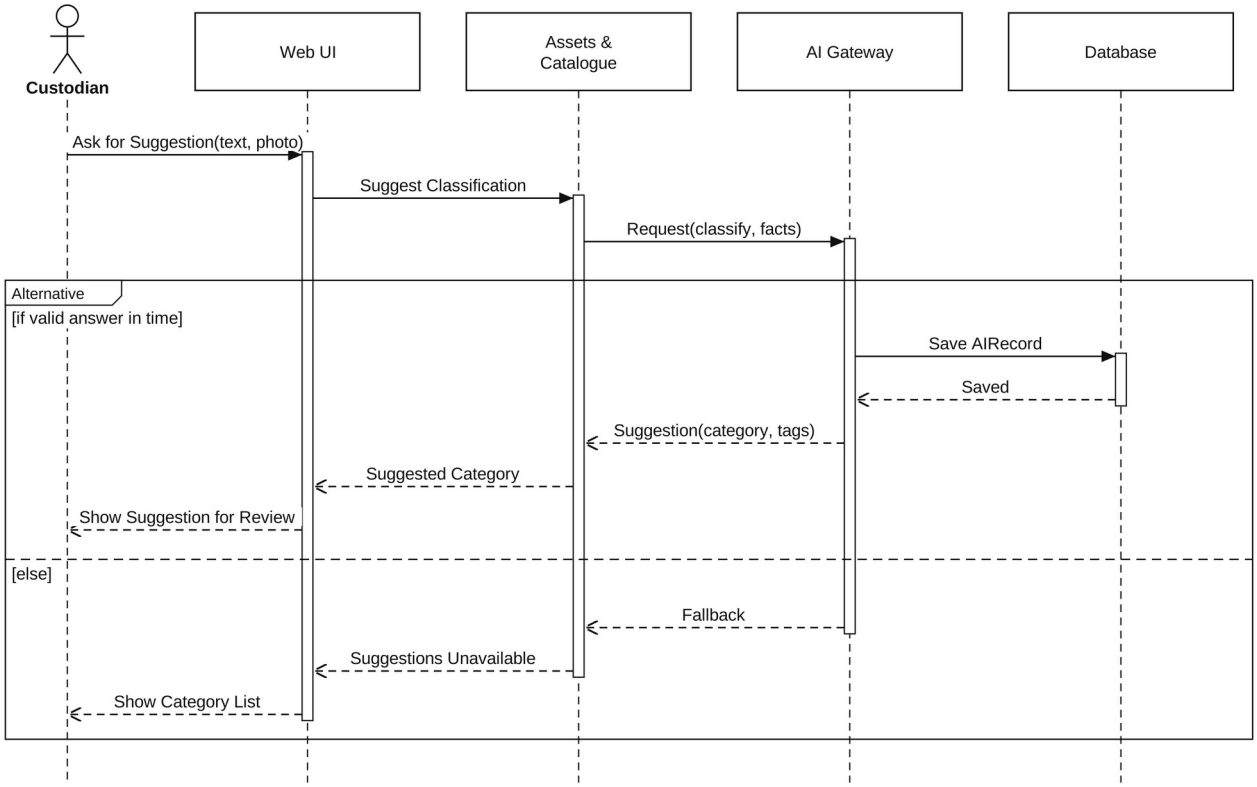


Figure 46. Sequence diagram: UC-31 Suggest asset classifications.

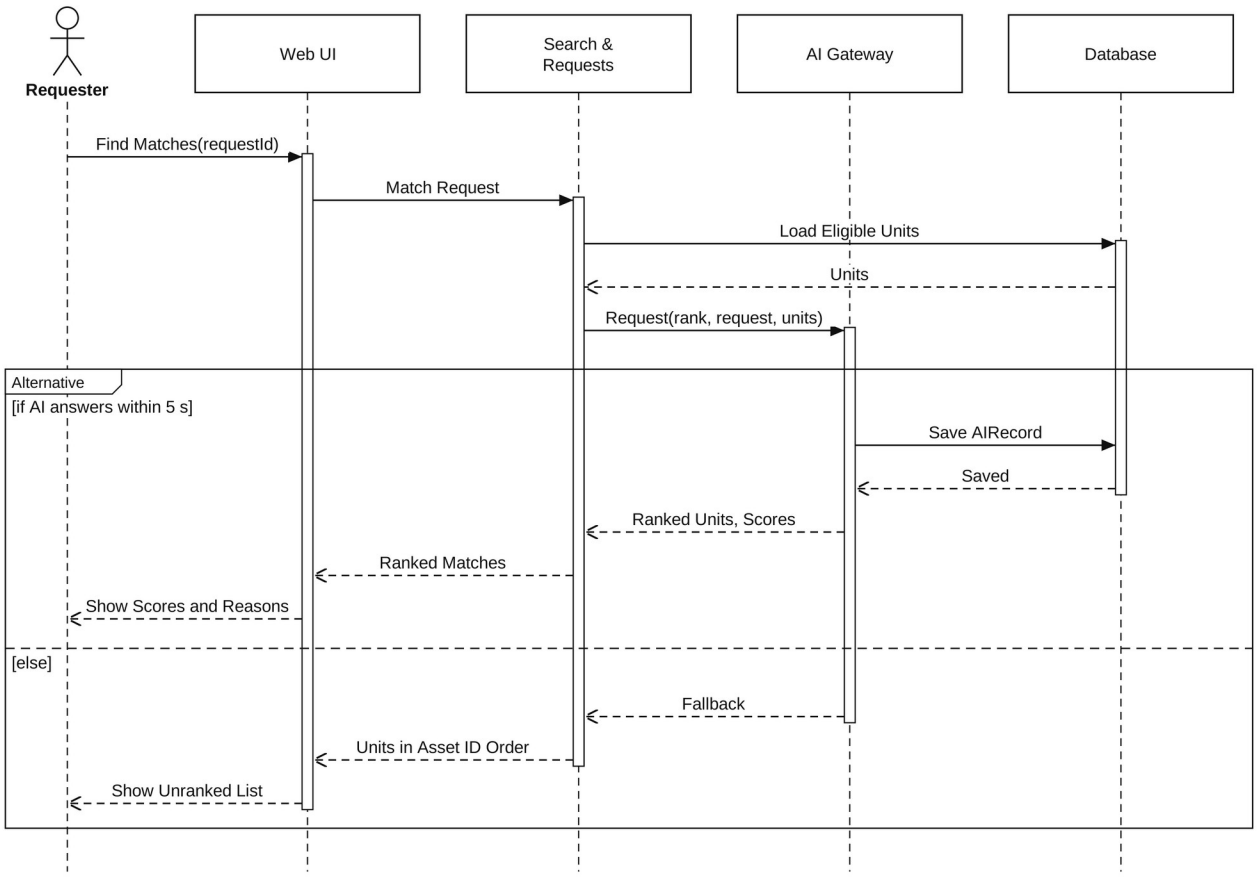


Figure 47. Sequence diagram: UC-32 Match resources semantically.

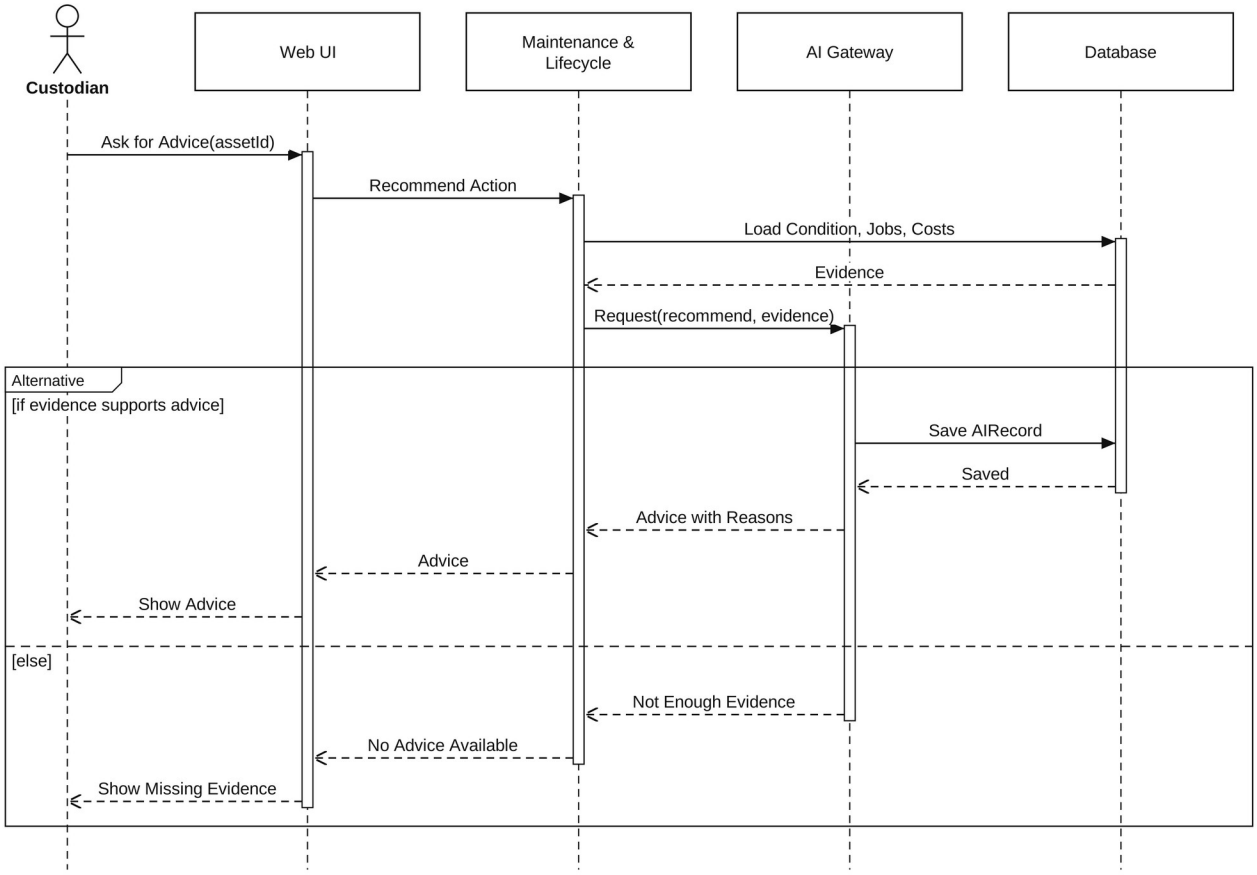


Figure 48. Sequence diagram: UC-33 Recommend a sustainable action (later increment).

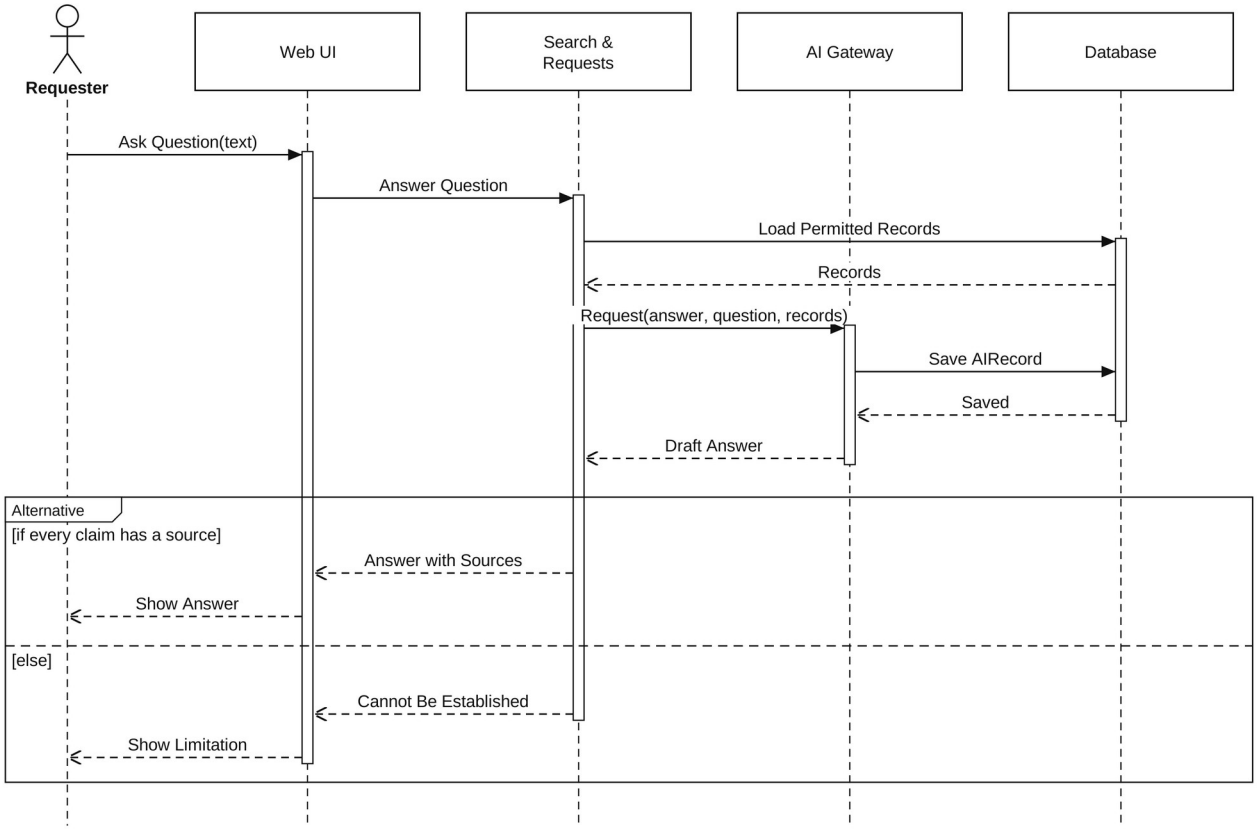


Figure 49. Sequence diagram: UC-34 Ask the resource assistant (later increment).

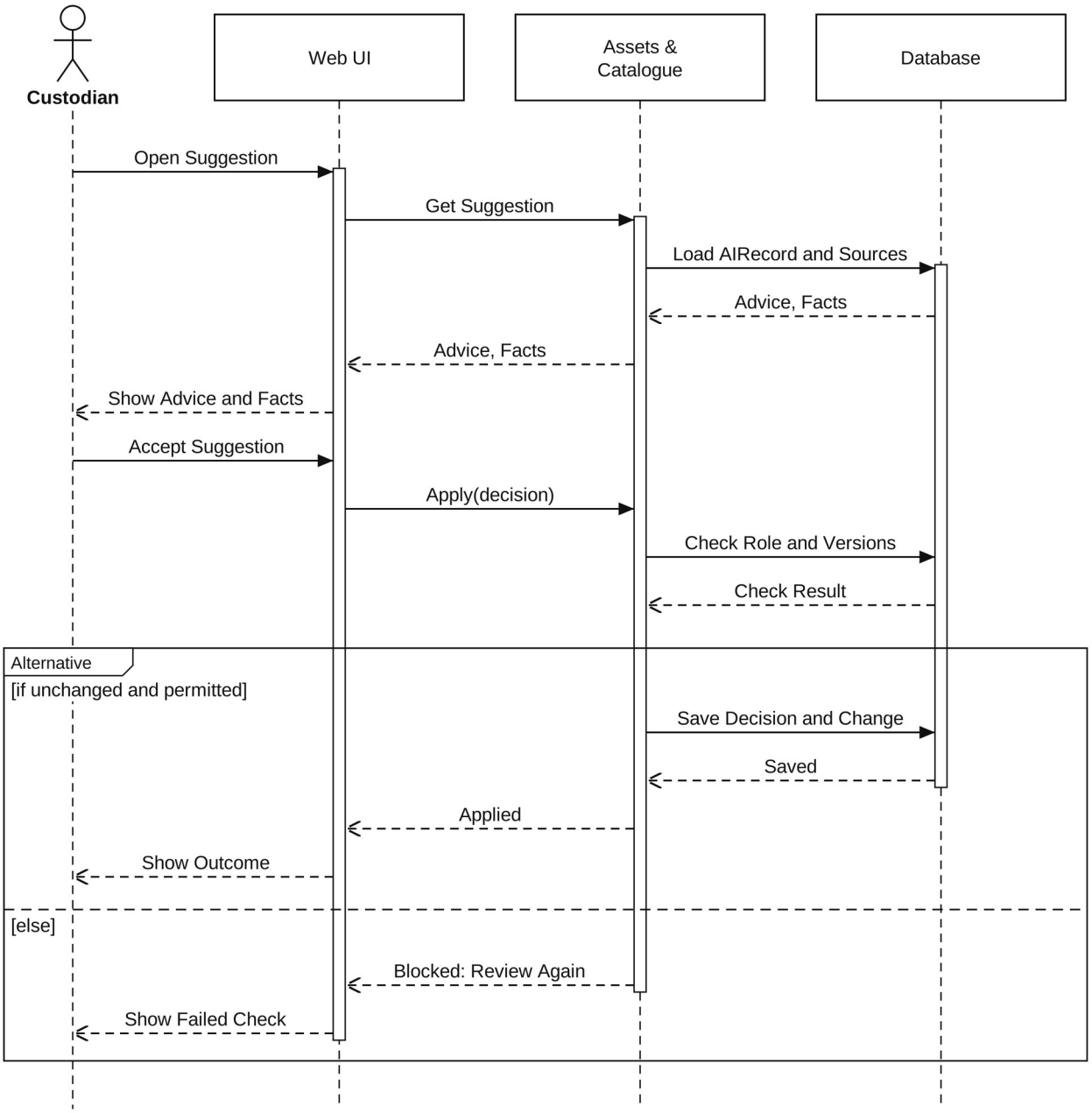


Figure 50. Sequence diagram: UC-35 Review an AI recommendation.

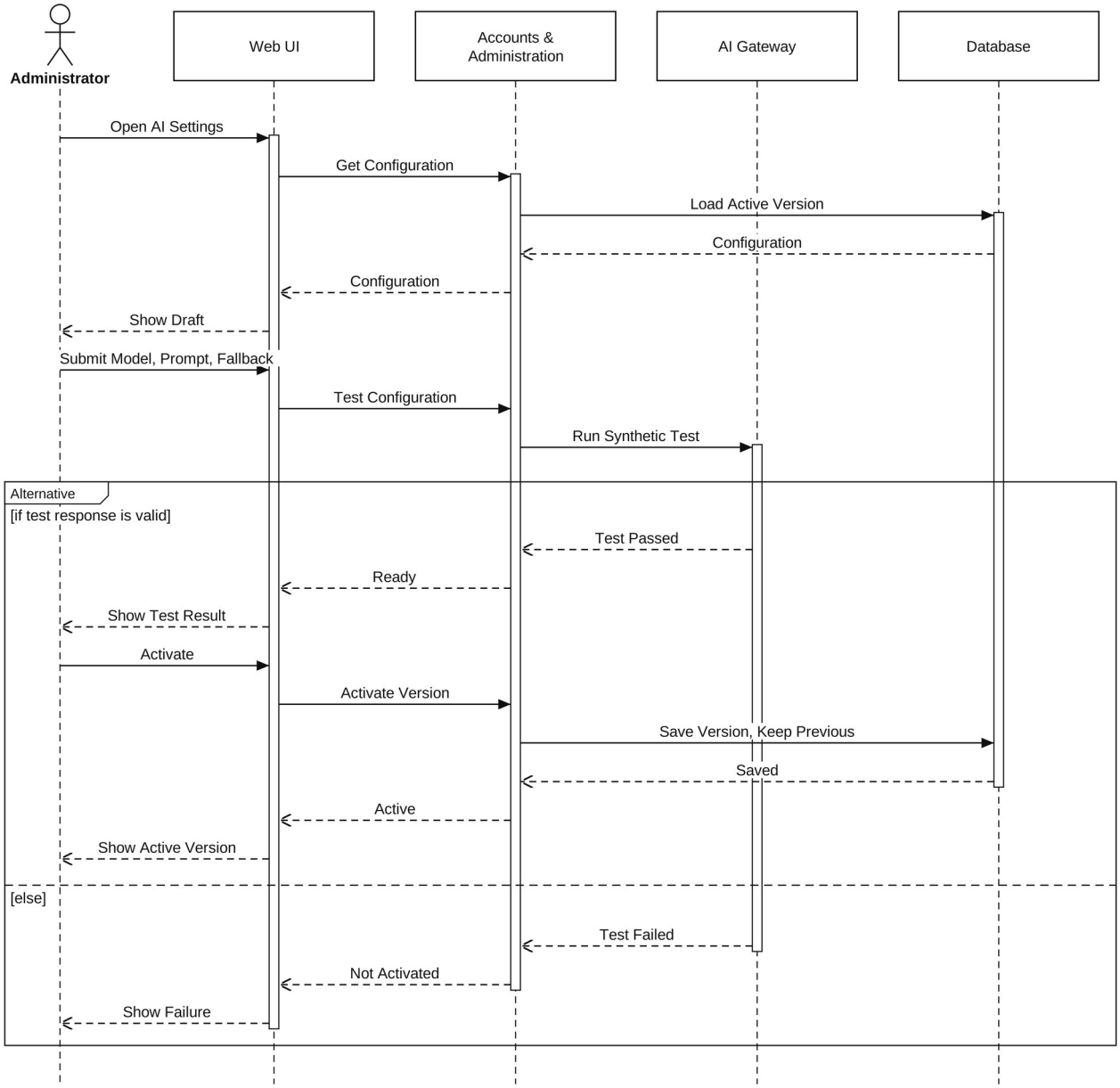


Figure 51. Sequence diagram: UC-36 Configure AI services.

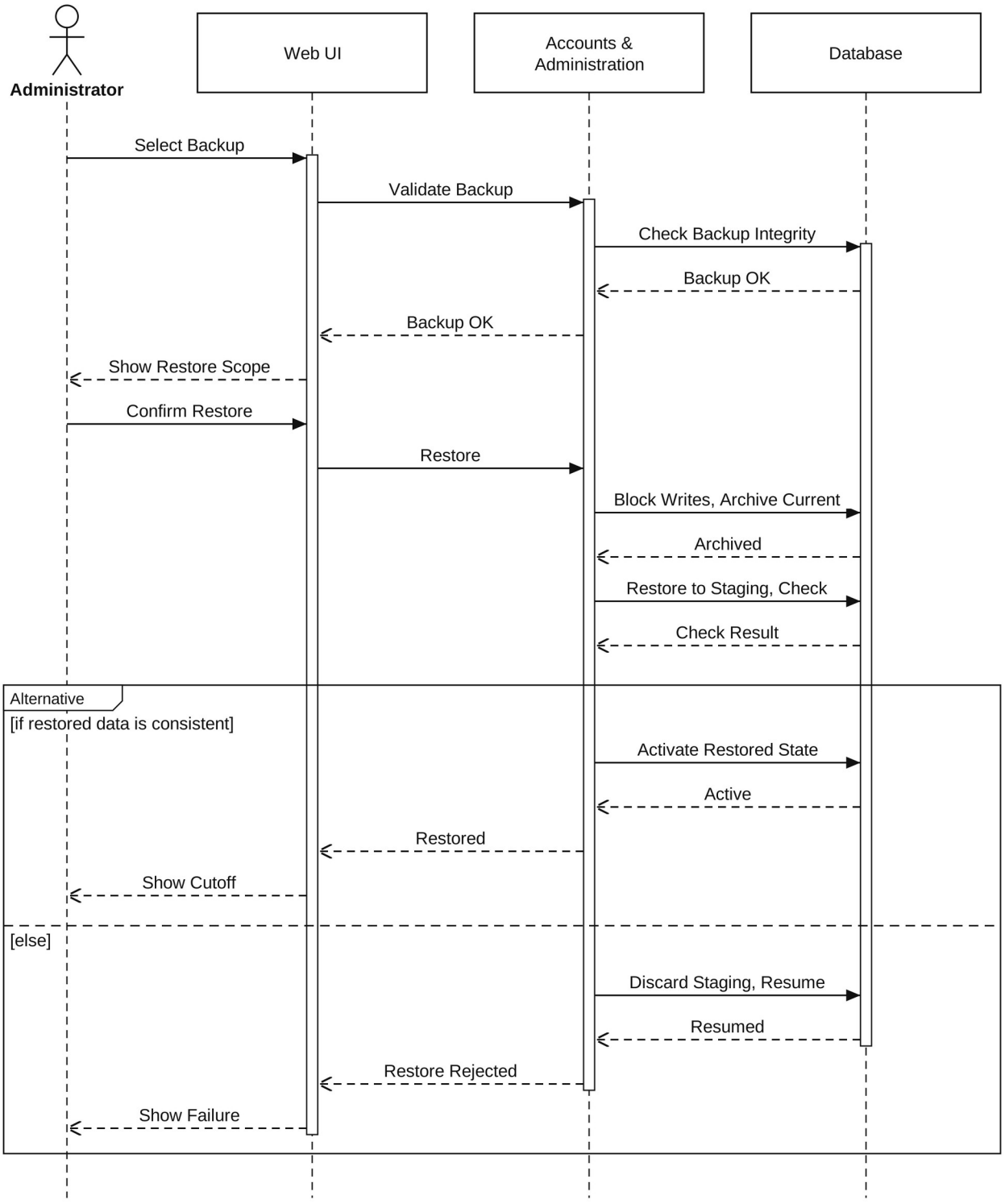


Figure 52. Sequence diagram: UC-37 Restore a prototype backup (later increment).

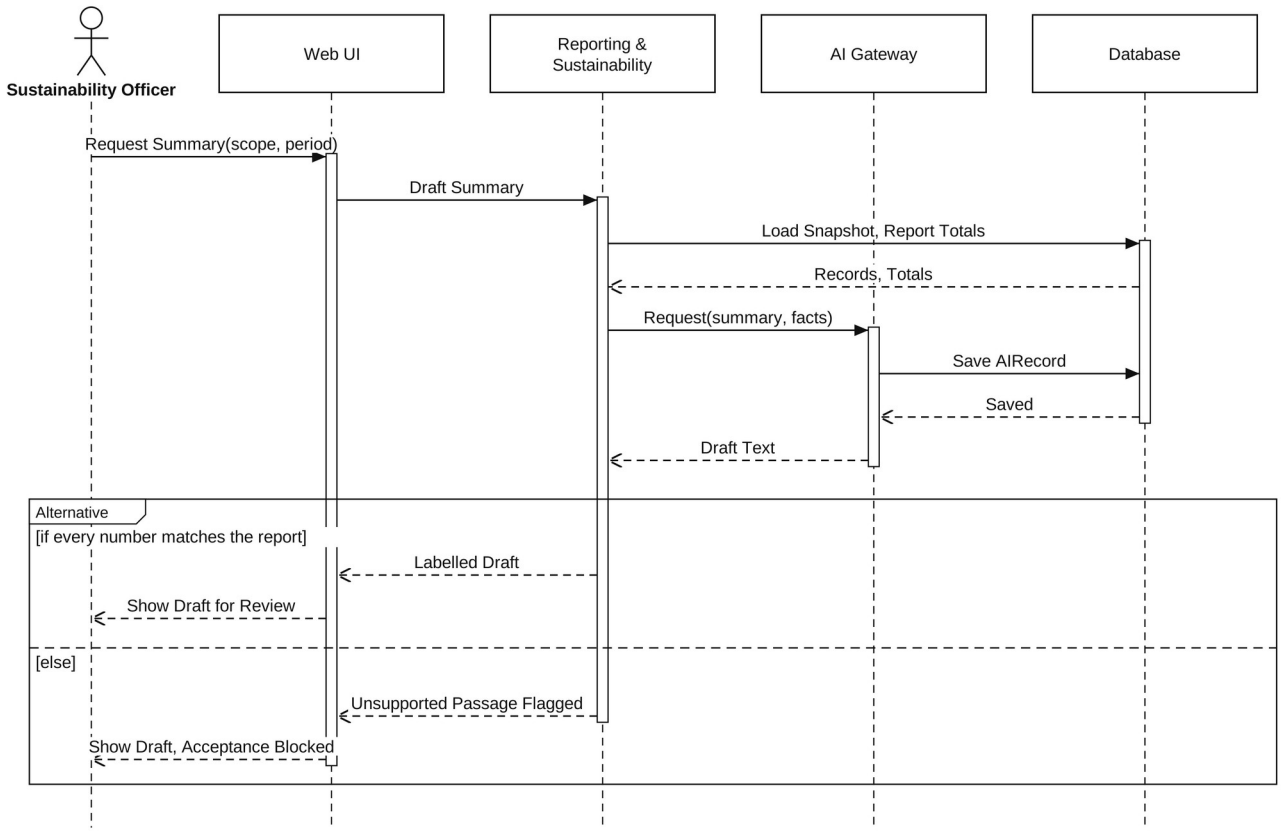


Figure 53. Sequence diagram: UC-38 Summarize recorded activity.

8.4 Activity diagrams

The two activity diagrams use one swimlane per actor so the reader sees who acts at each step. AI steps are drawn in the teal lane and never lead straight to a saved decision.

Figure	Flow	Decisions	Human gates
54	Request to chosen units	Fields valid? Ranked within 5 s? Matches suitable?	Gate 2
55	Reservation to receipt	All units available? Approve? All units match on receipt?	Gates 3, 4, 5 and 6

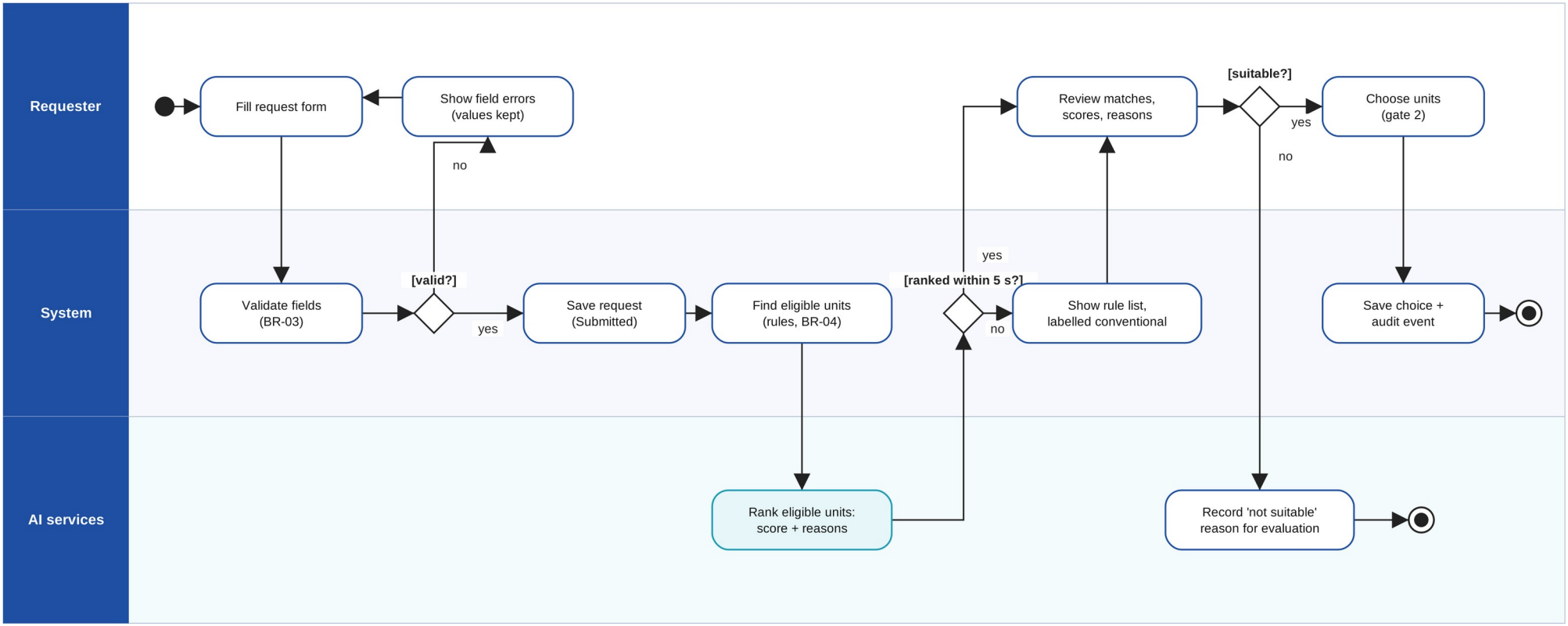


Figure 54. Activity diagram: request and matching.

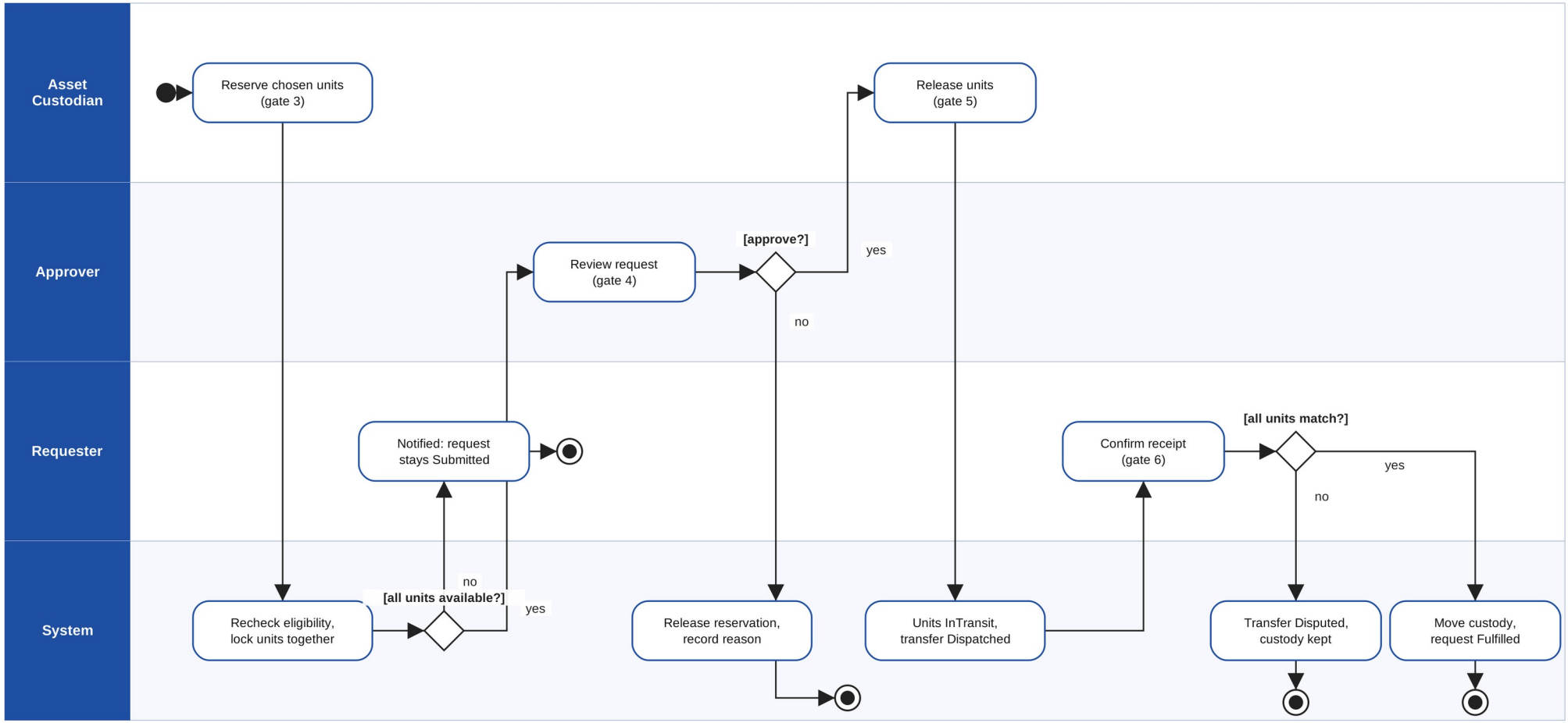


Figure 55. Activity diagram: reservation, approval, transfer and receipt.

8.5 State diagrams

State names are the Phase 3 status values, so the code, the database and the requirements use the same words. Each status is stored in one column and changed only by the service method named on the transition.

Figure	Record	States	Rule that the diagram enforces
56	Asset	Draft, Available, Reserved, InTransit, InUse, UnderMaintenance, Retired	Custody moves only when the receiver confirms all units (BR-07)
57(a)	Resource request	Draft, Submitted, UnderReview, Approved, Fulfilled, Rejected, Cancelled	A material change after approval creates a new version and needs a new approval (BR-06)
57(b)	Approval	Pending, Approved, Rejected, ReturnedForCorrection, Invalidated	The approver is never the requester (BR-02); the AI has no transition
58(a)	Transfer	Draft, Scheduled, Dispatched, Received, Disputed, Cancelled	A mismatch keeps the previous custody
58(b)	Reservation	Active, Consumed, Released	Units are locked together or not at all
58(c)	Lifecycle proposal (later increment)	Proposed, Approved, Rejected, Executed, Cancelled	Donation, recycling or disposal needs an independent approver and execution evidence (BR-10)

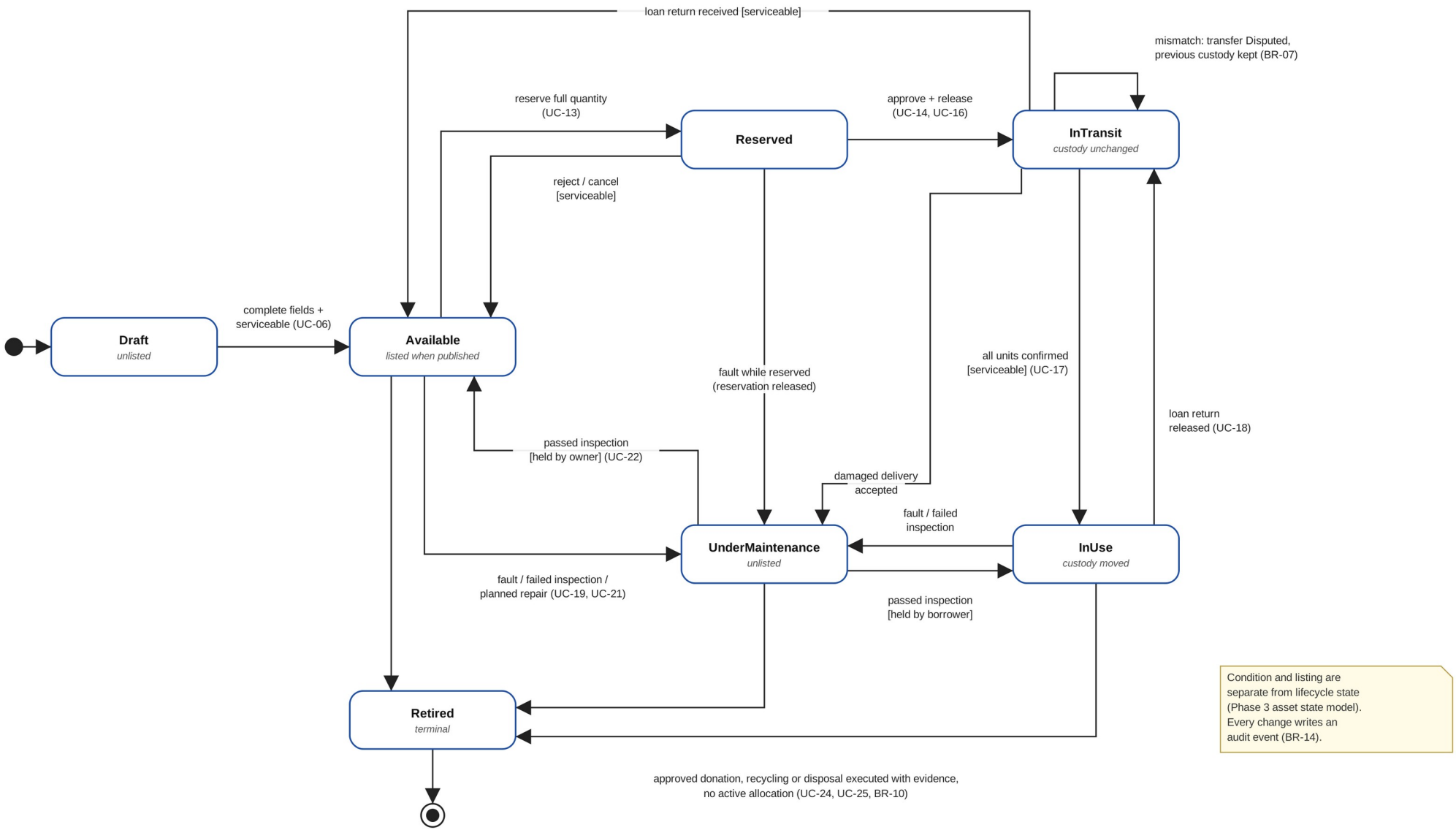


Figure 56. State diagram: asset.

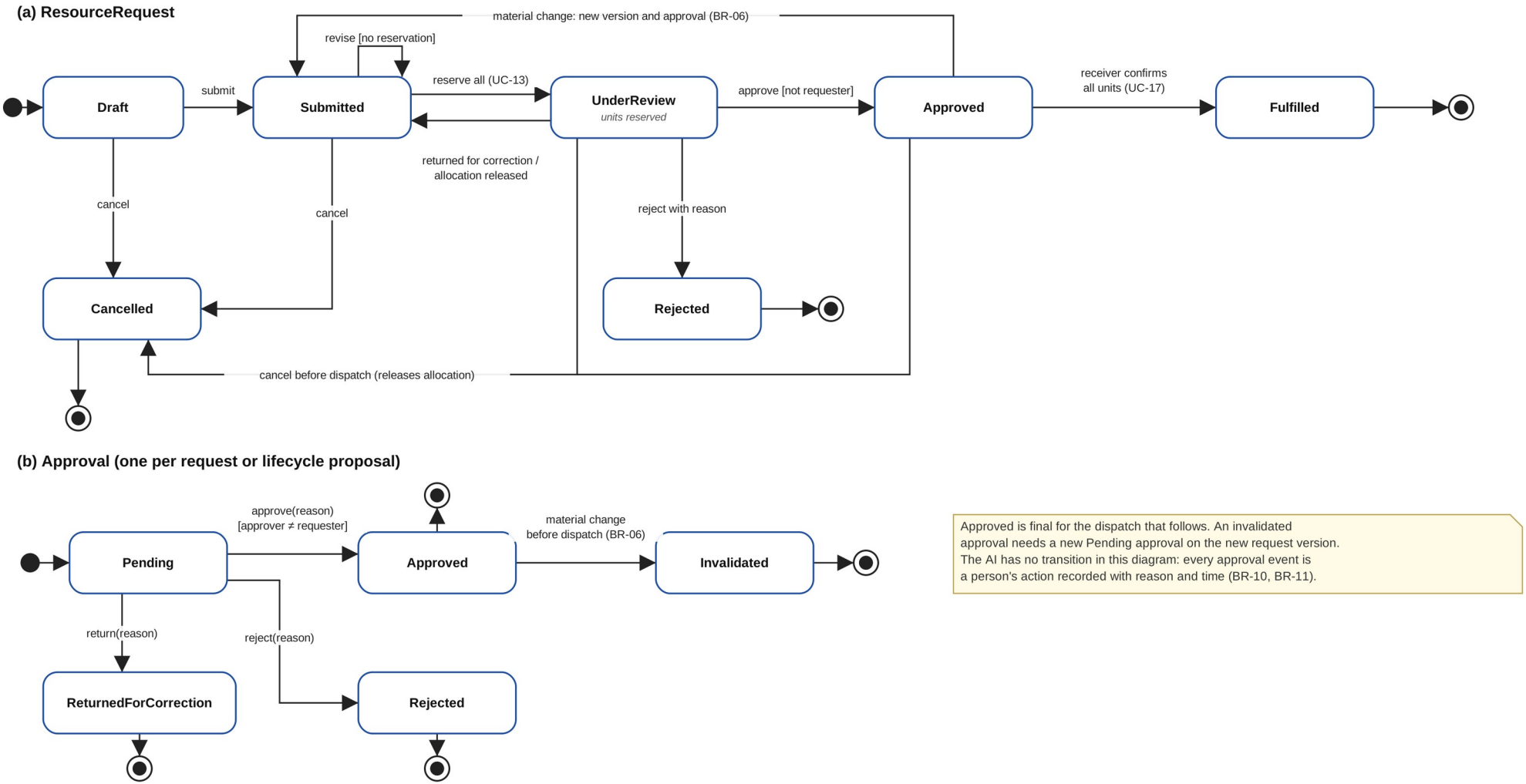


Figure 57. State diagrams: (a) resource request and (b) approval.

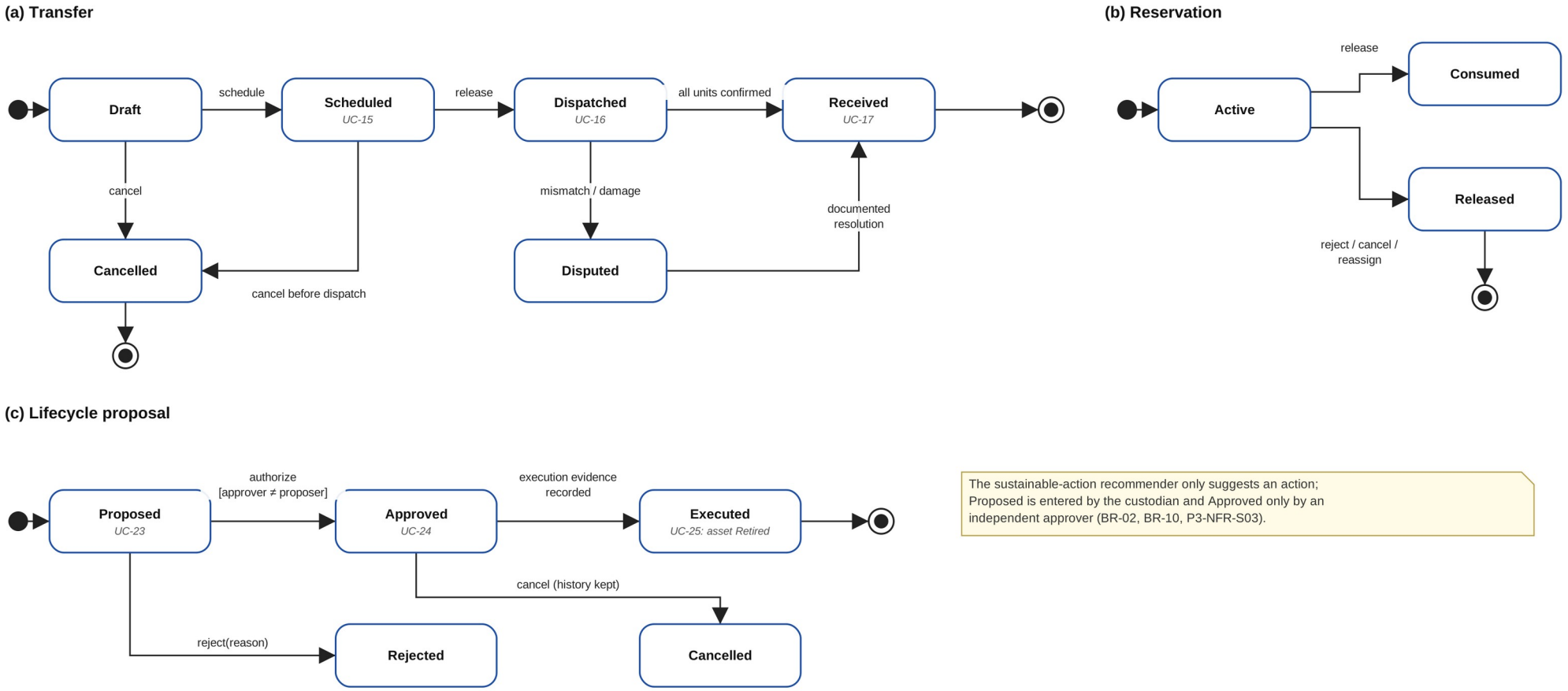


Figure 58. State diagrams: (a) transfer, (b) reservation and (c) lifecycle proposal (later increment).

8.6 Component diagram

Figure 59 shows how the architecture of Figure 4 will be built. Each provided interface is drawn as a lollipop and each dashed arrow is a «use» dependency.

Interface	Provided by	Used by	Main operations
IAIRequest	AI Gateway	All six application modules	request(task, facts) returns Suggestion
IAccess, IAudit	Security & Audit	All modules and the AI Gateway	can(user, action, record); log(event); logAIRecord(record)
INotify	Notifications	All modules	notify(user, message, linkedRecord)
IRepository	Database (Django ORM models)	All modules	Read and write D1 to D7 inside transactions
AIService	The six AI services	AI Gateway	run(facts) returns Suggestion
TextGenerator, EmbeddingModel	Adapters	AI services	generate(task, facts, schema, timeLimit); embed(text)

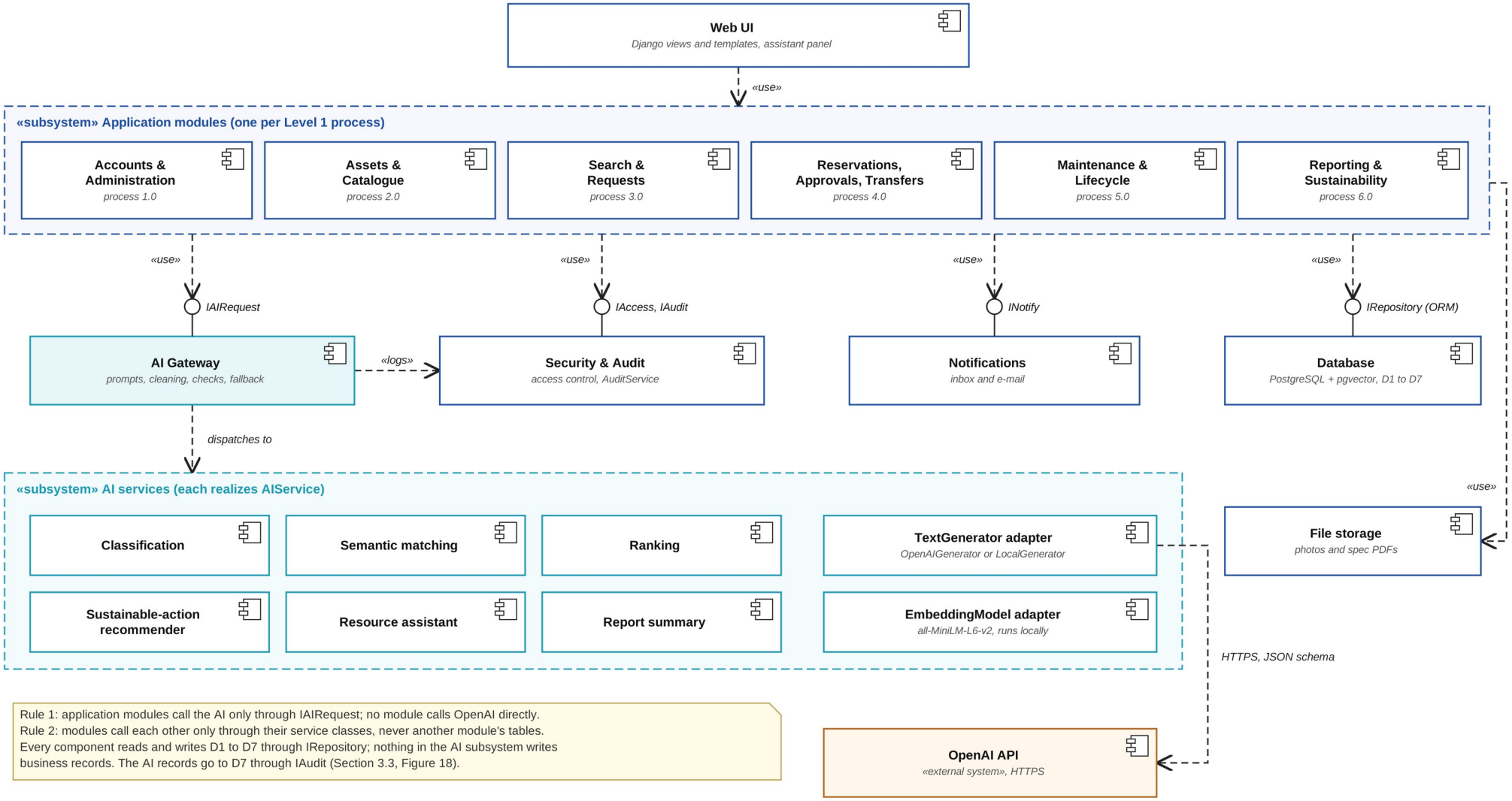


Figure 59. Component diagram.

8.7 Deployment diagram

The prototype runs on one demo PC, Saeed's laptop with an NVIDIA RTX 3050 Ti (4 GB), as decided in AD-1 and AD-2. The browser reaches Django over loopback only. Only the AI gateway's TextGenerator adapter connects to the internet.

Node	Runs	Connects to	Protocol
Demo PC: Python environment	Django app, background worker, prompts, MiniLM model files, .env	PostgreSQL, file system, OpenAI API	TCP 5432, local file I/O, HTTPS
Demo PC: PostgreSQL 16	circular_campus database with pgvector (D1 to D7)	Python environment	TCP 5432 (local)
Demo PC: background schedule	Jobs table poll every 30 s; nightly backup (later increment)	PostgreSQL, backup drive	local, USB
OpenAI API (cloud)	The model named in AI settings	AI gateway only	HTTPS through the official openai library
Backup drive (later increment)	Encrypted; 7 daily and 4 weekly copies	Demo PC	USB
GitHub repository	Source code and prompt templates	Demo PC	git pull to deploy

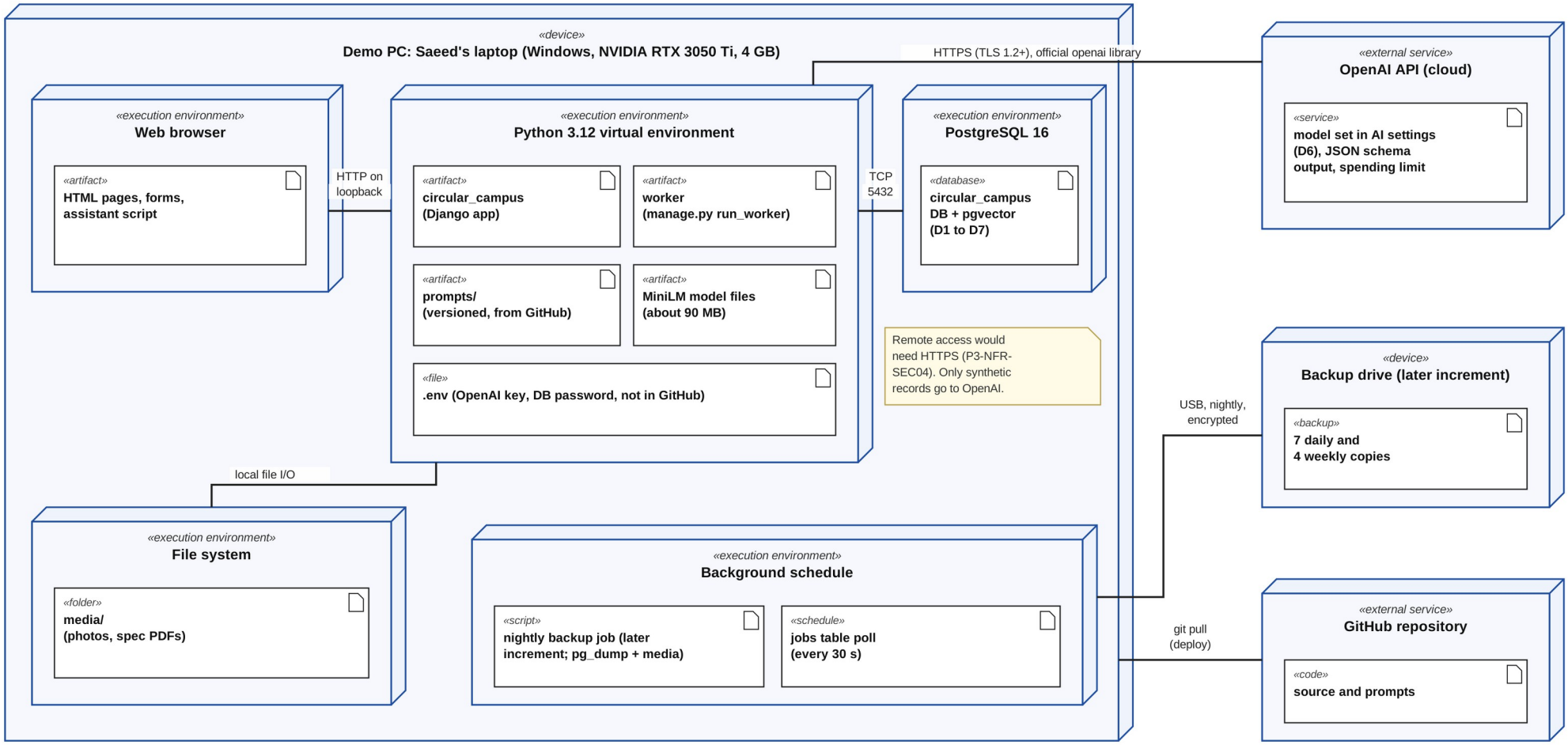


Figure 60. Deployment diagram.

8.8 AI recommendation workflow

Figure 61 shows the full life of one recommendation, from the saved request to the monthly review. Section 11 gives the score design and Section 13 gives the monitoring records.

1. **Filter.** Rules remove every unit that fails a mandatory condition: availability, quantity, safety, required specification and date. Similarity can never put an ineligible unit back (BR-04).
2. **Embed.** The request text becomes a vector of 384 values with the local MiniLM model. Asset vectors were computed when each asset was saved.
3. **Compare.** pgvector returns the cosine similarity between the request and each eligible unit.
4. **Score.** Five factor values between 0 and 1 are computed. A factor missing for any unit is dropped for all and the weights are rescaled. Ties are broken by Asset ID.
5. **Explain.** Each result gets its factor bars, the two strongest reasons, any missing inputs and the version numbers.
6. **Check and record.** The gateway checks the output and writes one AI record to D7.
7. **Decide.** The requester chooses units or rejects the list with a reason (gate 2). A top score below 70 is labelled "Weak match" and no unit is marked as the top pick.
8. **Monitor.** The choice is added to the AI record. Each month the Administrator reviews the metrics of Section 13 and may set new weights, which creates a new ranking version.

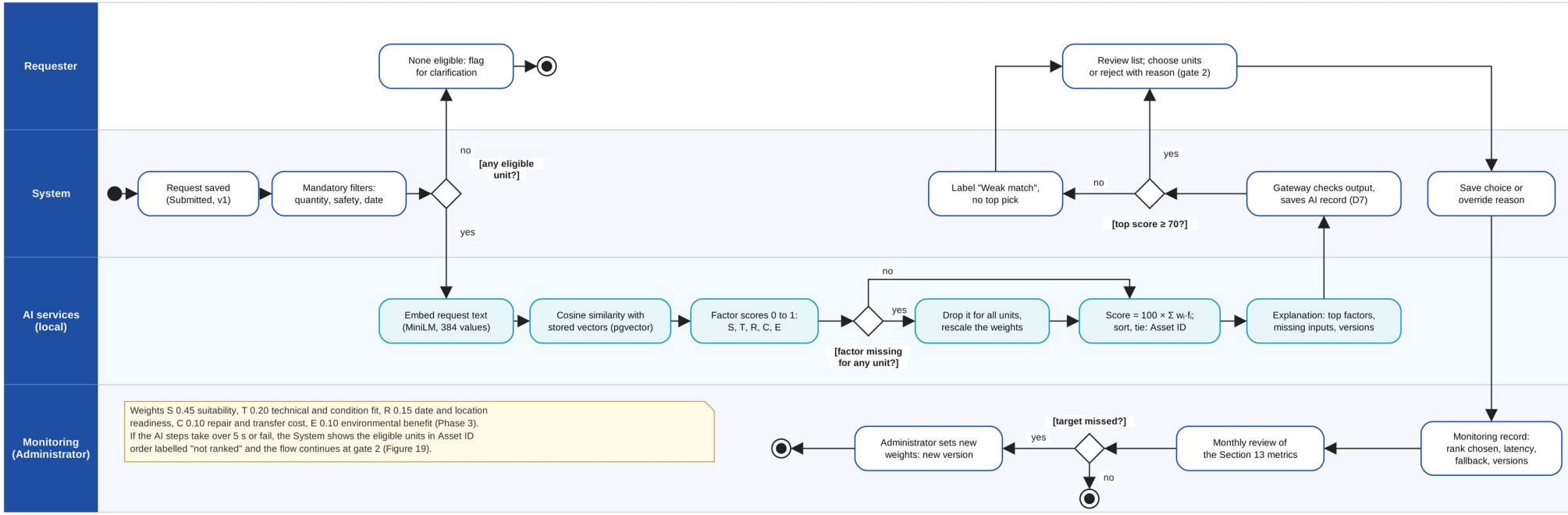


Figure 61. AI recommendation workflow.

9 Database Design

The database has 26 tables in PostgreSQL. Figure 62 is the entity-relationship diagram of the whole system. Figure 63 shows the same entities as tables with their primary keys (PK) and foreign keys (FK); every line in Figure 63 is one relationship of Figure 62. The class diagram (Figure 15) maps to these tables one to one: each Asset subclass is its own table that shares the assetId key, RequestSpecification is deleted with its request, and AssetCollectionItem links collections and assets.

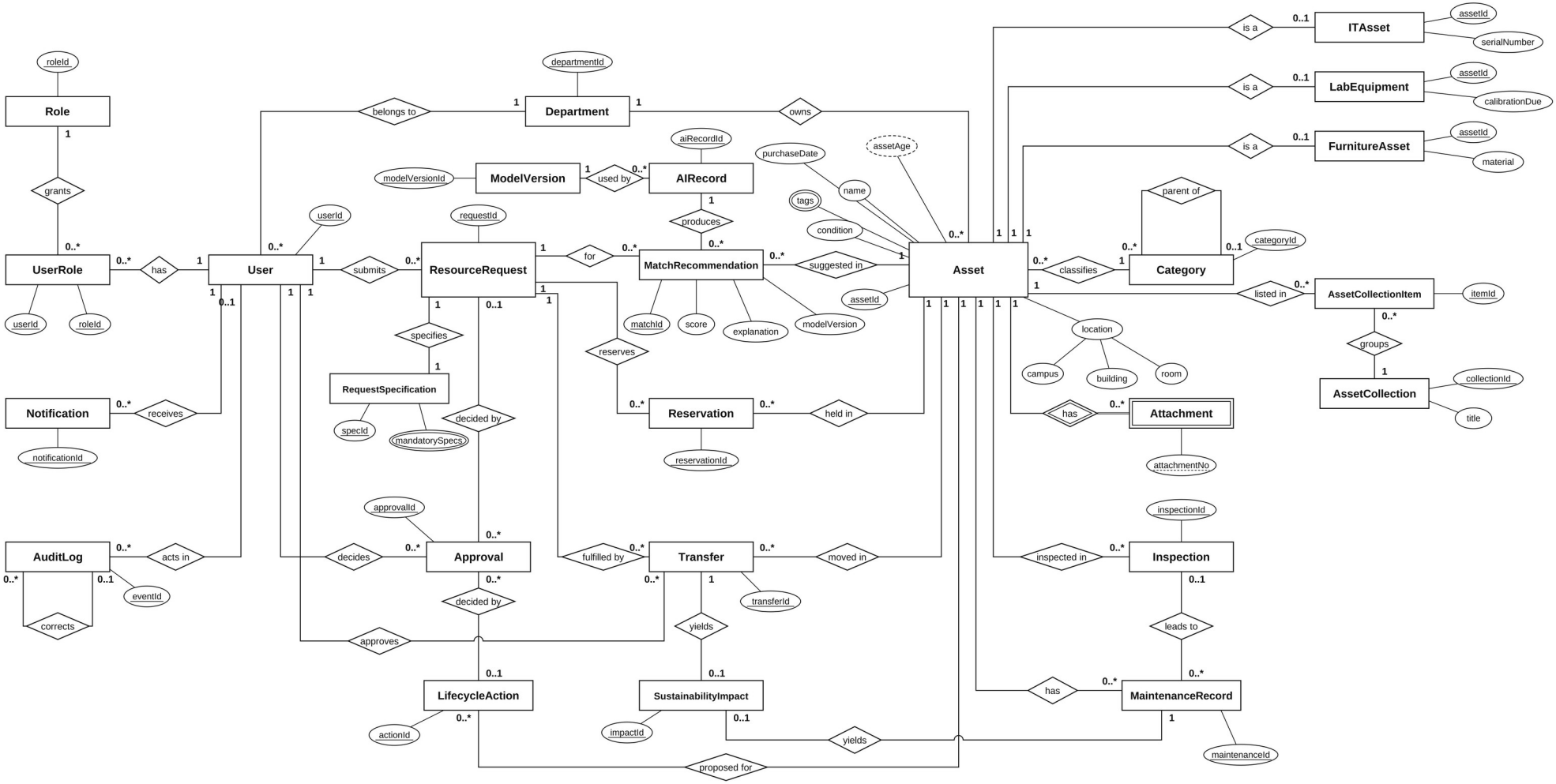


Figure 62. Entity-relationship diagram of the whole system.

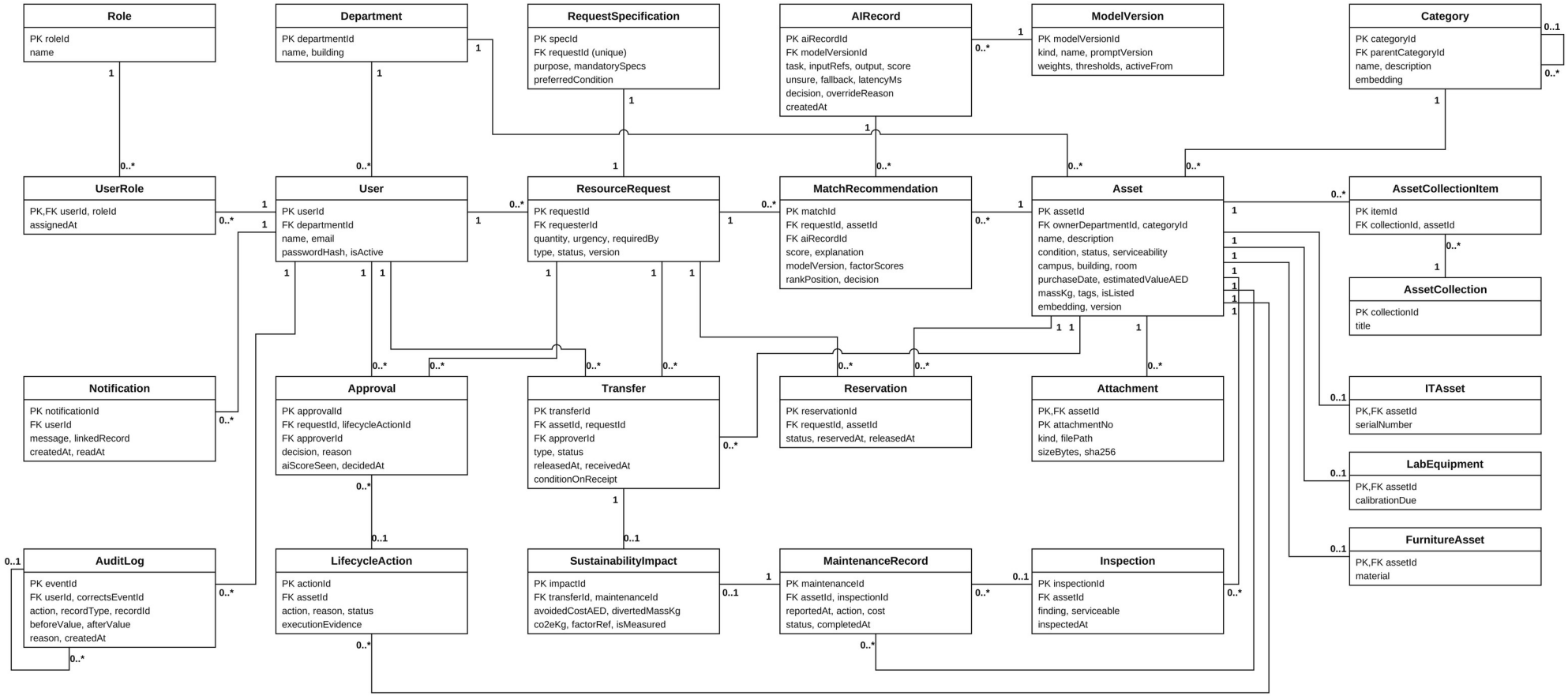


Figure 63. Entity-relationship diagram of the whole system: tables with primary and foreign keys.

9.1 Coverage of the brief

Brief entity	Table(s)	Store	Brief entity	Table(s)	Store
Users	User	D1	Transfers	Transfer	D4
Roles	Role, UserRole	D1	Inspections	Inspection	D5
Departments	Department	D1	Maintenance records	MaintenanceRecord	D5
Assets	Asset, ITAsset, LabEquipment, FurnitureAsset, Attachment	D2	Sustainability impacts	SustainabilityImpact	D5
Asset categories	Category	D6	Notifications	Notification	D1
Resource requests	ResourceRequest, RequestSpecification	D3	AI model versions	ModelVersion	D6
AI recommendations	MatchRecommendation	D3	Audit logs	AuditLog, AIRecord	D7
Reservations	Reservation	D4	(added) Lifecycle actions	LifecycleAction	D5
Approvals	Approval	D4	(added) Asset collections	AssetCollection, AssetCollectionItem	D2

9.2 Data dictionary

Primary keys are integers created by the database; status columns hold the state names of Section 8.5. Every table with a version column uses it for optimistic locking, so a stale screen cannot overwrite a newer record.

Table	Primary key	Purpose	Rules kept by the database
User	userId	A person who signs in	email unique; passwordHash only (Django PBKDF2); isActive false instead of delete
Role / UserRole	roleId / (userId, roleId)	The eight Phase 3 roles and who holds them	A user may hold several roles; separation rules are checked in the service layer (BR-02)
Department	departmentId	Owning and receiving departments	name unique
Asset	assetId	One physical item	status in the Figure 56 states; condition is New, Good, Fair or NeedsRepair; massKg and estimatedValueAED may be empty but never 0 by default; /assetAge is computed from purchaseDate, not stored
ITAsset	assetId (also FK)	IT equipment details	serialNumber unique; one row at most per asset; deleted with its asset
LabEquipment	assetId (also FK)	Laboratory equipment details	calibrationDue may be empty; one row at most per asset; deleted with its asset
FurnitureAsset	assetId (also FK)	Furniture details	material required; one row at most per asset; deleted with its asset
AssetCollection	collectionId	A named set of assets, for example for an event	title required; deleting a collection keeps its assets
AssetCollectionItem	itemId	One asset in one collection	(collectionId, assetId) unique; deleted with its collection
Attachment (weak)	(assetId, attachmentNo)	Photo or specification PDF	At most one photo and one PDF per asset; sha256 detects a changed file; deleted with its asset
Category	categoryId	Category tree used by classification	parentCategoryId may be empty for a top category; no cycles
ResourceRequest	requestId	What a department needs	quantity ≥ 1; requiredBy not in the past at submission; a new version on material change
RequestSpecification	specId	Purpose, mandatory specifications and preferred condition of one request	requestId unique (one per request); deleted with its request; mandatorySpecs is a list of text; preferredCondition is a Condition value
MatchRecommendation	matchId	One ranked candidate for one request	score 0 to 100; factorScores JSON with the five factors; decision empty until the requester acts
ModelVersion	modelVersionId	Model, prompt, ranking weights and thresholds in use	Rows are never edited; a change adds a row with a new activeFrom (P3-NFR-Q11)
Reservation	reservationId	One unit held for one request	Only one Active reservation per asset (partial unique index)
Approval	approvalId	A person's decision on a request or lifecycle proposal	Exactly one of requestId or lifecycleActionId (check constraint); approverId differs from the requester or proposer
Transfer	transferId	Movement of one tracked asset for one request	assetId, requestId and approverId are required; the receiving department is the request's department
Inspection	inspectionId	Recorded condition check	serviceable is true or false, never empty
MaintenanceRecord	maintenanceId	Fault, repair and its cost	cost (AED) ≥ 0; completedAt set only in Completed
LifecycleAction	actionId	Proposed donation, recycling or disposal	executionEvidence required before Executed
SustainabilityImpact	impactId	Benefit of one completed reuse or repair	Exactly one of transferId or maintenanceId; avoidedCostAED may be negative; co2eKg empty when no WARM factor fits
Notification	notificationId	Inbox message	readAt empty until read
AuditLog	eventId	Who changed what and why	Insert only (BR-14): Prototype 1 has no update or delete operation; the database-level insert-only right is added in a later increment
AIRecord	aiRecordId	One AI call and the person's answer to it	Insert only, except decision and overrideReason, which are filled once (Section 13)

10 Interface Contracts

A contract fixes what a caller sends, what it gets back, how long it waits and what happens on failure. Modules call only AIGateway.request(task, facts); the gateway calls the services and the adapters. Every answer has the same envelope, so the screens handle all AI results the same way.

10.1 The Suggestion envelope

Field	Type	Meaning
task	text	classify_asset, rank_matches, recommend_action, answer_question or draft_summary
value	JSON	The task result; its schema is given in 10.2
score	number or empty	The task's own score (similarity 0 to 1, or match score 0 to 100); never called a probability
unsure	true or false	True when the score is below the task threshold of Section 4
reasons	list of text	Short reasons shown to the user
missing	list of text	Inputs that were not available, such as repair cost or mass
fallback	true or false	True when the result came from the conventional path
versions	object	modelVersionId, prompt version, ranking version, input record versions
aiRecordId	integer	The AI record written for this call (Section 13)

10.2 Contracts per task

Task (caller)	Input facts	value returned	Time limit	On error	Checks before return
classify_asset (Assets & Catalogue)	name, description, photo and PDF if present (no personal data)	{categoryId, tags[], material, sourceUsed: text | photo | pdf}	5 s text; 20 s with photo or PDF	fallback = true; form shows the category list	categoryId exists in Category; unsure if top similarity < 0.60 or lead < 0.10
rank_matches (Search & Requests)	request version, eligible unit IDs (already filtered)	[{assetId, score, factors{S,T,R,C,E}, rank}]	5 s	fallback = true; units in Asset ID order, "not ranked"	Every assetId was in the input list; scores 0 to 100; ties ordered by Asset ID
find_similar (used by rank_matches)	request text, unit IDs	[{assetId, similarity}]	inside the 5 s above	keyword search, labelled conventional	similarity −1 to 1; units below 0.40 flagged weak
recommend_action (Maintenance & Lifecycle)	condition, inspection result, repair cost, replacement baseline, mass, safety flags	{options[{action, reasons[]}], recommended, confidence: High | Medium | Low}	2 min (background)	"No advice available. Assess the options manually."	No option that breaks a safety flag; missing inputs listed
answer_question (assistant panel)	question, records the user may read, approved policy records	{answer, sources[], draftAction or empty}	15 s	"Assistant unavailable. Use Search or New request."	Each source is a record the user may read; a draftAction opens the normal form
draft_summary (Reporting & Sustainability)	report query result and period	{text, numbersUsed[]}	30 s, 2 retries (background)	"Draft summary unavailable"; tables still shown	Every number in text equals a numbersUsed value from the query, or publishing is blocked

10.3 Adapter contracts

Operation	Input	Output	Errors
TextGenerator.generate	task, facts marked as data, JSON schema, time limit	JSON that matches the schema	Timeout: no retry. Network error: one retry. Schema mismatch: treated as unsure, never saved as fact
EmbeddingModel.embed	text up to 256 words	384 numbers	Model not loaded: matching and classification use their fallback

11 Recommendation Score and Explanation

The score ranks units that already passed every mandatory rule; it never makes an ineligible unit eligible. The weights are the Phase 3 starting values and are an assumption to evaluate, not a finding.

11.1 Factors

Factor (label on screen)	Weight	Value between 0 and 1	If missing
S: Function and purpose	0.45	(cosine similarity − 0.40) ÷ 0.50, kept between 0 and 1	Cannot be missing; without the vector the fallback runs
T: Condition fit	0.20	Half from condition (1 if at least the preferred grade, 0.5 one grade lower, 0 lower still; grades New, Good, Fair, NeedsRepair) and half from the share of preferred, non-mandatory specifications met	Preferred condition not given: condition half counts as 1
R: Ready by date	0.15	1 if the unit is ready at least the urgency slack before the required date (High 5 days, Normal 2 days, Low 0); otherwise 0.5 + 0.5 × days early ÷ slack	Cannot be missing; date feasibility is a mandatory rule
C: Repair + transfer cost	0.10	(lowest cost in the set + 50) ÷ (this unit's cost + 50); equal costs give equal values	Dropped for all units in this comparison
E: Environmental benefit	0.10	This unit's supported CO2e or diverted mass ÷ the largest in the set, same boundary only	Dropped for all units in this comparison

11.2 Formula

Score = 100 × (wS·S + wT·T + wR·R + wC·C + wE·E), rounded to a whole number. If a factor is dropped, each remaining weight is divided by the sum of the remaining weights, so they again add up to 1. Units with equal scores are ordered by Asset ID, never by department.

Worked example (Figure 8)	S	T	R	C	E	Score
Dell P2419H, all five factors known	0.91	0.90	0.80	0.70	0.65	100 × (0.45×0.91 + 0.20×0.90 + 0.15×0.80 + 0.10×0.70 + 0.10×0.65) = 84
Same unit if E were missing for any unit	0.91	0.90	0.80	0.70	dropped	Weights become 0.50, 0.22, 0.17, 0.11; score = 87

11.3 Explanation shown with every result

Part	Content	Built from
Score and label	84 / 100, "Strong match" at 70 or more, "Weak match" below 70	Formula above
Factor bars	Five bars with weight and value; a dropped factor shows "not compared"	factorScores
In plain words	Two strongest reasons as sentences, for example "same function, all three have HDMI"	Fixed sentence templates, no language model
Warnings	Partial quantity, missing inputs, weak match	Rules and the missing list
Versions	Embedding model, ranking version, asset and request versions, time	versions in the Suggestion

12 Sustainability Calculations

A benefit is counted only for a completed event (a received transfer or a completed repair) with a validated baseline, never because an item was listed or approved. Each result is stored in SustainabilityImpact with its inputs, so a report can show how it was calculated.

Measure	Calculation	Stored in	Counting rule	Phase 3
Avoided purchase (AED)	replacement baseline − repair cost − transfer cost	avoidedCostAED	A negative result is kept, not turned into zero; a loan return adds nothing	P3-XR-09
Reuse count and reuse rate	Count of distinct completed reuse events; rate = 100 × assets reused by the cut-off ÷ assets first published in the period	Derived from Transfer and Asset	Zero assets published means the rate is shown as unavailable; the cohort period and cut-off are printed	P3-XR-10
Waste diversion (kg)	massKg of each reused or recycled asset, counted once per period	divertedMassKg, isMeasured	Only with a stated disposal counterfactual; reuse, repair and recycling are not added into one total	P3-XR-11
Life extension	Service after the expected retirement date	Asset and MaintenanceRecord dates	No expected date means unavailable	P3-XR-11
Carbon (kg CO2e)	(massKg ÷ 907.185) × WARM factor in MTCO2E per short ton × 1000, minus supported repair and transfer emissions	co2eKg, factorRef	Factor from the U.S. EPA WARM model [1] with version, material category, unit and limits recorded; no default value per asset	P3-XR-12

12.1 Worked example

These are synthetic test values from the Phase 2 report [3], not KU records.

Step	Value
Replacement baseline per event	AED 800
Minus repair cost and transfer cost	− AED 150 − AED 50
Net avoided purchase per event	AED 600
Twenty qualifying events	AED 12,000
Minus Phase 2 first-year project cost	− AED 9,120
Net result	AED 2,880

The reports module recalculates these numbers from the stored inputs every time, and the report summary generator may only quote numbers that come from this calculation (Section 10.2).

13 Model-Monitoring Records

Every AI call writes one AIRecord (Figures 62 and 63). The person's decision is added to the same row later. These rows are the only source for the monitoring figures, so the numbers can be checked against the audit trail.

Field	Content	Why it is kept
task, modelVersionId	Which service ran with which model, prompt and ranking version	Results from different versions are never mixed (P3-NFR-Q11)
inputRefs	Record IDs and their version numbers	A changed asset or request makes an unreviewed suggestion stale (P3-XR-07)
output, score, unsure	What the AI returned and how sure it was	Shows what the person saw
fallback, latencyMs	Whether the conventional path ran and how long it took	Time-limit and availability figures
decision, overrideReason	Accepted, corrected or rejected, and why	Acceptance and override figures (P3-FR-35.2)
createdAt	When the call was made	Monthly grouping; the AuditLog event of the same action names the person

13.1 Metrics and targets

The targets below cover the AI features in Prototype 1. They are targets to evaluate in Phase 7, not guaranteed results.

Metric	Calculated from AIRecord as	Target to evaluate in Phase 7	If missed
Top-3 match rate	Labelled test requests with an acceptable unit in the top three	16 of 20; all 10 ineligible cases excluded	Review weights and rubrics; new ranking version
Classification accuracy	Correct category in the 30 fixed examples	24 of 30	Review category descriptions and prompt
Summary numbers	Numbers in drafts that equal the query result	All numbers in 10 of 10 drafts	Block publishing; review prompt
Timeouts and fallbacks	fallback = true ÷ all calls	Below 5% of calls	Check the service; raise the issue in the team review
Latency	95th percentile of latencyMs per task	Within the Section 4 time limit	Move the task to background if possible
Acceptance rate	Accepted ÷ all decided suggestions, per task and month	Watched, no fixed target	Read the override reasons
Override reasons	Grouped overrideReason text	Watched	Add a rule or change a factor

Later increment: the resource assistant target is correct permitted answers or drafts in 8 of 10 fixed tasks, with every write confirmed.

13.2 Review routine

When	Who	What
Each month	Administrator with the team	Reads the metrics page; any change to weights, thresholds or prompts gets a new ModelVersion row
After every version change	Section owner of that service	Re-runs the fixed evaluation sets before the new version is switched on
Whenever an input record changes	System	Marks unreviewed suggestions on that record as stale

AI-Use Statement

We used Claude, an AI assistant made by Anthropic, to draft diagram layouts and the code that draws them, the mockup screens, and first drafts of the tables and text from our Phase 2 and Phase 3 reports and the Phase 4 lab material. Each section owner reviewed and edited their section, checked every requirement and use-case ID against the Phase 3 report, and checked the balancing tables against the diagrams. The team is responsible for the content.

References

[1] U.S. Environmental Protection Agency, Waste Reduction Model (WARM). https://www.epa.gov/waste-reduction-model

[2] Group 6, Circular Campus Phase 3 Requirements Report, COSC 336, 2026.

[3] Group 6, Circular Campus Phase 2 Report, COSC 336, 2026.

[4] COSC 336 lecture slides, Phase 4 and 5: Sequence, ER and Class Diagrams.

[5] OpenAI, API reference: structured outputs. https://platform.openai.com/docs